



Interpretability research of deep learning: A literature survey

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ABSTRACT

Deep learning (DL) has been widely used in various fields. However, its black-box nature limits people's understanding and trust in its decision-making process. Therefore, it becomes crucial to research the DL interpretability, which can elucidate the model's decision-making processes and behaviors. This review provides an overview of the current status of interpretability research. First, the DL's typical models, principles, and applications are introduced. Then, the definition and significance of interpretability are clarified. Subsequently, some typical interpretability algorithms are introduced into four groups: active, passive, supplementary, and integrated explanations. After that, several evaluation indicators for interpretability are briefly described, and the relationship between interpretability and model performance is explored. Next, the specific applications of some interpretability methods/models in actual scenarios are introduced. Finally, the interpretability research challenges and future development directions are discussed.

1. Introduction

With the rapid development of Artificial Intelligence (AI) technology, Deep Learning (DL) has become a critical technology that promotes progress in fields such as intelligent transportation [1,2], security monitoring [3,4], and big data [5,6]. According to statistics, DL has achieved unprecedented accuracy in multiple tasks, including image processing and speech recognition, significantly improving work efficiency and decision-making quality. However, although DL has shown powerful performance in practical applications, its exceptional performance is often based on extensive data sets and fine experimental tuning. More critically, people currently cannot accurately comprehend useful features learned by models, nor can they clearly understand the basis of the final decision. A complete mathematical theory system is lacking to guide model improvement [7–9]. Thus, this leads to the model-tuning process relying only on trial-and-error experience. This black-box characteristic results in a lack of trust in the application level of the model, which leads to the concern that unexplainable models are being used in real-world scenarios.

The model interpretability is particularly important, especially in critical areas such as medical image analysis, intelligent driving, and security attack and defense. For example, in medical image analysis [10, 11], expert diagnostic results often rely on Machine Learning (ML), but clinicians have much less trust in such undecipherable models. In intelligent driving [12,13], although the models accurately identify road

obstacles, the model cannot explain wrong decisions well. Concerns about life safety have caused considerable obstacles to the application of intelligent driving. Most notably, model non-interpretability presents many potential dangers in security attacks and defense [14,15], significantly reducing the trustworthiness between humans and machines. Therefore, improving model transparency and interpretability has become an urgent priority.

Furthermore, it is difficult for people to explain and predict problems where large models have many parameters, posing challenges to resolving security problems. For instance, a trained model may display good performance and generalization capabilities. However, in an adversarial sample attack network, the network's prediction results can be changed arbitrarily by making tiny perturbations to the input data [14–16]. For example, the MAP-Elites algorithm [17] uses Convolutional Neural Networks (CNNs) that perform well on ImageNet and MNIST datasets. When combining evolutionary algorithm concepts, it randomly generates images that are difficult for humans to recognize, yet the model can identify them as specific objects with 99.99 % confidence. Therefore, even though DL shows good performance in many tasks, trust [18–20], ethics [21–23], and AI bias [24–26] still need to be taken into account. Two keywords, "deep learning" and "interpretability," are also searched in this review. The results revealed that only four papers were published in 2014, followed by 152 papers in 2018, and a remarkable 1894 papers in 2023, and the data comes from <https://web.ofscience.clarivate.cn/wos/alldb/basic-search>. Thus, it is sufficient to

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1566-2535/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Table 1

The investigation perspectives of some interpretability reviews.

Dimension	Research focus	Key contributions	Challenges
Methodological theory	Model-agnostic explanations, mechanistic approaches, comprehensive definitions.	Hassija et al. [20]: Local Interpretable Model-agnostic Explanations (LIME) and Shapley Additive Explanations (SHAP) for model trust and regulatory compliance. Bereska and Gavves [27]: Causality and reverse engineering for AI safety. Murdoch et al. [28]: Theoretical framework for interpretability.	Implementation complexity, scalability issues.
Practical application	Specific agricultural scenarios, medicine diagnosis, reinforcement learning.	Paudel et al. [29]: Feature importance analysis in crop yield. Teng et al. [30]: Various visual explanations for clinical decision-making. Glanois et al. [31]: Structured methods and hierarchical strategies.	Varied evaluation standards, challenges in cross-field comparisons and generalizability.
Evaluation and challenges	Interpretability assessment, methodology and impact evaluation.	Carvalho et al. [32]: Propose quantitative evaluation methods. Gilpin et al. [33]: Meta-explanation and new evaluation perspectives. Chakraborty et al. [34]: Comparison of DL interpretability methods' performance across tasks.	Needs to unify assessment standards, strengthen methods' reliability and stability.

demonstrate that numerous scholars have researched the DL's interpretability in recent years.

So, what interpretable research or Explainable Artificial Intelligence (XAI) is available? Prior scholarly reviews on interpretability research are surveyed, as illustrated in Table 1. Generally, these reviews are categorized into methodological theory, practical application, and evaluation challenges.

These reviews offer valuable insights and theoretical frameworks for follow-up research. It is worth noting that many reviews only address one dimension, lacking a comprehensive investigation. In this review, interpretability from multiple perspectives will be analyzed and redefined, and interpretability methods will be re-categorized and discussed in detail, various interpretability evaluation indicators and practical application scenarios will be listed, and so on. In addition, code libraries of interpretability methods are given to help readers combine theory and practice. Specifically, the following Research Questions (RQs) are addressed:

- RQ1: How can the lack of interpretability in DL models be discovered?
 RQ2: How do scholars arrive at the definition of interpretability?
 RQ3: How should the proposed methodologies be categorized?
 RQ4: What are the common interpretable evaluation indicators?

RQ5: How can the trade-off between interpretability and model performance be balanced?

RQ6: What practical application scenarios require interpretability?

RQ7: What are the current challenges and future developments in interpretability?

In reply to the above RQs, the following sections are elaborated. To answer RQ1, Section 2 briefly describes several classical multi-layer models in DL's different fields. It is also pointed out that the DL model is full of black-box properties by deriving the propagation process of the neural network. Towards RQ2, Section 3 describes that scholars have analyzed a combination of perspectives to obtain some interpretability definitions, and it also redefines interpretability and emphasizes its importance. To address RQ3, Section 4 re-categorizes interpretability methods and describes some methods' principles. In response to RQ4, Section 5 discusses several mainstream interpretability evaluation indicators. In reply to RQ5, Section 6 argues that a good trade-off must be between model interpretability and accuracy, robustness, fairness, transparency, etc. To respond to RQ6, Section 7 presents some practical application scenarios. For RQ7, Section 8 points out current challenges and future development directions. Finally, Section 9 summarizes the research implications of this review. Simultaneously, in Fig. 1, the review's general structure is given.

2. Deep learning

DL is an essential branch of ML, which aims to learn universal patterns from large amounts of data and apply them to unknown data samples. Traditional ML algorithms generally rely on manual feature extraction or rule-based, which is notably practical for simple tasks. However, with the advent of the information age, ML can no longer handle massive, complex data such as images, text, and audio. Therefore, the DL model emerged (e.g., CNNs, RNNs), which can automatically learn the characteristics and patterns from data.

2.1. Typical deep learning models

The most typical DL models can be categorized into several typical network structures [35], including CNNs [36], RNNs (Recurrent Neural Networks) [37], SAE (Stacked Auto-Encoders) [38], and DBN (Deep Belief Networks) [39].

CNNs have demonstrated significant efficacy in image processing [40], effectively extracting features from images, and they consist of convolutional, pooling, and fully connected layers. CNNs achieve efficient feature learning and spatial hierarchical representation by automatically extracting features using convolutional layers. RNNs, on the other hand, are well-suited for processing sequential data, with two prominent models being Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) [41]. LSTM and its variants have extensive application scenarios [42]. RNNs capture temporal dependencies in sequential data, making them ideal for tasks such as time series prediction and language modeling. The Mamba network [43] has recently achieved a decisive breakthrough in DL. Compared to Transformers and CNNs, Mamba can potentially elevate the entire AI industry to a higher level.

SAE and DBN are primarily utilized for unsupervised learning tasks, enabling the automatic learning of data's high-level representations. The SAE comprises multiple layers of sparse autoencoders, where the hidden layer output from each autoencoder serves as the input for the subsequent layer [44]. This structure ultimately connects to a classifier (e.g., softmax). SAE is capable of compressing data into a lower-dimensional latent space, thereby reducing the complexity of the data while preserving the most significant information. This capability is particularly advantageous for high-dimensional data. This model has many practical applications, especially in image processing [45,46], speech recognition [47] and finance [48]. DBN are generative graph

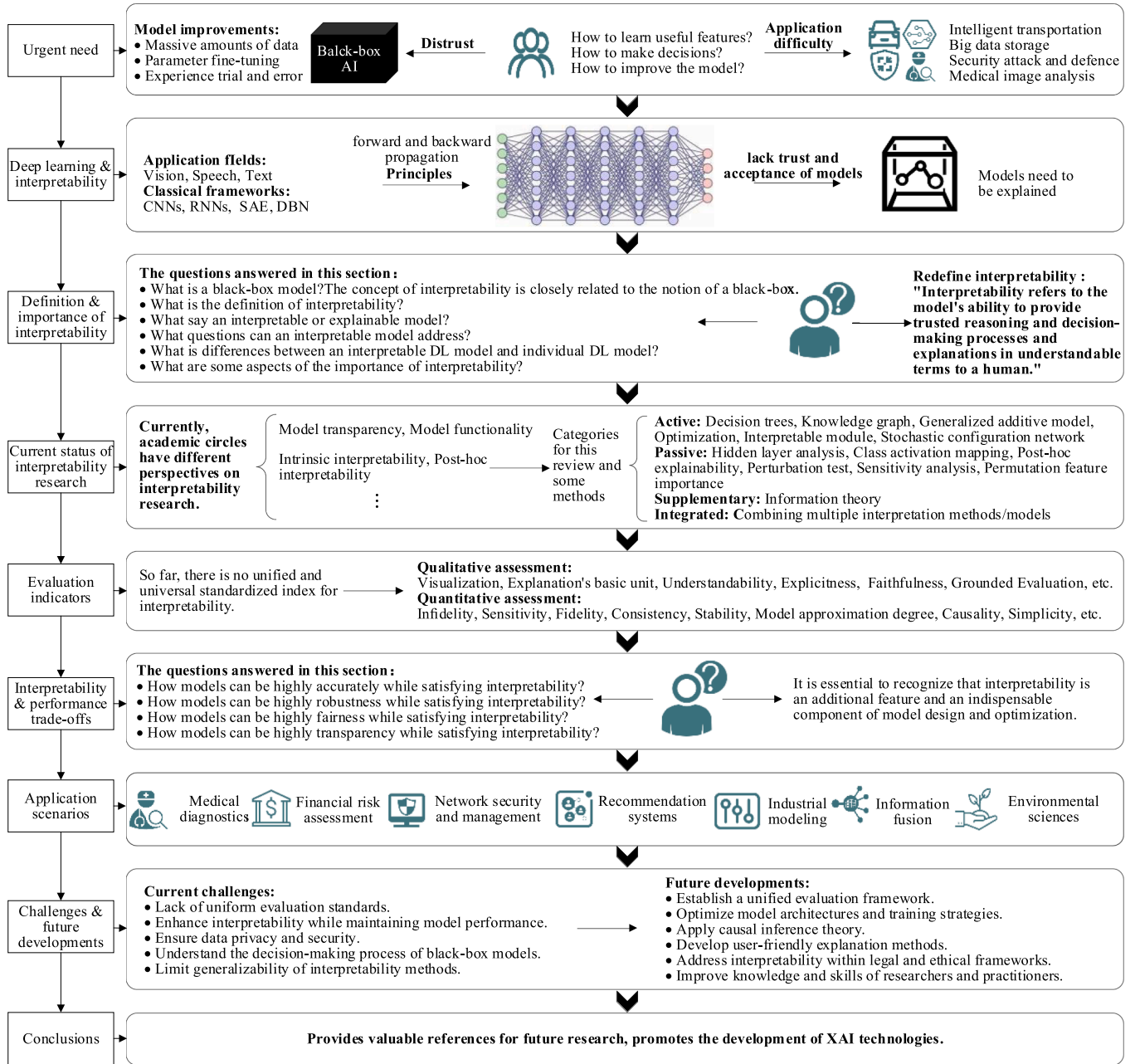


Fig. 1. Framework for reviewing deep learning interpretability research.

Table 2

Main features, advantages and disadvantages of four deep learning models.

Model	Characteristics	Advantages	Disadvantages
CNNs	Grid-like data processing, use for image and video recognition.	Spatial feature capture, efficient, layer sharing.	Sequential data struggle, high computational demand.
RNNs	Specialized in sequential data, can handle varying input lengths.	Handles variable lengths, effective for time-series.	Gradient issues, long-term dependency challenge.
SAE	Unsupervised, multi-layer autoencoders.	Dimensionality reduction, feature extraction.	Time-consuming training, Limited scalability.
DBN	Probabilistic, Hierarchical representations.	Complex distribution learning, pre-training.	Slow training, requires fine-tuning.

models stacked with multiple restricted Boltzmann machine layers, and DBN is also the first DL model to be successfully trained [49]. Each network layer is independently trained through unsupervised learning to discover high-order features in the input data. DBN also has many applications. For instance, Chen et al. [50] proposed an efficient algorithm for detecting anomalous network behavior based on DBN and LSTM. Wang et al. [51] proposed a complexity-based DBN structure optimization algorithm and applied it to the wastewater treatment process. Finally, as shown in Table 2, we summarize these four DL models' main characteristics, advantages, and disadvantages.

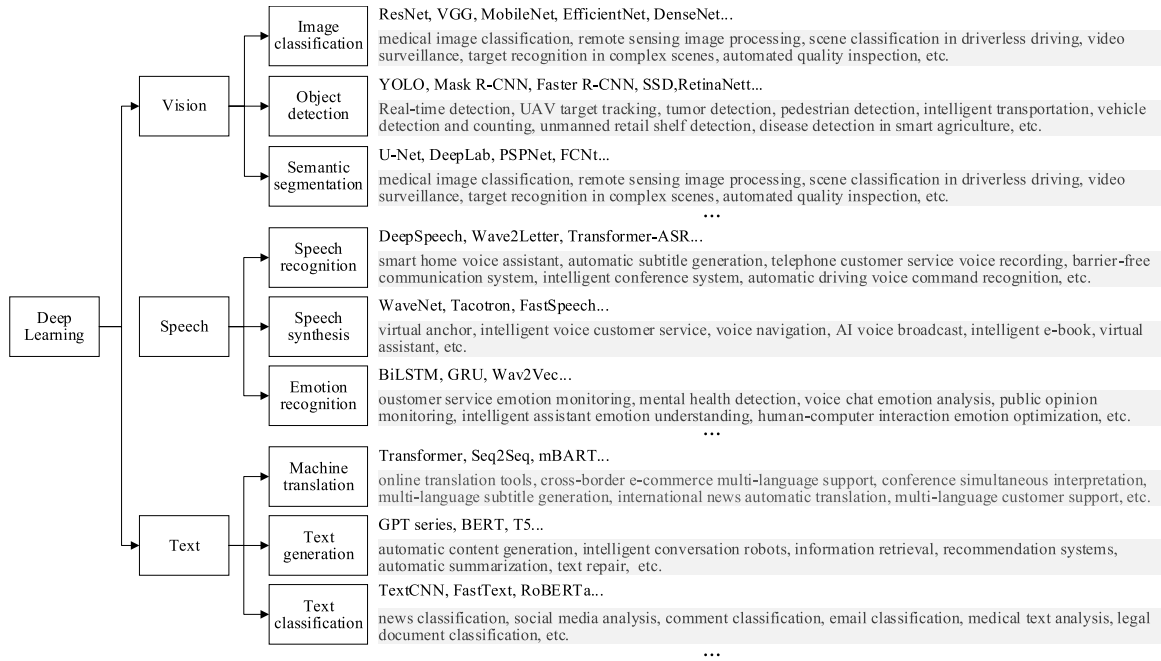
2.2. Principles of forward and backward propagation

The complex biological neural systems inspire the construction of the DL model. The network is made up of hundreds of millions of interconnected neuron nodes. Each node represents a specific output function f , and the connection between each two nodes represents a weight W

Table 3

Neural network forward and backward propagation processes.

Forward Propagation
1. Input layer to first hidden layer: 1) Calculate the weighted sum for each neuron: $z^{[1]} = W^{[1]}X + b^{[1]}$, X is the input vector, $W^{[1]}$ is the weight matrix of the first layer, and $b^{[1]}$ is the bias vector of the first layer. 2) Apply activation function: $a^{[1]} = \sigma(z^{[1]})$, $\sigma(\cdot)$ represents the activation function (such as ReLU, Sigmoid, etc.), $a^{[1]}$ is the output of the first hidden layer. 2. Calculation from hidden layer to hidden layer (layer l to layer $l + 1$): Repeat the same calculation for each hidden layer, $z^{[l+1]} = W^{[l+1]}a^{[l]} + b^{[l+1]}$, $a^{[l+1]} = \sigma(z^{[l+1]})$, $W^{[l+1]}$ and $b^{[l+1]}$ are the weight matrix and bias vector of the $l + 1$ th layer respectively, a is the activation value of layer l. 3. Calculation of output layer: The output of the last layer is the predicted output of the neural network: $\hat{y} = \sigma(W^{[L]}a^{[L-1]} + b^{[L]})$, $\hat{y}$ is the predicted output of the model, and L is the total number of layers of the network.
Backward Propagation
1. Calculate the loss function: Define the loss function $J(\hat{y}, y)$, such as cross-entropy loss or mean square error. 2. Gradient calculation of output layer: Calculate the error of the output layer: $\delta^{[L]} = \frac{\partial J}{\partial a^{[L]}} \odot \sigma'(z^{[L]})$, $\delta^{[L]}$ is the error of the output layer, $\sigma'(z^{[L]})$ is the derivative of the output layer activation function, $\odot$ represents element-wise product. 3. Propagation of hidden layer gradients: Calculate the error for each hidden layer: $\delta^{[l]} = (W^{[l+1]})^T \delta^{[l+1]} \odot \sigma'(z^{[l]})$, $\delta^{[l]}$ is the error of layer l. 4. Calculate the gradient: Calculate the gradient of the loss function for each layer weight and bias: $\frac{\partial J}{\partial W^{[l]}} = \delta^{[l]} (a^{[l-1]})^T$, $\frac{\partial J}{\partial b^{[l]}} = \delta^{[l]}$, $\frac{\partial J}{\partial W^{[l]}}$ is the gradient of the weight of the l-th layer, and $\frac{\partial J}{\partial b^{[l]}}$ is the gradient of the bias of the l-th layer. 5. Parameter update: Update weights and biases using gradient descent or other optimization algorithms: $W^{[l]} := W^{[l]} - \alpha \frac{\partial J}{\partial W^{[l]}}$, $b^{[l]} := b^{[l]} - \alpha \frac{\partial J}{\partial b^{[l]}}$, α is the learning rate.

**Fig. 2.** Typical deep learning application scenarios and features.

that transmits signals through the connection. During the forward propagation, the data x is fed into the hidden layer $l = 1, 2, 3, \dots, L - 1$ from the input layer. After being weighted $z^{[l]}$, the output $a^{[l]}$ is obtained through the function f , and it is transmitted to the neurons of the next hidden layer or as the output layer prediction result $\hat{y}$. In the back-propagation process, the loss function calculates the error between the prediction result $\hat{y}$ and the actual value y , and then the error is back-propagated. The chain rule calculates the gradients $\partial J / \partial W$ and $\partial J / \partial b$ of each layer's parameters, thereby continuously updating the weights W and biases b , making the prediction results closer to the actual values. This iterative process learns the nonlinear relationships and abstract features of complex data layer by layer, which can provide a practical

capability for processing complex tasks. The forward and backward propagation processes are illustrated in Table 3 below.

2.3. Typical application of deep learning

In recent years, the continued popularity of Deep Neural Networks (DNNs) has led to remarkable advancements in various fields, which have achieved massive success in many applications, such as Vision (e.g. visual analytics), Speech (e.g. speech recognition), and Text (e.g. text processing). In Fig. 2, some examples are given, and in the following paragraphs, small fields are introduced in detail.

In visual analytics, some current algorithms have been able to be

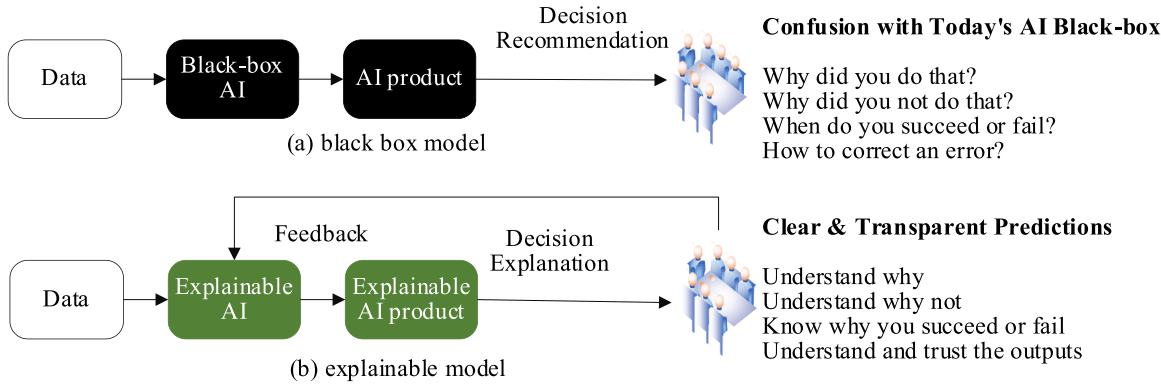


Fig. 3. The difference between black box AI models and explainable AI models.

maturely applied in practical Computer Vision (CV) scenarios, such as image classification [52], object detection [53], image segmentation [54], etc. For example, Wang et al. [55] introduced a novel DL method, BrantNet, which can detect open biomass burning, and aims to protect the environment and people's health. Ma et al. [56] proposed an improved remote sensing image segmentation model, SWINT-RESNet. The purpose is to identify complex regions, including buildings, low vegetation cover areas, etc.

With the persistent development of DL, speech recognition has reached a higher level, and the mainstream model can be generally categorized into seq2seq and HMM structures. It covers various typical models, such as DCN, DMN, DNN-HMM, etc., for broad applications in various scenarios [57], such as intelligent home, voice broadcast, intelligent customer service, etc. Surprisingly, speech recognition technology can also enhance the convenience of household garbage collection [58], significantly enhancing our expectations of its potential contribution to environmental health. Furthermore, speech recognition has significantly enhanced speech-to-text conversion accuracy [59]. It allows researchers to conduct more exploration in text processing tasks.

Text processing is essential in Natural Language Processing (NLP), which involves analyzing, understanding, and generating textual data. With the emergence of Large Language Models (LLMs) [60,61], text processing development has undergone a qualitative change to some extent. Some search engines, like ChatGPT, Claude, and Coze, have strong text retrieval capabilities. Amazingly, some scholars are researching the correlation between ChatGPT and biological/environmental sciences [62,63]. Currently, DL is also gradually applied in some domains of higher security risk, including medical diagnosis [64], autonomous driving [65], and financial analysis [66].

The application of these algorithms needs to be based on the prerequisite of ensuring the safety of people's lives. These algorithms are mostly built on DNNs, so DL interpretability must be explored. Because interpretability is directly related to the security of DL models, especially in fields such as autonomous driving, medical diagnosis, and financial analysis, algorithms' transparency is crucial to security. It can solve the DL black box problem, visualize model decision-making, and allow humans to understand the model behavior intuitively in the future.

Despite significant progress in DL, the decision-making and learning mechanisms, the forward and backward propagation processes in the neural networks' multi-layer structures, and how they contribute to superior performance remain unknown. Thus, it leads to the subject of the DL black-box nature. This lack of transparency limits the model's wide application in various fields. Users and decision-makers are often skeptical of incomprehensible models which lack trust and acceptance of models.

3. Definition and importance of interpretability

With the advancement of AI, particularly in DL, interpretability is closely linked to the notion of black-box models. Understanding the relationship between these two concepts is essential, as it fosters trust and acceptance of AI models in real-world applications.

3.1. Black-box model and interpretability

Black-box (empirical) models refer to phenomena with rarely known internal laws [67]. As shown in Fig. 3, these models' (Black-box AI)

Etymology	Definition: "Interpretability" combines "interpret" (to explain) and "-ability" (capability).	Logic	Definition: Logical conditions required for explanations.
	Key Point: Indicates the characteristic or ability of something to be explained or understood.		Key Points: • Deductibility: The explanation must logically follow from given information. • General Laws: Must include universal principles. • Empirical Content: Should be testable.
Philosophy	Definition: Philosophical understanding of explanations.	Human Cognition	Definition: Role of human factors in understanding ML models.
	Key Points: • Focus on "why" questions. • Context-specific nature of explanations.		Key Points: • Human interpretation and verification of model decisions. • Integration of cognitive science in interpretive ML.

Fig. 4. Several perspectives about the interpretability meanings.

Table 4
Difference between "Interpretability" and "Interpretation".

Nouns	Means
Interpretation	According to the literature [18], "Interpretation" is often called an interpretation algorithm. Computational explanations or attributions elucidate how deep models make decisions [74,75].
Interpretability	"Interpretability" pertains to the intrinsic properties of deep models and evaluates the extent to which the inference results of these models are predictable or comprehensible to humans [76].

inner workings frequently remain opaque to the user. The emphasis is typically placed on the model's inputs and outputs in model building, while specific implementation details are often overlooked. Although these models can achieve high-accuracy predictions, they lack interpretability. It complicates understanding the reasons for these highly accuracy prediction results.

With DL's widespread application across various fields [68,69], black-box models, such as CNNs and RNNs, have become mainstream. However, the internal decision-making processes often remain opaque, which is a challenge in some application scenarios. Consequently, researchers have proposed new methods/models to enhance the transparency and understandability of black-box models, called model interpretability analysis. By elucidating black-box model internal mechanisms, this research clarifies the decision-making process. It fosters user trust in the models' predictions, addressing the shortcomings in understanding and trust associated with black-box models and encouraging adoption in practical applications.

3.2. Multidimensional perspectives on interpretability

What is "Interpretability?" [70] In Fig. 4, the etymology of "interpretability" traces back to the Latin "interpretari," which means to explain or clarify. The suffix "-ability" derives from the Latin "habilitatem," signifying talent or capability. Therefore, "interpretability" denotes the property or ability of something to be explained or understood. In AI, interpretability is frequently characterized by understandability, transparency, and clarity.

From the logical perspective, Hempel and Oppenheim [71] assert that explanations must meet specific logical criteria, including logical derivation, the inclusion of universal laws for derivation, and empirical content. Philosophically, Van Fraassen [72] highlights the significance of the "why" in explanations, noting that such explanations are context-dependent rather than exclusively scientific. Concurrently, Narayanan et al. [73] underscore the necessity of incorporating human factors and cognitive science in understanding and validating the

predictions or decisions of ML models. These researchers argue that even though these algorithms are technically intricate, the value depends on the capacity to provide comprehensible explanations. Thus, a foundational understanding of interpretability is established, whether viewed through etymology, logic, philosophy, or human cognition, enhancing the comprehension of "interpretability" in DL.

3.3. Definitions of interpretability

What is the definition of interpretability? Before researching the definition of interpretability, it is essential to distinguish between the concepts of "Interpretability" and "Interpretation" to avoid confusion. Table 4 shows the difference between "Interpretability" and "Interpretation."

Currently, there are still many different views on the definition of interpretability. As illustrated in Fig. 5, At the 2017 International Conference on Machine Learning, Been Kim, a research scientist at Google Brain, discussed the concept of interpretability [76], first defining interpretability as "Interpretation is the process of giving explanations to Human." She also pointed out that, besides the application perspective, making ML interpretable promotes its more equitable use. She argued that a tool as powerful as ML should not be understood and used by only a few. It should benefit a broader range of people and become available to all.

Harvard University, Doshi-Velez et al. [76] also analyzed the interpretability of ML systems based on eight key features. It proposed a definition: "In the context of ML systems, we define interpretability as the ability to explain or to present in understandable terms to a human."

In 2019, reference [77] was co-authored by several research organizations, including TECNALIA, UPV/EHU, BCAM in Spain, ENSTA in France, etc. Based on some features and concepts, such as transparency and understandability. This analysis redefined the concept: "Interpretability refers to a passive characteristic of a model referring to the level at which a given model makes sense for a human observer. This feature is also expressed as transparency." The target audience is crucial in defining explainability, as different audiences may require different levels of explanation and understanding.

In 2021, a chief professor at the University of Birmingham, Aleš Leonardis et al. [78], proposed a comprehensive definition of interpretability after thoroughly considering various factors, including understandability, logical decision rules, etc. Interpretability was defined as "Interpretability is the ability to provide explanations in understandable terms to a human." The phrase "understandable terms" in this definition highlights the importance of presenting explanations. That is, interpretability should be easily comprehensible to human users.

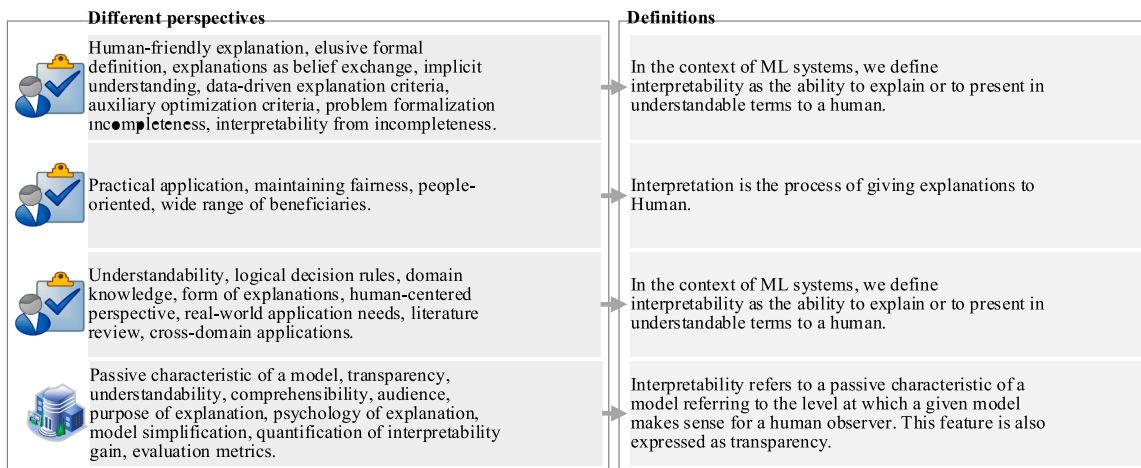


Fig. 5. Interpretability definitions from different scholars and research organizations.

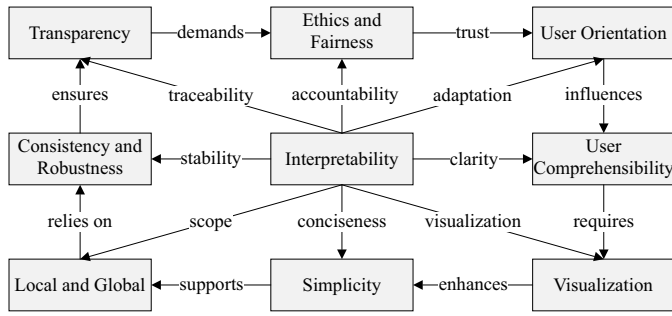


Fig. 6. Multi-perspective analyses redefine interpretability.

Table 5

Essential questions for interpretable models.

Specific aspects	Questions	Examples
Principles behind model decision making	Why does the model arrive at a particular prediction or decision?	Lundberg and Lee [79] assert that the primary objective is to establish a framework that facilitates the understanding and explaining the model's predictive decisions.
Address important features	Which input features significantly influence the model's predictions or decisions?	Shrikumar et al. [80] introduced a method known as Deep Learning Important Features (DeepLIFT), a decomposition technique for elucidating the relationship between specific input and output predictions of neural networks.
Settles model behavior	How does the model make decisions under varying input conditions?	Montavon et al. [81] introduced a novel method known as deep Taylor decomposition, which elucidates the influence of individual input elements on the classification decisions made by a multi-layer neural network.
Solves model uncertainty	In what circumstances might the model be deemed unreliable?	Lakshminarayanan et al. [82] proposed a deep ensemble-based approach to estimate and evaluate the model's prediction uncertainty. The author also discussed scenarios where the model may become unreliable due to distribution shifts and unforeseen circumstances.

Interpretability multidimensional analyses by different scholars can reveal the inherent link between model interpretability and human comprehension. Specific analyses can be developed from the following perspectives:

- 1) **User orientation.** The content and form of the explanation must satisfy the expectations of humans with prior knowledge.
- 2) **User comprehensibility.** The model must demonstrate the reasoning process in comprehensible terms to humans.
- 3) **Visualization.** The model's output can be intuitively presented through graphs, charts, rules or other tools to help humans understand its decision-making process or results.
- 4) **Simplicity.** Explanations should be clear and concise, avoiding overly complex terminological descriptions.
- 5) **Local and global.** The model's behavior on a single data point can be interpreted (locally), while the model's behavior on the overall dataset can also be interpreted (globally).
- 6) **Consistency and robustness.** Explanations are consistent across data samples or scenarios, which can respond to perturbations or changes in the data and remain reliable.
- 7) **Transparency.** The model's decision-making process is traceable.

8) **Ethics and fairness.** The ability to reveal and account for potential bias or discrimination ensures fair and ethical decision-making process and outputs.

Based on the above eight perspectives on the characteristics, this review redefines interpretability: "**Interpretability refers to the model's ability to provide trusted reasoning and decision-making processes and explanations in understandable terms to a human.**" Fig. 6 illustrates the connection between the eight perspectives and how interpretability relates to these eight perspectives.

3.4. Interpretable dl models

According to the interpretability definition mentioned above, an interpretable or explainable model is one whose operation and decision-making processes can be comprehended and articulated by humans. Such models typically possess a transparent structure that enables individuals to trace each step from input to output. For instance, traditional ML models, such as linear regression and logistic regression, are frequently regarded as interpretable due to explicit and traceable decision-making processes.

What questions can an interpretable model address? In Table 5, this review proposes that it encompasses four specific aspects.

Interpretable DL models significantly differ from conventional DL models in several key aspects. Firstly, interpretable DL models offer greater transparency, making their internal operations and decision-making processes more comprehensible to humans. In contrast, traditional DL models, such as DNNs, are often labeled black-box models due to the complexity and opacity of their decision-making processes [83]. Secondly, interpretable DL models can assess the importance of input features, enabling users to identify which features most influence decision-making and whether this capability is less accessible in conventional DL models. Furthermore, some interpretable DL models can deliver local explanations for specific inputs, elucidating the model's behavior in response to particular data points, a feature that conventional DL models typically lack. Finally, interpretable DL models enhance transparency and interpretability and are more likely to foster user trust and acceptance in practical applications [84].

3.5. Importance of interpretability

Meanwhile, scholars generally recognize the importance of interpretability in the following aspects:

- 1) **Avoid catastrophic consequences** [85]. Interpretability helps to avoid serious social problems caused by model poor transparency, including negative impacts related to health, freedom, racial prejudice, and security.
- 2) **Build trust and comply with regulations** [86]. Interpretability helps to meet regulatory and policy requirements for AI systems, avoid legal and economic risks caused by illegal operations, and build user's trust in these systems.
- 3) **Model debugging and improvement** [87]. Interpretable models can help developers identify and solve model problems and deficiencies so that the model can be debugged and improved to enhance performance and accuracy.
- 4) **Promote transparency and user engagement** [88]. Interpretable models can provide clear explanations for decision-making, making it easier for users to understand and trust the systems' working principles, which promote user participation in the system's decision-making process.
- 5) **Detect and correct bias** [89]. Interpretability helps to detect biases and injustices in modeling so that steps can be taken to correct them.
- 6) **Improve scientific applications and insights** [90]. Interpretability helps to reveal underlying structures and relationships in

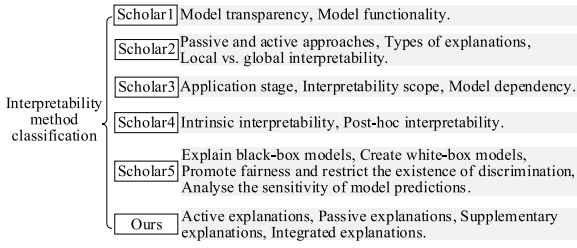


Fig. 7. Some categorizations of interpretable methods.

data, provides an understanding of neural networks, and offers valuable insights for scientific research and applications.

4. Current status of interpretability research

4.1. Interpretability method's classifications

Currently, academic circles have different dimensions/perspectives on interpretability research. Thus, the classifications of interpretability methods proposed by various scholars are different. For example, in Fig. 7, Chakraborty et al. [34] categorized prior research into two dimensions of interpretability: model transparency and model functionality. Model transparency is defined in terms of three parameters: simulatability, decomposability, and algorithm transparency, whereas model functionality is defined in terms of textual description, visualization, and local explanation.

Zhang et al. [78] also categorized interpretability into three dimensions: passive and active approaches, types of explanations, and local vs. global interpretability. In the first dimension, the passive interpretation does not require network structure or optimization process modifications, and conversely, the active interpretation method necessitates alterations during training, modifying the network architecture or optimization process to enhance interpretability. The second dimension, types of explanations, is intended to make it possible to observe differences in the degree of clarity of the various interpretations produced by different methods. In the last dimension, local explanations usually use the information at the target input, but global explanations try to generalize to as wide a range of inputs as possible.

Chen et al. [91] also classified interpretability technology into three dimensions: application stage, interpretability scope, and model dependency. They divided it into ante-hoc and post-hoc according to the application stage dimension. Ante-hoc (usually self-explanatory) and post-hoc (apply to black-box models) methods are applied during and after the model training process. Interpretability scope refers to the model output's scope that needs to be interpreted, including global and explanation. Global interpretation is explained based on a full view of the model structures and parameters, and local interpretation explains each prediction individually. Model dependency refers to whether the interpretable ML technique can be applied to any ML model or specific models, including model-agnostic and model-specific. Model-agnostic techniques can be applied to any ML model, and Model-specific techniques can dig into the specific characteristics or architecture of the ML model.

In addition, Du et al. [92] categorized interpretable ML into two main categories: intrinsic interpretability and post-hoc interpretability. Both can be further classified into global interpretability and local interpretability. Linardatos et al. [93] categorized interpretability methods into four groups: explain complex black-box models (related to black-box pre-trained ML models), create white-box models (create interpretable and easily understandable from human models), promote fairness and restrict the existence of discrimination (the most widely used ML fairness method), analyze the sensitivity of model predictions (attempts to represent each input variable with a single value, or is perturbed by a small but sufficient number of interferences).

Table 6
Characteristics of interpretability methods.

Categories	Methods	Characteristics
Active	1. Decision trees.	1. Ability to handle nonlinear relationships and insensitive to outliers and missing values.
	2. Knowledge graph.	2. Simple structure, strong interpretation, high accuracy, logical reasoning.
	3. Generalized additive model.	3. Each prediction result has a nonlinear expression.
	4. Optimization.	4. Introduce activation regularization or improve the loss function and use related optimization methods to explain.
	5. Interpretable module.	5. Introduce interpretable modules and modify the original network structure.
	6. Stochastic configuration network.	6. Configure hidden layer node parameters through a supervision mechanism.
Passive	7. Hidden layer analysis.	7. Generate human-understanding images through visualization methods to explain hidden meanings.
	8. Class activation mapping.	8. Use heatmaps to visualize the region of interest in the trained model.
	9. Post-hoc explainability.	9. Apply to existing models after completing training to provide model interpretability.
	10. Perturbation test.	10. Use carefully designed perturbation inputs to impact model predictions.
	11. Sensitivity analysis.	11. Identify the most influential factors by systematically changing input variables and observing and quantifying the impact of those changes on output results.
	12. Permutation feature importance.	12. By randomly shuffling the values of a single feature, the impact of this perturbation on model performance is evaluated to quantify the importance of that feature.
Supplementary	13. Information theory.	13. Integrate concepts related to information theory into DL to obtain more interpretive information.
Integrated	14. Improve understanding and interpretation of model decision-making processes by combining multiple interpretation methods or models.	

Current interpretability research is indeed experiencing a period of contention among various schools of thought. Many scholars categorize interpretability research according to different dimensions, facilitating a more nuanced understanding of this field. In the meantime, the black-box problem associated with contemporary DL models can be addressed. Based on the above others' reviews, this review found that the interpretability of DL models largely hinges on whether they involve modifications to the target model. The primary aim of the review is to elucidate the principles of these methods for readers through a detailed examination of the literature and methodologies of objective interpretability research. This review aims to introduce relevant evaluation indicators and application scenarios. Readers will be better equipped to absorb this knowledge by organizing relevant discussions from various perspectives.

Therefore, the review divides interpretability methods into four categories: active, passive, supplementary, and integrated explanations. Active and passive interpretation depends on whether the model structure is modified/optimized before the model is trained to make the model somewhat interpretative or an inherently transparent model is designed. Consequently, a brief overview of the range of interpretability methods covered in this review is given in Table 6. Subsequently, these methods are introduced in detail in subsections.

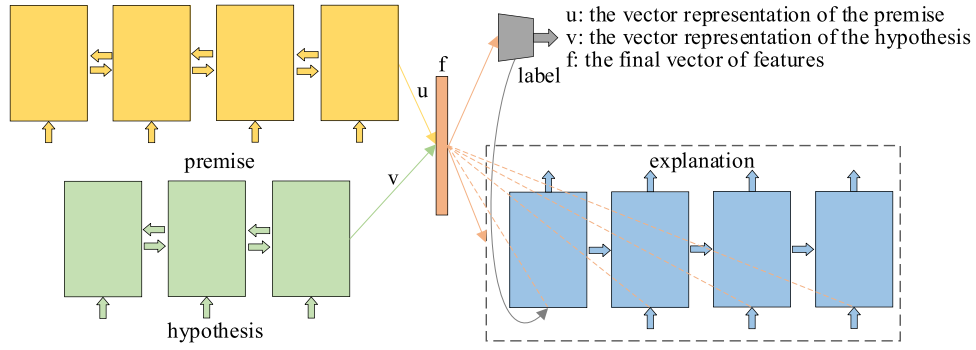


Fig. 8. Architecture illustration of the self-explanatory model BiLSTM-Max-PredExpl.

4.2. Active explanations

Active explanations refer to active intervention explanations. Active explanations involve intervening in the model's structure before training or designing transparent model structures, which can perform tasks while enhancing interpretability and making the output more intelligible for humans. As shown in Fig. 8, it is a standard self-interpretation model, and the authors augmented the BiLSTM-Max model with an explanation generator to create a new model, BiLSTM-Max-PredExpl [94].

4.2.1. Decision tree

A Decision Tree (DT) [95] is an algorithm widely used to process numerical and categorical data and capture the interactive information between features. A series of "if-then" decision rules in its tree structure intuitively display the decision-making process and reasoning logic, which provide a basis for a profound understanding of the model's decision-making and predictions.

Given a DT with N nodes, each node i has a condition M_i and a decision output y_i , and the condition constitutes a logical assessment of an input feature, such as $x_j > a$. Where the input feature component x_j is represented, and a is a threshold. T can be denoted as:

$$T(X) = \sum_{i=1}^N y_i \cdot \prod_{j=1}^{M_i} I(C_{ij}(X)) \quad (1)$$

where $T(X)$ represents DT's output for the input sample X , $C_{ij}(X)$ denotes the judgment result of the j -th condition at node i when applied to the input sample X , $I(\cdot)$ signifies the indicator function, 1 if the condition in parentheses is true, 0 otherwise.

In recent years, many scholars have used DT with DL to achieve interpretability [96–98]. For instance, within simple regions of the input space, the method of Classification And Regression Tree (CART) [99] employs simple geometric interpretation to maintain constant function values. This method is not only directly visualizable but also can be visualized in low dimensions of the corresponding partitions. However, visualization in high dimensions has certain limitations in improving interpretability. CERD (Continuous/discrete Rule Extractor via Decision tree induction) [100], as another typical case, connects the output layer y of each category to the input layer x through layer-by-layer extraction and applies the C4.5 algorithm [101] in N hidden layers. Based on the hidden layer h_{k-1} , the decision rule $R_{h_{k-1} \rightarrow h_k}$ ($k = 2, 3, 4, \dots, N$) of the hidden layer h_k is acquired, and all the rules are merged into $R_{h_N \rightarrow y}$, $R_{h_{N-1} \rightarrow N}, \dots, R_{h_1 \rightarrow h_2}, R_{x \rightarrow h_1}$. Ultimately, the output rule set $R_{x \rightarrow y}$ that describes the network is obtained according to the input of the network model. CART can construct tree structures similar to the original model, but the generation scale is too large, and its computationally time-consuming nature limits its scalability.

Consequently, to solve the problems of excessive scale and poor prediction performance of a single DT, Wu et al. [102] introduced a novel regularization approach, Regional Tree Regularization (RTR),

Table 7

The basic forms of average decision path length for Global/L1/L0.

Methods	Basic forms of APL
Global [103]	$\Omega^{\text{global}}(\theta) \triangleq \text{APL}(\{x_n\}_{n=1}^N, f(\cdot, \theta))$
L1	$\Omega^{\text{regional-L1}}(\theta) \triangleq \sum_{r=1}^R \text{APL}(X_r, f(\cdot, \theta))$
L0	$\Omega^{\text{regional-L0}}(\theta) \triangleq \max_{r \in \{1, 2, \dots, R\}} \text{APL}(X_r, f(\cdot, \theta))$

Table 8

Expression formulas for two tree supervision losses.

Types	Basic forms
Hard-TSL	$L_{\text{hard}} = L_{\text{CE}} + \frac{1}{N} \sum_{i=1}^N L_{\text{CE}}^{(i)}$
Soft-TSL	$L_{\text{soft}} = L_{\text{CE}} + \frac{1}{N} \sum_{i=1}^N p_i L_{\text{CE}}^{(i)}$

aiming to improve model interpretability and this method employs the Average Decision path Length (APL) to evaluate interpretability. Table 7 lists several basic forms of APL for tree regularization. Where $f(\cdot, \theta)$ denotes the target neural network, $\{x_n\}_{n=1}^N$ represents the input dataset, $X_1, X_2, \dots, X_R$ signifies the area set covering the entire input space.

In predefined input space regions, the region tree regularization method improves the model interpretability by utilizing multiple DTs while maintaining or enhancing the model's performance. However, this method has a severe impact on the accuracy.

The Neural-Backed Decision Trees (NBDTs) method [104] solves the trade-off problem between accuracy and interpretability. The method is implemented in four steps: inference, building induced hierarchies, labeling decision nodes with Wordnet, and fine-tuning with Tree Supervision Loss (TSL). The method enables to provide better interpretability while maintaining or improving accuracy. In addition, NBDTs are also able to identify fuzzy labels in the dataset through path entropy, further enhancing model interpretability. The TSL proposed in the article includes Hard-TSL and Soft-TSL, as shown in Table 8. Here, L_{CE} represents the original cross-entropy loss, $L_{\text{CE}}^{(i)}$ denotes the cross-entropy loss for the i -th internal node, and p_i indicates the probability of the i -th internal node.

Besides, Sagi and Rokach [105] introduced a new model, Forest Based Tree (FBT), which transforms a decision forest into a more interpretable decision tree. The significant contribution of FBT lies in broadening the scope of models that achieve a balance between predictive accuracy, computational speed, and interpretability. Moreover, unlike other approaches, FBT can be effectively utilized with large decision forests without hyperparameter optimization. The implementation process of FBT is outlined as follows:

When working with a data set, $D = \{(x_i, y_i) | D\} = n, x_i \in R^m, y_i \in 1, \dots,$

Table 9

Summary of advantages and disadvantages of decision tree-based interpretability methods.

Advantages	Disadvantages
Easy to understand and interpret	Not applicable to linear relationships
Able to handle nonlinear relationships	Poor stability
Possesses good interpretability	Not smooth decision boundaries
Insensitive to outliers and missing values	Easy to overfit

Table 10

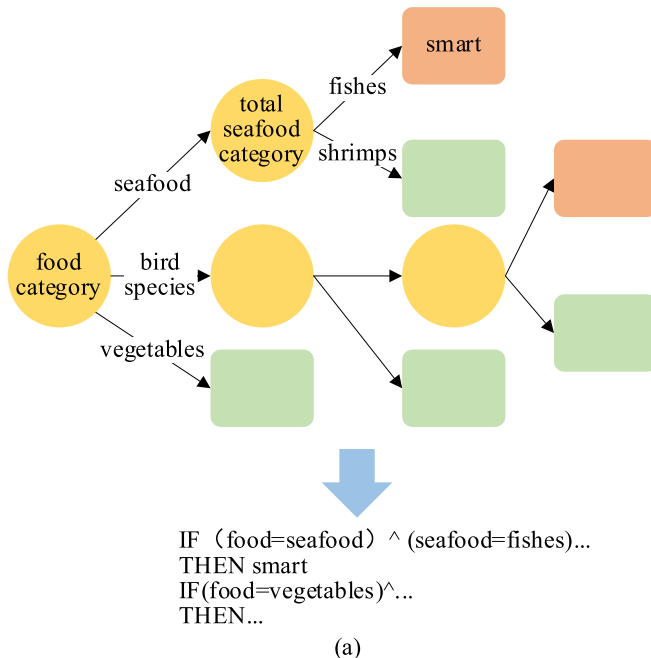
Decision tree-based interpretability methods characterization and provision of code links.

Methods	Characteristics and code links
DT [95]	Data mining, classification prediction, non-parametric, decision tree model, optimize tree structure. https://github.com/igrigorik/decisiontree.git
CART [99]	Recursive partitioning, optimal feature selection, decision tree, classification, regression. Recursive partitioning, optimal feature selection, decision tree, classification, regression https://github.com/Erikfater/Decision_tree-python.git
NBDTs [104]	Neural-backed, decision tree, interpretability, accuracy enhancement, dataset debugging. https://github.com/alvinwan/neural-backed-decision-trees.git
FBT [105]	Decision forest transformation, interpretability, rule conjunctions, tree structure. https://github.com/sagyome/forest_based_tree.git
FTF [107]	Fuzzy clustering, fuzzy decision trees, ensemble classifiers, flexibility, robustness. https://github.com/ZhaoqingLiu/FuzzyTrees.git

$c\}$, that includes n samples, m features, and c different categories, the decision forest combines K additive functions to convert m -dimensional feature vectors into c -dimensional probability vectors:

$$\phi(x_i) = \frac{\sum_{k=1}^{|T|} t_k(x_i)}{|T|}, t_k \in R^c \quad (2)$$

where T is a set of DT contained in the decision forest, the proposed method aims to construct a new tree such that:



$$\forall x_i, \hat{t}(x_i) \approx \phi(x_i) \quad (3)$$

The FBT method comprises two main stages. The first stage involves constructing a set of rule connectives from a given decision forest, while the second stage focuses on organizing these connectives into a tree structure for efficient prediction of unseen instances.

In the first stage, the hierarchical structure of the decision tree t_i contained in the decision forest T is ignored, and the tree is regarded as a set of regular connectives CS_i . Each connective word $(c_{ij}, \hat{y}_{ij})$ in CS_i is a logical rule sequence c_{ij} mapped to $\hat{y}_{ij}$, which is a K -dimensional vector, where K is the number of labels, and each unit saves the probability of the corresponding class. The two conjunctive sets CS_1 and CS_2 can be merged by applying a cartesian product, where each conjunction $(c_{1j}, \hat{y}_{1j})$ is merged with the conjunction $(c_{2j}, \hat{y}_{2j})$ to create a new conjunction $(c_{1j} \wedge c_{2j}, \hat{y}_{1j} + \hat{y}_{2j})$.

In the second stage, an algorithm is introduced to solve the problem of organizing a set of connectives into a tree structure. The goal is to enable efficient classification and provide a mechanism to associate the most suitable subset of connectives with unused ones. The method allows for matching instances, even when no connective matches a given instance exactly.

DT-based interpretability methods have been further developed, including but not limited to CERD [106], Fuzzy Frees and Forests (FTF) [107], mixed-integer linear optimization [108], etc. Some advantages and disadvantages of DT methods are summarized in Table 9.

Based on the introduction of methods in scholars' papers, the characteristics of DT and its variants are summarized in Table 10, along with code links for readers' convenience.

4.2.2. Knowledge graph

In 2012, Google derived the Knowledge Graph (KG) from Metaweb to provide relevant results during the search process. With the advancement of DL, KG has gradually become one of the core technologies in multiple cognitive intelligence fields, such as information retrieval [109], question-answering systems [110], and image vision [111]. KG is a directed graph that contains entity nodes E and edges R

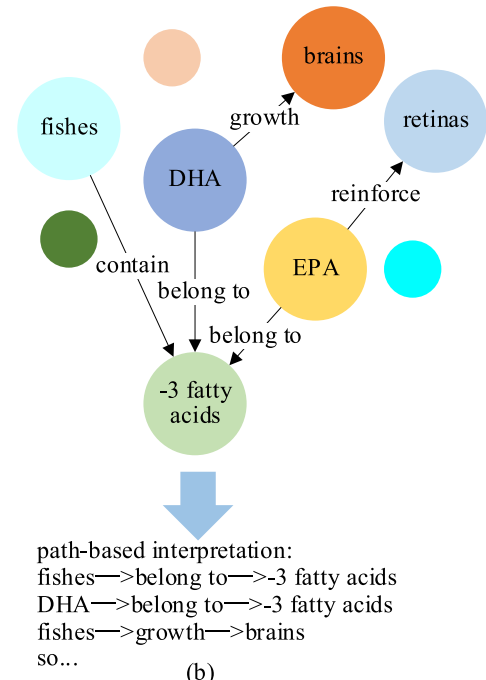


Fig. 9. Comparison of decision tree rules and knowledge graph path method.

representing the relationships between entities. The fundamental unit, a triplet (h, r, t) , consists of a head entity h , a tail entity t , and a relation r connecting the head and tail. For example, the triplet $(\text{Madrid}, \text{capital}, \text{Spain})$ is represented as "Madrid is the capital of Spain."

$$KG = \{(h, r, t) | h, t \in E, r \in R\} \quad (4)$$

The heterogeneous graph structure of the KG exhibits superior interpretability compared $KG = \{(h, r, t) | h, t \in E, r \in R\}$ to traditional data structures. For instance, in Fig. 9, DT(a) and KG(b) are used to extract rules summarizing the causes for food-related health outcomes, and (b) is more interpretable than (a). This article introduces KG's path and embedding methods to study model interpretability.

(1) Path method

In recent years, KG has been widely used in recommendation systems. As a classical approach, the Knowledge-aware Path Recurrent Network (KPRN) [112] uncovered the connection path $p(u, i) = [u \rightarrow^{r_1}, \dots, \rightarrow^{r_{l-1}} i]$ between user $\{u_t\}_{t=1}^M$ and item $T = \{i_t\}_{t=1}^I$ by mining relationships in KG, which provides a wealth of supplementary information for user-item interactions. Empower recommendation systems with reasoning capabilities and explainability. KPRN consists of an embedding layer, an LSTM [113], and a weighted pooling layer.

Firstly, the path's entity types and specific values are projected into embedding vectors, explicitly incorporating the semantics of relationships into path representation learning.

Subsequently, an LSTM network is employed to capture the combined semantics of entities and relationships. A series of LSTM units are utilized to process the sequence of elements in the path and generate the final state representing the entire path.

Following this, a weighted pooling operation is designed to aggregate the prediction scores of multiple paths.

Finally, output is the final interaction score of the user and target item.

In the step of weighted pooling operation, the KPRN can distinguish the strengths of different paths in connecting user and item, thus endowing the model with a certain degree of interpretability, as represented below:

$$g(s_1, s_2, \dots, s_K) = \log \left[\sum_{k=1}^K \exp \left(\frac{s_k}{\gamma} \right) \right] \quad (5)$$

$$\hat{y}_{ui} = \sigma(g(s_1, s_2, \dots, s_K))$$

where g denotes the weighted pooling function, s_k represents the predicted score for the k -th path, γ is the hyperparameter controlling each index weight, and $\hat{y}_{ui}$ is the predicted interaction score between user u and item i .

Noteworthy, within a large-scale KG, the KPRN model may lose substantial information when automatically extracting all paths between user and item. The method leads to ambiguous information conveyed, which is not conducive to interpretability. Consequently, Huang et al. [114] introduced the Path-enhanced Recurrent Network

(PeRN) model, which aims to enhance the performance and interpretability of recommendation systems by integrating KG. The PeRN integrated the RNNs encoder with a meta-path-based entropy encoder, improving recommendations' accuracy and interpretability and reducing the cold-start problem.

Firstly, PeRN devised a bidirectional search algorithm to efficiently extract paths between user and item entities from the KG.

Secondly, a neural network based on bidirectional LSTM encodes path information and converts it into prediction scores.

Thirdly, within the entropy encoder, information gain based on meta-paths is utilized to allocate weighted scores to each path, distinguishing the contributions of different paths in user-item interactions.

Finally, the path and weight scores in the entropy encoder are combined to obtain prediction results through a weighted pooling layer. The cross-entropy loss function and L2 regularization train the model and optimize the prediction results. The bidirectional LSTM is represented as follows:

$$\begin{aligned} \vec{h}_{t+1} &= \text{LSTM}([\alpha_1, \alpha_2, \dots, \alpha_{t+1}]), \\ \overleftarrow{h}_{t+1} &= \text{LSTM}([\alpha_{t+1}, \dots, \alpha_2, \alpha_1]) \\ h &= \vec{h}_{t+1} \oplus \overleftarrow{h}_{t+1} \end{aligned} \quad (6)$$

In the entropy encoder, the end-to-end PeRN model enhances the interpretability of the recommendation system through the entropy encoder. Applying a meta-path avoids the limitation of indistinguishable paths in user-item interactions. The calculation of information entropy is illustrated as follows:

$$\text{Ent}(D) = - \sum_{j \in \{0,1\}} P(D=j) \log_2 P(D=j) \quad (7)$$

where $\text{Ent}(D)$ notes the information entropy of event D , and $P(D=j)$ represents the probability of event D being equal to j .

The weighted pooling and optimization objectives are expressed as follows:

$$\begin{aligned} \hat{y}_k &= \sigma \left(\sum_{i=1}^{|P_k|} w_i s_i \right) \\ L &= \sum_{a \in A} (y \log \hat{y} + (1-y) \log(1-\hat{y})) \end{aligned} \quad (8)$$

where $\hat{y}_k$ denotes the predicted score of user-item interactions, w_i signifies the path's weighted score, s_i represents the path's score, L stands for the cross-entropy loss function, y indicates the observed user-item feedback, and $\hat{y}$ refers to the predicted score.

Furthermore, Chang et al. [115] introduced the Power-Link method, which, by combining path amplification loss with mutual information-based loss, can efficiently generate explanatory paths that interpret the Graph Neural Networks (GNNs) prediction results for KG completion. Yang et al. [116] proposed the Knowledge Guided Interpretable Disease Prediction (KGxDP), which utilizes the extracted paths as prompts for ChatGPT to generate user-friendly and comprehensible

Table 11

Experimental results of Query2box on three standard KGs.

Method	Avg	1p	2p	3p	2i	3i	ip	pi	2u	up
FB15k										
Q2B	0.484	0.786	0.413	0.303	0.593	0.712	0.211	0.397	0.608	0.33
GQE	0.386	0.636	0.345	0.248	0.515	0.624	0.151	0.310	0.376	0.273
GQE-DOUBLE	0.384	0.630	0.346	0.250	0.515	0.611	0.153	0.320	0.362	0.271
FB15k-237										
Q2B	0.268	0.467	0.24	0.186	0.324	0.453	0.108	0.205	0.239	0.193
GQE	0.228	0.402	0.213	0.155	0.292	0.406	0.083	0.17	0.169	0.163
GQE-DOUBLE	0.23	0.405	0.213	0.153	0.298	0.411	0.085	0.182	0.167	0.16
NELL995										
Q2B	0.306	0.555	0.266	0.233	0.343	0.48	0.132	0.212	0.369	0.163
GQE	0.247	0.418	0.228	0.205	0.316	0.447	0.081	0.186	0.199	0.139
GQE-DOUBLE	0.249	0.417	0.231	0.203	0.318	0.454	0.081	0.188	0.2	0.139

Table 12

Summary of advantages and disadvantages of path-based and embedding-based methods.

Categories	Typical methods	Advantages	Disadvantages
Path	KPRN [112], PeRN [114], Power-Link [115], KGxDP [116], PGPR [117]	Simple structure, strong interpretability.	Not suitable for complex structures.
Embedding	Query2box [122], Query2Geom [124], GENI [125], SDFormer [126]	High accuracy, strong logical reasoning.	Complex structure, non-intuitive explanation.

Table 13

Feature summary and code link collection about the typical knowledge graph approach.

Methods	Characteristics and code links
KPRN [112]	KG, path representation, sequential dependencies, RNN, weighted pooling, explainability. https://github.com/eBay/KPRN.git
PeRN [114]	KG enhancement, RNN, metapath entropy encoding, explainability, accuracy, cold-start reduction. https://github.com/dhatGuy/PERN-Store.git
KGxDP [116]	Knowledge-guided, interpretable disease prediction, path reasoning, medical KG, admission history, graph attention network. https://github.com/SCUT-CCNL/KGXDP.git
PGPR [117]	Reinforcement learning, KG Reasoning, explainable recommendation, soft reward strategy, user-conditional action pruning, multi-hop scoring function. https://github.com/orcax/PGPR.git
Query2box [122]	Box embeddings, logical reasoning, KG, vector space, disjunctive normal form. http://snap.stanford.edu/query2box
Query2Geom [124]	Geometric computation, parameter reduction, interpretability enhancement, attention deviation metric, semantic-geometric alignment and mismatch. https://github.com/Jeffrey-Sardina/Query2Geom.git

natural language explanations for the results. Xian et al. [117] proposed a method known as Policy-Guided Path Reasoning (PGPR), which integrates KG into the recommendation system, leveraging structured information in KG to enhance recommendation accuracy and interpretability. The entities and relationships in the KG offer abundant semantic information, which aids in understanding the logic behind the recommendations.

(2) Embedding method

Embeddings, as described in [118–120], involve mapping entities and relationships in KG into a continuous vector space, thus facilitating computation while retaining model structure information [121]. Query2box [122], one of the classic methods, has demonstrated significant potential in breaking down human-machine barriers in question-answering systems and possesses strong interpretability. Previous methods model "query" as a single point in vector space [123], but a complex "query" may correspond to a vast set of answer entities, and it is incredibly unreasonable to represent it directly as a point. It is reassuring that Query2box employs a "box" form for embedding, which is defined as follows:

$$\text{Box}_p \equiv \{v \in \mathbb{R}^d : \text{Cen}(p) - \text{Off}(p) \leq v \leq \text{Cen}(p) + \text{Off}(p)\} \quad (9)$$

where $p = (\text{Cen}(p), \text{Off}(p)) \in \mathbb{R}^{2d}$, $\leq$ denotes element-level inequality relations; $\text{Cen}(p) \in \mathbb{R}^d$ represents the center of the "box," $\text{Off}(p) \in \mathbb{R}^d \geq 0$ indicates the positive offset of the "box."

Query2box is a model for reasoning about query tasks with arbitrary operator symbols, providing a reasonable way of natural representation between "boxes." Furthermore, the embedding dimension is proportional to the number of entities in the KG during processing parsing.

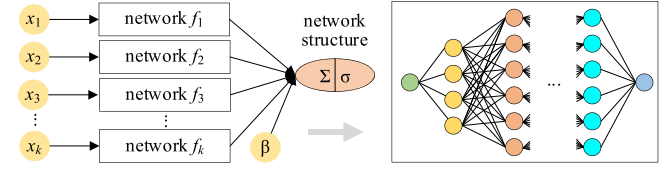
**Fig. 10.** Structure of the neural additive model.

Table 11 experimental results demonstrate that the model performs well on three standard KGs (FB15k, FB15k-237, NELL995).

It is frustrating that Query2box's attention mechanism utilizes "latent intersections." These intersections focus on semantic properties in the embeddings that are not expressed in the geometry of "boxes," thereby constraining the model's interpretability. Based on this, Jeffrey Sardina et al. [124] proposed the Query2Geom model. This model uses geometric operations to replace the attention mechanism module of Query2box, resulting in fewer model parameters and, thus, enhanced interpretability.

In addition, Amador-Domínguez et al. [125] introduced the GENI interpretability framework for KG embedding predictions, which can generate insights and explanations for the KG embedding models. Li et al. [126] have proposed a Shallow-to-Deep Feature Interaction for Knowledge Graph Embedding (SDFormer), taking into account the interpretability of sub-feature tags for entities and relationships and learning shallow-to-deep interaction information between entities and relationships at a more fine-grained level. Table 12 summarizes the typical methods for path-based and embedding-based approaches.

Some scientists have introduced based on the KG approach in the above works. A summary of some characteristics and compiled code links for methods/models are provided in Table 13 for better reader understanding.

4.2.3. Generalized additive model

As an extension of the regression family, the Generalized Additive Model (GAM) proposed by Hastie [127] can be used to model any regression problem. The model not only understands predictions by examining smoothing functions, model parameters, hypothesis testing, and diagnostic plots but also explains the interactions between variables as follows:

$$g(E(y)) = \sum_{i=1}^n f_i(x_i) + \beta \quad (10)$$

where $g(\cdot)$ denotes the connection function, $x = (x_1, x_2, \dots, x_n)$ represents the input n features, and f_i is a univariate shape function corresponding to $E(f_i) = 0$ each.

GAM ensures that each predictor variable has a relatively straightforward expression of nonlinear relationships, which has good interpretability. Nevertheless, when confronted with nonlinear relationships in more complex data, GAM still has certain limitations in capturing the interactive relationships between variables. Building on this, Caruana et al. [128] proposed the Generalized Additive Models with Pairwise Interactions (GA²Ms) model. This model can more comprehensively consider nonlinear relationships and interactive relationships between variables, which can better adapt to complex data structures. This model ensures that the fitting ability is improved without excessively sacrificing the model interpretability, and its form is as follows:

$$g(E(y)) = \sum_{i=1}^n f_i(x_i) + \sum_{i \neq j} f_{ij}(x_i, x_j) + \beta \quad (11)$$

In recent years, researchers have applied GAM to DL for interpretability studies. For instance, Agarwal et al. [129] introduced the Neural Additive Model (NAM) to enhance the DL's interpretability. NAM learns a linear combination of networks; each network has separate input

Table 14
Characteristics and code links of GAM and its improved methods.

Methods	Characteristics and code links
GAM [127]	Smooth functions, local scoring algorithm, nonparametric regression, automatic effect revelation. https://github.com/dswah/pyGAM.git
GA ² Ms [128]	Interpretability, modularity, accuracy, medical prediction. https://github.com/yinlou/mltk.git
NAM [129]	Interpretability, neural networks, additive models, high expressiveness, intuitive understanding. https://github.com/AmrMKayid/nam.git
IGANN [130]	Interpretability, gradient boosting, neural networks, high predictive performance. https://github.com/MathiasKraus/igann.git
GAMI-Net [132]	Interpretability, neural network, GAM, structured interactions, sparsity constraint, heredity constraint, marginal clarity. https://github.com/ZebinYang/gaminet.git
NODE-GAM [133]	Interpretability, DL, GAM, differentiable DTs, high-risk domain application. https://github.com/zzzace2000/nodegam.git

features, and each f_i is obtained through a neural network, as illustrated in Fig. 10.

The differentiability and composability of NAM make it easier to expand and more accurate than GAM. Concurrently, Kraus et al. [130] introduced the Interpretable Generalized Additive Neural Networks (IGANN). The model leverages Gradient Boosting (GB) and customized neural networks to achieve high predictive performance while maintaining interpretability. The prediction function $\hat{y}$ of IGANN is defined as follows:

$$\hat{y} = \underbrace{a \cdot x + b}_{\text{linear component}} + \underbrace{\sum_{l=1}^L s_l f_l(x)}_{\text{non-linear base functions}} \quad (12)$$

where $a \in R^d$ represents the parameters of the linear component and $b \in R$ is the intercept.

Concurrently, the base function with parameter $\theta = (\alpha^1, \dots, \alpha^d, w^1, \dots, w^d)$ is defined as:

$$f_{\theta}(x) = \sum_{k=1}^d \sum_{j=1}^N \alpha_j^k \sigma(x_k w_j^k) \quad (13)$$

Similarly, it should be noted that the proposed base functions can be regarded as sparse single-hidden-layer neural networks, that is, for each $\theta \in R^{2dN}$, there exist $\bar{\alpha} \in R^{N^2}$, $\bar{w}^j \in R^{N^2}$, and $j \in \{1, \dots, N^2\}$ such that:

$$f_{\theta}(x) = \sum_{j=1}^{N^2} \bar{\alpha}_j \sigma(\langle x, \bar{w}^j \rangle) \quad (14)$$

Furthermore, Nori et al. [131] proposed an explainable boosting machine, a novel interpretability algorithm. This method automatically detects and includes interaction terms between features to enhance the model's accuracy while preserving its interpretability. Yang et al. [132] proposed an interpretable neural network architecture, Generalized Additive Models with Structured Interactions (GAMI-Net). The architecture employs three key interpretability constraints to enhance model

interpretability: sparsity, heredity, and marginal clarity. Chang et al. [133] designed a neural GAM (NODE-GAM) that can be extended to large-scale data sets and maintain high interpretability. This model solves the problems of differentiability and scalability of GAM in deep neural networks.

Table 14 compiles characteristics and code links for GAM and its variants based on the techniques introduced by scientists in their research to facilitate a better understanding for readers.

4.2.4. Model optimization

Optimizing the network architecture, which can enhance the model's transparency and interpretability, is one of the essential directions in interpretability research, such as introducing constraint restrictions and improving loss functions.

(1) Impose constraint restrictions

In active explanations, imposing constraint restrictions is a common approach to interpreting models. As shown in Fig. 11, introducing regularization technology [134] during model training, which involves randomly dropping some nodes, helps enhance the model's generalization ability and prevents overfitting. This method can be understood as pruning certain nodes in a large neural network model to obtain a smaller model, thereby improving its interpretability.

On this basis, Wu et al. [135] proposed the "activation regularization" technique, which compels the output of the neural network's activation functions to meet a preset target pattern during the training phase, thus aiding in understanding network behavior.

These target patterns include time-varying and time-invariant patterns. Time-varying patterns treat the activated grid as a series of spatiotemporal region divisions, with the intrinsic meaning of each area abstracted into various concepts. For instance, phone-dependent patterns encourage the modeling of different phonetic units. In each hidden layer, each point in grid space is associated with a phone p and is denoted as s_p , so Gaussian contours can be used to define time-varying target patterns:

$$g_{ij}^{(l)} = \frac{\exp\left(-\frac{1}{2\sigma^2} \|s_{ij} - \hat{s}_{p_t}\|_2^2\right)}{\sum_{m,n} \exp\left(-\frac{1}{2\sigma^2} \|s_{mn} - \hat{s}_{p_t}\|_2^2\right)} \quad (15)$$

The above formula $\hat{s}_{p_t}$ represents the "correct" position of the phone in the grid space at time t and σ controls the sharpness of the pattern surface. The Gaussian mean represents the phone's position within the grid. Consequently, for each phone, this profile covers the area in the surrounding grid. The result is that different grid regions respond to various calls, and the behavior of the activation function can be interpreted based on its position in the grid.

Whereas time-invariant patterns do not require the extraction of multiple "labels" from the data but can be used to describe the general and desirable properties of network activation patterns, defined as follows:

$$G_t^{(l)} = G^{(l)}, \forall t \quad (16)$$

Ultimately, by utilizing a regularization function $R(\theta)$ to correlate activation output $\tilde{H}_t^{(l)}$ with the target pattern $G_t^{(l)}$, an interpretable

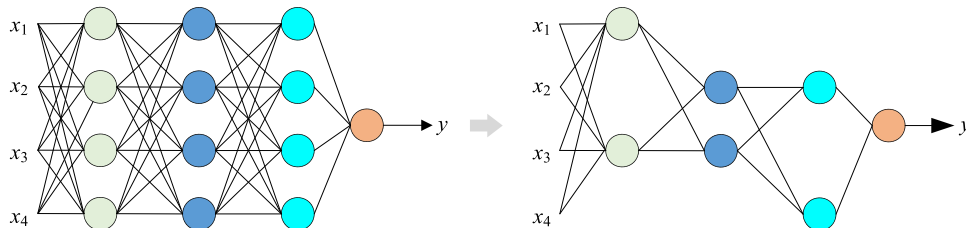


Fig. 11. Sparsify the model through the regularization technique.

method of activation regularization is derived:

$$R(\theta) = \frac{1}{T} \sum_t \sum_l D(\tilde{H}_t^{*(l)}, G_t^{(l)}) \quad (17)$$

Based on traditional regularization, scholars have proposed various variants to enhance model interpretability. For instance, Wu et al. [103] introduced a new model complexity penalty function—tree regularization. Through this method, the deep time series model maintains high prediction accuracy, and the class probability prediction can be closely simulated by a decision tree with few nodes, thereby enhancing model interpretability. Szolnoky et al. [136] introduced a new framework, Model Gradient Similarity (MGS), which employs MGS to quantify and understand the effects of different regularization methods on neural network generalization ability and gradient behavior. Liu et al. [137] proposed a new method called the Jacobian norm and Selective Input Gradient Regularization (J-SIGR). The technique utilizes Jacobian regularization to enhance model robustness and introduces regularization of perturbation-based saliency maps to achieve interpretability in predictions.

(2) Improve loss function

Moreover, the model's loss function is improved and possesses a certain degree of interpretability. In video captioning, Dong et al. [138] introduced explanation loss, thereby making the features learned by the model more consistent and more accessible to understand with externally provided semantic information. The explanation loss is defined as the difference measure between the video feature vector V and the semantic vector s , and its formula is as follows:

$$LI(V, s) = \left\| f\left(\frac{1}{n} \sum_{i=1}^n v_i\right) - s \right\|_2^2 \quad (18)$$

where f represents an arbitrary function that maps the video feature vector space to the topic space. Specifically, the meaning of this expression is to map the average feature vector of all frames to the topic space and then sum its squared Euclidean distance from the semantic vector s . Such explanation loss aims to push the model to match the feature representation to the semantic topic as much as possible during the learning process.

In research on interpretability-aware industrial anomaly detection, Jiang et al. [139] enhanced the model interpretability, constraining the decision boundaries of deep models so that they can be accurately simulated by a simple decision tree with fewer nodes. This approach can be flexibly applied to various models, producing interpretable anomaly detection results. The method backpropagates the CNNs based on scores for specific classification types to obtain the interpretability-aware loss L_{IA} for the input image x encoded as latent variable z . During this process, the gradient z_i of each z entry is backpropagated according to the feature map A_j of the encoder's j -th layer, where $i = 1, \dots, d$. This process generates the Gradient-weighted Class Activation Mapping (Grad-CAM) [140] attention map M_j , which can be obtained through a linear combination of A_j with ReLU activation:

$$M_j = \text{ReLU}\left(\sum_k \alpha_{ij}^k A_j^k\right) \quad (19)$$

where k represents the channel index and $\alpha_{ij}^k \in \mathbb{R}$ is obtained by applying the Global Average Pooling (GAP) operation to the gradients of z_i concerning A_j^k .

$$\alpha_{ij}^k = \text{AvgPool}\left(\frac{\partial z_i}{\partial A_j^k}\right) \quad (20)$$

Implementing activation regularization techniques and improved loss function methods makes the model easier to understand and explain during training, thus effectively enhancing the model's transparency

Table 15

Characteristic summaries of model optimization methods.

Methods	Characteristics and code links
Dropout [135]	Regularization, random dropping, overfitting prevention, generalization enhancement. https://github.com/mdenil/dropout.git
Tree regularization [103]	DT control, overfitting reduction, generalization improvement. https://github.com/dtak/tree-regularization-public.git
MGS [136]	Model gradient similarity, regularization metric, generalization ability, mechanism understanding. https://github.com/google/neural-tangents.git and https://github.com/google/jax.git
J-SIGR [137]	Jacobian normalization, selective input gradient regularization, enhanced robustness, improved interpretability. https://github.com/Lywu-github/jJ-SIGR.git

and interpretability.

Based on the introduction of methods in some scientists' articles, the characteristics and some code links corresponding to model optimization are summed up, as shown in Table 15, to facilitate readers' learning.

4.2.5. Interpretable modules

Within the realm of DL, research on model interpretability based on interpretable modules has already been applied to various tasks such as image recognition, visual question answering, and object detection, thereby propelling further development in DL interpretability.

The capsule network proposed by Sabour et al. [141] is regarded as a novel type of neural network. In capsule networks, the output is represented by length and direction activity vectors representing wholes or parts, with length representing confidence and direction representing instantiation parameters.

Firstly, capsule neurons perform affine transformations on input vectors v_1 and v_2 , resulting in u_1 and u_2 .

Subsequently, a weighted sum is applied to u_1 and u_2 via $s = c_1 u_1 + c_2 u_2$, where c_1 and c_2 are obtained through a dynamic routing mechanism.

Ultimately, the output vector v is obtained through the squashing function $v = \left(\frac{\|s\|^2}{1 + \|s\|^2}\right) \left(s / \|s\|\right)$.

This dynamic routing mechanism enables each capsule to encode specific semantic concepts, making the network highly transparent and interpretable.

La Rosa et al. [142] proposed a self-explanatory module, Memory Wrap, for image classification tasks. The module enhances model performance and explains the decision-making process, focusing on instance explanation and counterfactual scenarios. Specifically, given an input x_i , the current prediction $f(x_i)$, and the encoding matrix M_{S_i} of the memory set, each is associated with a weight $m_j^i \in M_{S_i}$ related to w_j . Dividing M_{S_i} into three disjoint sets:

$$M_{S_i} = M_e \cup M_c \cup M_z \quad (21)$$

where M_e comprises the sample encodings predicted by the network to belong to the same category $f(x_i)$ and is associated with the weight $w_j > 0$, denoted as:

$$M_e = \left\{ f_1(x_{m_j}^i) \mid f(x_i) = f(x_{m_j}^i) \right\} \quad (22)$$

Whereas M_c contains the sample encodings that the network predicts to belong to different categories $f(x_i)$ and is associated with weight $w_j > 0$, denoted as:

$$M_c = \left\{ f_1(x_{m_j}^i) \mid f(x_i) \neq f(x_{m_j}^i) \right\} \quad (23)$$

M_z encompasses all other samples and is related to weight $w_j = 0$. However, M_z does not contribute to the decision-making process and, therefore, cannot be employed to explain. Conversely, the weights for

Table 16

Validation accuracy for the CLEVR dataset (Visual Question Answering) and CLEVR-Ref dataset (Referential Expression Grounding).

Trained on	Expert layout	VQA accuracy (%)	REF accuracy (%)
VQA	yes	96.6	n/a
REF	yes	n/a	96.0
VQA+REF	yes	96.5	96.2
VQA	no	93.0	n/a
REF	no	n/a	93.4
VQA+REF	no	93.9	95.4

M_e and M_c are positive can be used to extract instance-based explanations. The higher the weight w_j corresponding to the sample $x_{m_j}^i \in M_{S_i}$, the more similar the encoding of input x_i is to the encoding of the sample $x_{m_j}^i$. Thus, it can be considered a candidate instance for the instance-based explanation.

Hu et al. [143] introduced an interpretable model, the Stack Neural Module Network or Stack-NMN, which aims to explain high-complexity Visual Question Answering (VQA) and Referential Expression Grounding (REF) tasks. This model uses a stacked neural module network to achieve the interpretability of neural calculations. Initially, the layout controller decomposes the reasoning task T into a series of subtasks $T_1, T_2, \dots, T_n$, $T = T_1 \cup T_2 \cup \dots \cup T_n$, and transforms the input question Q into a soft layout $L = f(Q)$. Subsequently, the controller utilizes a set of neural modules $M = \{M_1, M_2, \dots, M_n\}$ to process the collection of subtasks $T = \{T_1, T_2, \dots, T_n\}$. Ultimately, a differentiable memory stack S is used to store and retrieve the intermediate outputs of each module during execution. As shown in Table 16, the modular structure improves model interpretability and achieves higher accuracy in VQA and REF tasks.

Chen et al. [144] introduced the Information Maximizing Generative Adversarial Networks (InfoGAN) based on information maximization, which extends the information theory of disentangled representations through GAN and achieves completely unsupervised learning. It can be visualized from Fig. 12 that InfoGAN decomposes the input noise vector Z into stochastic noise Z and latent vector C , then employs the generator G to produce synthetic data $G(Z, C)$. Subsequently, the generator G and discriminator D are updated based on mutual information to contrast real data X_{real} with the synthetic data X_{fake} . Through this process, the maximization of the obtained mutual information is achieved, which results in more interpretable representations.

Wu et al. [145] proposed end-to-end integration of a general top-down grammar model (Interpretable And-Or Graph, IAOG) with a bottom-up convolutional network, aiming to learn qualitatively interpretable models in object detection. The AOGParsing operator is introduced in this model, which can replace the region of interest pooling operator widely used in region-based CNNs. This method demonstrates detection performance comparable to the state-of-the-art region-based CNN systems, offering a new perspective on the accuracy and transparency of DL models in object detection.

Meanwhile, Gu [146] proposed a novel graph capsule network, Graph Capsule Networks (GraCapsNets), for object recognition. GraCapsNets employs a graph pooling method based on multi-head attention, replacing the iterative routing mechanism in traditional capsule networks to enhance model interpretability. This method effectively generates explanations for single classifications and unexpectedly

enhances classification performance. The number of parameters is reduced, and adversarial robustness is improved while retaining other advantages of capsule networks, such as decoupled representation and robustness to affine transformations.

Customized interpretability modules explain specific tasks, promoting a deep understanding of the decision-making process. The specially designed interpretability module enhances the overall explicability of the system, thus making the model's decision-making more comprehensible and interpretable. However, ensuring compatibility between the network and the interpretability module may increase the complexity of model design and tuning. Furthermore, interpretability modules designed for specific tasks may have limited applicability to other tasks, and thus generalizability may be limited.

Based on the introduction of these methods in some scholars' articles, some characteristics and code links corresponding to the interpretable modules are summarized, as shown in Table 17, to facilitate readers' learning.

4.2.6. Stochastic configuration network

Stochastic Configuration Network (SCN) is an emerging incremental neural network model that adjusts hidden layer node parameters through a supervisory mechanism [147–149]. Compared to traditional networks, SCN possesses advantages such as stochastic configuration of parameters, adaptive adjustment of parameter ranges, fast learning, and good generalization capabilities. The core feature of SCN is configuring the hidden layer node parameters through a supervision mechanism. It implies that it is possible to track the parameter values of each node, thereby understanding the model's decision-making process on various tasks. At this level, SCN has a certain interpretability. Random configurations of node parameters can result in different model instances, which makes interpreting the model's overall behavior more complex. In this respect, SCN's interpretability may be somewhat limited.

Therefore, the parameter configuration of the SCN can be considered a local explanation. However, regarding global explanation, the SCN lacks interpretability, and the model needs to be associated with other methods to facilitate its interpretation. Considering the particularity of

Table 17

Links to characteristics and codes of some interpretable modules.

Methods	Characteristics and code links
CapsNet [141]	Dynamic routing, capsule networks, hierarchical structure, activity vectors, image recognition. https://github.com/naturomics/CapsNet-Tensorflow.git
Memory Wrap [142]	Self-interpretable module, sparse content attention, small data enhancement, performance improvement, interpretability. https://github.com/KRLGroup/memory-wrap.git
Stack-NMN [143]	Neural modularization, automatic sub-task decomposition, compositional reasoning, enhanced interpretability. https://github.com/ronghanghu/snmn.git
InfoGAN [144]	InfoGAN, unsupervised learning, information maximization, disentangled representation, mutual information. https://github.com/openai/InfoGAN.git
IAOG [145]	Interpretability, unfolding latent structures, unsupervised learning, R-CNN, extractive rationale generation. https://github.com/msracver/Deformable-ConvNets.git
GraCapsNets [146]	Graph capsule networks, multi-head attention, graph pooling, interpretability, adversarial robustness. https://github.com/JindongGu/GraCapsNet.git

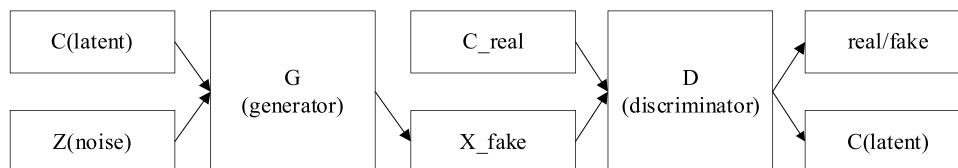


Fig. 12. Basic structure of the information-maximizing generative adversarial networks model.

this network, it is analyzed in separate sections.

Specifically, Zhou et al. [150] introduced the stochastic configuration fuzzy inference system with interpretable fuzzy rules to improve the interpretability and performance of fuzzy inference systems, thereby automatically determining the appropriate model structure.

The method employs Interpretable Linguistic Fuzzy Rules (ILFRs) to establish the fuzzy system, endowing the system with clear semantic interpretability. The form of ILFRs is as follows:

$$R_l : \text{if } x_{i1} \text{ is } B_{l1} \text{ and } x_{i2} \text{ is } B_{l2} \text{ and } \dots \text{ and } x_{im} \text{ is } B_{lm}, \\ \text{then } y = \Phi(x_i)^T a_l, \quad l = 1, 2, \dots, L \quad (24)$$

For instance, there are two fuzzy rules, the input sample of each is x_i , and the two features of the five fuzzy sets with language labels (very low, low, medium, high, and very high) are:

$$C(:, :, 1) = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}, C(:, :, 2) = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{pmatrix} \quad (25)$$

Consequently, these 2 ILFRs can be expressed as:

R1: If x_{i1} is very low or low, x_{i2} is medium, and x_{i3} is high or very high, then $y = \Phi(x_i)^T a_1$.

R2: If x_{i1} is very high, x_{i2} is very low, and x_{i3} is medium or high, then $y = \Phi(x_i)^T a_2$.

Meanwhile, Wang et al. [151] proposed a more compact model, Evolving Stochastic Configure Network (E-SCN) based on SCN, which enhances the interpretability of constraint from an evolutionary perspective through a sampling mechanism. The designed constraint aims to make the cosine value between vector $e_{L-1,q}(X)$ and vector $g_L(X)$ approach 1 and 0. The interpretation of the designed constraints in SCN is expressed through inequality transformations and vector operations as follows:

$$\begin{aligned} \frac{[e_{L-1,q}^T(X)g_L(X)]^2}{g_L^T(X)g_L(X)} &\geq (1-r-\mu_L)e_{L-1,q}^T(X)e_{L-1,q}(X) \Rightarrow \\ \frac{\|e_{L-1,q}(X)\|^2 \|g_L(X)\|^2 \cos^2 \langle e_{L-1,q}(X), g_L(X) \rangle}{\|g_L(X)\|^2} &> (1-r-\mu_L) \|e_{L-1,q}(X)\|^2 \Rightarrow \\ \cos^2 \langle e_{L-1,q}(X), g_L(X) \rangle &> 1-r-\mu_L \Rightarrow \begin{cases} 1 > \cos \langle e_{L-1,q}(X), g_L(X) \rangle > \sqrt{1-r-\mu_L} \\ 0 > \cos \langle e_{L-1,q}(X), g_L(X) \rangle > -\sqrt{1-r-\mu_L} \end{cases} \end{aligned} \quad (26)$$

Compared to the traditional incremental random vector functional link network and SCN, E-SCN possesses higher compactness, faster training efficiency, and more robust generalization capabilities. Most importantly, E-SCN provides a theoretical explanation for the specially designed constraints in SCN, which contributes to a better understanding of SCN's advantages and offers insights for further optimizing SCN in the future.

Li et al. [152] proposed the Fuzzy Stochastic Configuration Network (F-SCN). This model integrates the fuzzy inference system Takagi-Sugeno (TS) into the SCN to improve fuzzy inference capabilities. The parameters of TS are determined by the distribution characteristics of the network's input-output data, thereby improving interpretability. The definition of TS is as follows:

$$y = \frac{\sum_{k=1}^K \omega^k(x) y_k}{\sum_{k=1}^K \omega^k(x)} = \frac{\sum_{k=1}^K \prod_{i=1}^d \mu_{A_i^k}(x_i) \xi_k(x_1, x_2, \dots, x_d)}{\sum_{k=1}^K \prod_{i=1}^d \mu_{A_i^k}(x_i)} \quad (27)$$

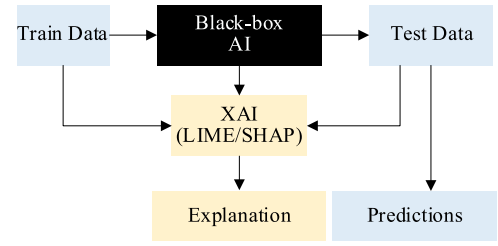


Fig. 13. Schematic diagram of the passive explanation process.

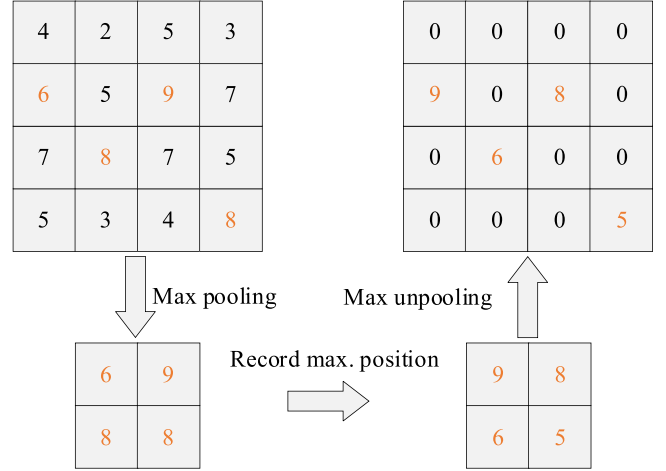


Fig. 14. The operation of the "depooling-deactivation-deconvolution" operation.

where A_i^k represents the corresponding fuzzy variable, $k = 1, 2, \dots, K$ denotes the number of fuzzy rules, $i = 1, 2, \dots, d$ indicates the dimension of input variables, y_k represents the output value of the k -th fuzzy rule, $\xi_k(x_1, x_2, \dots, x_d)$ signifies the first-order polynomial of the input vector, $\omega^k(x)$ indicates the firing strength of the k -th fuzzy rule.

At present, although interpretability research of SCN has made certain advancements, there are still many fields that have not been fully explored. The global interpretability of SCN is still a challenge, especially in areas involving CNN architectures, semi-supervised learning, unsupervised learning, multi-view learning, fuzzy neural networks, and recurrent neural networks, where research on explainability holds immense potential for development.

4.3. Passive explanations

Passive explanation is an interpretive method applied to already trained models, which extracts logical rules or comprehensible patterns by analyzing the model's structure, weights, and outputs, among other

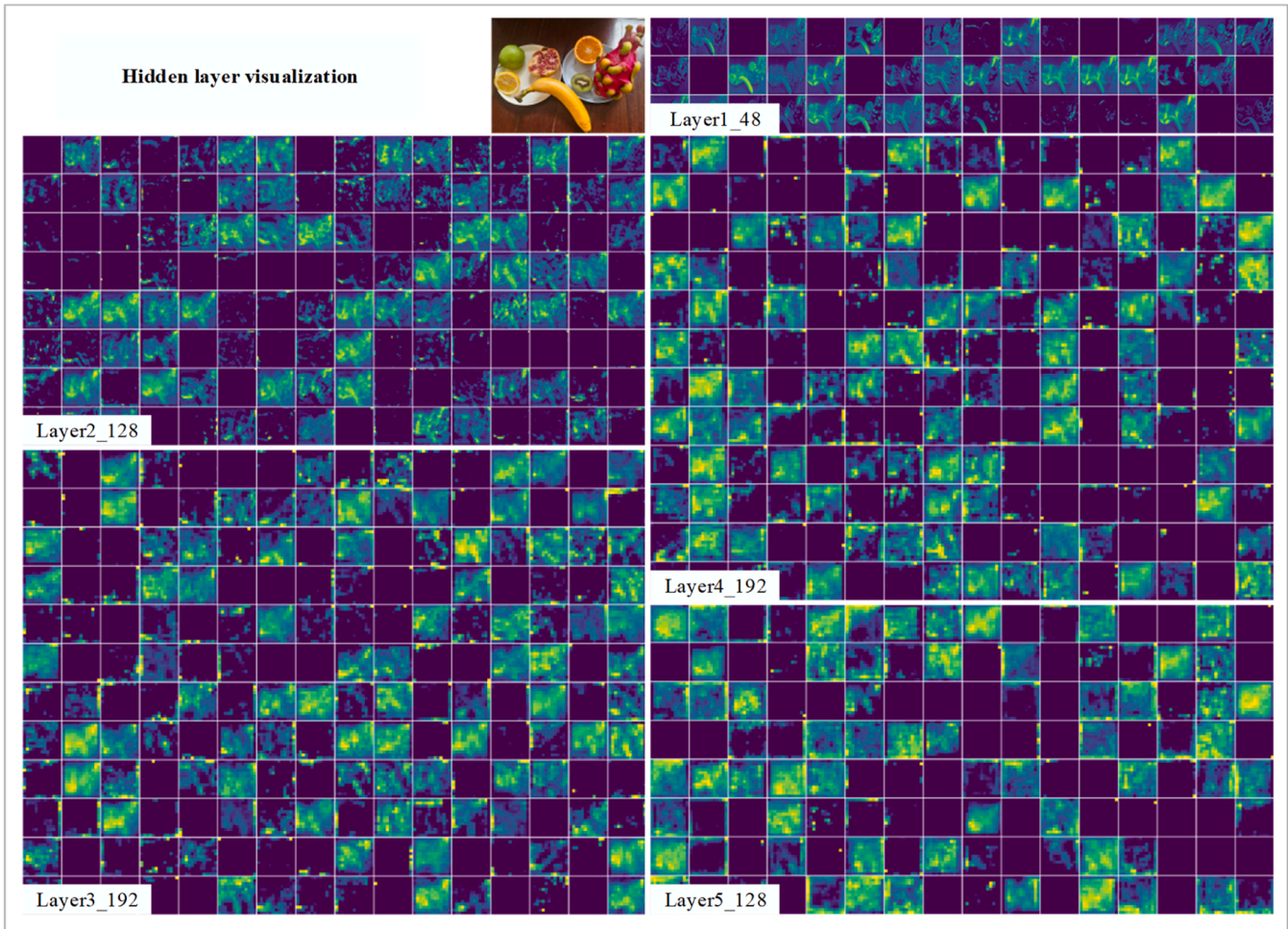


Fig. 15. The feature maps of AlexNet layers 2 to 5.

information. Fig. 13 shows the implementation flow of passive interpretation.

4.3.1. Hidden layer analysis

Utilizing interpretability methods based on hidden layer analysis, researchers employ visualization techniques to examine the hidden layers of a neural network. This process transforms the layers into images that are understandable to humans and convey practical significance, thereby illustrating the concepts learned by each layer in the neural network. In typical CNNs, a convolutional process usually consists of three steps: "convolution–activation–pooling." Conversely, the "depooling–deactivation–deconvolution" operations used in Fig. 14 aid in understanding the features learned at each layer of CNNs [153].

For instance, Zeiler and Fergus [153] conducted a feature visualization analysis on the 2 to 5 layers of AlexNet [154] using the ImageNet dataset. At the same time, this review also selects a fruit image for visual analysis of the hidden layer feature map. The projection in Fig. 15 exhibits the hierarchy of features in the network. The source code is available at <http://t.csdnimg.cn/xnMht>.

Bau et al. [155] introduced the "Network Dissection" framework, which aims to quantify the interpretability of hidden layers in CNNs. The framework achieves this goal by assessing the degree of alignment between individual hidden units and a set of semantic concepts. Zhang et al. [156] have learned a graphical model called an explanatory graph, which reveals the mixed patterns of object parts represented by each filter in pre-trained CNNs. By automatically disentangling different part patterns from each filter and constructing an explanatory graph, each node in the graph represents a part pattern, and each edge encodes the

Table 18

Classification error on ILSVRC validation set.

Networks	top-1 val. Error(%)	top-5 val. Error(%)
VGGnet-GAP	33.4	12.2
GoogLeNet-GAP	35.0	13.2
AlexNet*-GAP	44.9	20.9
AlexNet-GAP	51.1	26.3
GoogLeNet	31.9	11.3
VGGnet	31.2	11.4
AlexNet	42.6	19.5
NIN	41.9	19.6
GoogLeNet-GMP	35.6	13.9

co-activation relationships and spatial relationships between the patterns. Rauber et al. [157] employ the t-Distributed Stochastic Neighbour Embedding (t-SNE) algorithm for the first time to reduce the dimensionality of high-dimensional hidden layer data. It aims to illustrate the relationships among samples learned by artificial neural networks and emphasize visual feedback's critical importance for improving neural network design. The source code is available at <https://github.com/pavlin-polcar/openTSNE.git>.

The hidden layer analysis method is used to fully mine the information of the internal hidden layers in the network, enhancing the effectiveness of explanations. However, the absence of specific meanings in some hidden layers leads to a certain degree of subjectivity in the interpretation results.

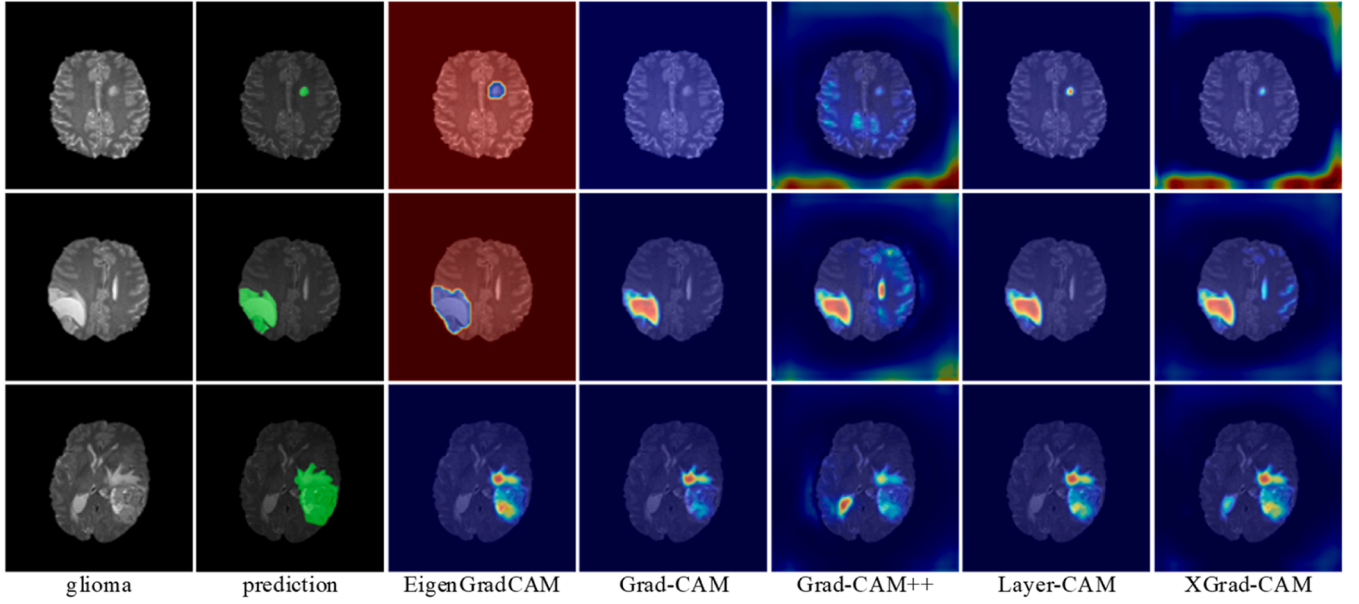


Fig. 16. Visualization of glioma regions using some class activation mapping methods.

4.3.2. Class activation mapping

Class Activation Mapping (CAM) [158] is a commonly used visualization technique. CAM utilizes the GAP in CNNs for linear weighted processing to generate a heat map the same size as the original image, thereby illustrating the contribution distribution to the prediction output. The highlighted areas in the heatmap reflect regions in the original image with higher network responses and more significant contributions, intuitively explaining the importance of each location for a specific category. The activation heatmap M_c for category c is defined as:

$$M_c(x, y) = \sum_k w_k^c f_k(x, y) \quad (28)$$

where $f_k(x, y)$ represents the activation of unit k at the spatial location (x, y) in the final convolutional layer and w_k^c denotes the weight corresponding to unit k for category c .

Although CAM can generate interpretative heatmaps for CNN prediction outputs, its application often requires modification and retraining of the model structure, necessitating a balance between interpretability and accuracy. As shown in Table 18, the authors have summarized the classification performance of the original network and the GAP network. The results indicate that removing additional layers from various networks resulted in a slight performance in most cases, with the decline being approximately 1–2 %.

In addition, CAM can only analyze the final convolutional layer output of the convolutional neural network but cannot analyze the intermediate layers. Furthermore, the CAM is mainly suitable for image classification tasks but is less effective when dealing with regression problems. Selvaraju et al. [128] introduced Grad-CAM to overcome the above issues.

Firstly, Grad-CAM calculates weights by global averaging of gradients:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \quad (29)$$

where Z signifies the number of pixels in the feature map, y^c denotes the gradient for the score of class c , and A_{ij}^k represents the pixel value at the (i, j) location in the k -th feature map.

Subsequently, the forward activation maps are subjected to a weighted combination and are passed through the ReLU activation

function to obtain the final forward activation map:

$$L_{Grad-CAM}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right) \quad (30)$$

Grad-CAM has been widely applied in tasks such as image classification, object detection, and question-answering systems, covering models such as CNNs and Transformers. However, Grad-CAM also has certain drawbacks:

- ① When multiple similar objects are in the image, Grad-CAM can only generate one heat map.
- ② Gradient values at different locations are treated equally after GAP.
- ③ There are problems such as gradient saturation, gradient vanishing, and gradient noise.
- ④ Channels with larger weights do not necessarily contribute the most to the class prediction score.
- ⑤ Only backpropagation is considered, but the impact of forward prediction is not considered.
- ⑥ Both the coarse-grained heat maps generated by deep layers and the fine-grained heat maps generated by shallow layers may not be sufficiently accurate.

In response to problems ①②, Chattopadhyay et al. [159] proposed Generalized Gradient-based Visual Explanations for Deep Convolutional Networks (Grad-CAM++). The method considers that multiple parts of the same type of samples or objects in an image may be distributed across different feature maps. Since these parts may occupy different sizes of space, objects that occupy larger space will have greater weights in the feature maps, and therefore, the effect will be more obvious in the final heat map. Conversely, feature maps that occupy a smaller space may disappear in the heat map. To solve this problem, researchers introduce additional weights and apply a second weighting to the gradient maps.

$$w_k^c = \sum_i \sum_j \left[\frac{\frac{\partial^2 y^c}{(\partial A_{ij}^k)^2}}{2 \frac{\partial^2 y^c}{(\partial A_{ij}^k)^2} + \sum_a \sum_b A_{ab}^k \left\{ \frac{\partial^3 y^c}{(\partial A_{ij}^k)^3} \right\}} \right] \cdot \text{relu} \left(\frac{\partial y^c}{\partial A_{ij}^k} \right) \quad (31)$$

Table 19

Summary of advantages and disadvantages of some class activation mapping methods.

Methods	Advantages	Disadvantages
CAM [158]	1. Input of any size, strong robustness, and few parameters.	1. The model needs to be modified for retraining.
Grad-CAM [140]	2. Visualization of CNNs of any architecture.	2. Subtle details are not prominently displayed.
Grad-CAM++ [159]	3. More suitable for homogenous multi-objective situations.	3. A lot of background information.
Score-CAM [160]	4. Get rid of gradient dependence and reduce noise in the background.	4. Possible limitations when generalizing to new scenarios.
Layer-CAM [161]	5. Obtain coarse spatial location and fine-grained object details.	5. Noise may still be present in shallow feature maps.
Ablation-CAM [162]	6. Direct masking of feature maps results in more accurate localization.	6. Computationally time-consuming.
Ablation-CAM++ [163]	7. Use for weakly supervised semantic segmentation or weakly supervised localization.	7. Still takes a long time to compute.
IS-CAM [164]	8. Use for semantic segmentation or object detection tasks.	8. May be sensitive to small changes in input data.
SS-CAM [165]	9. Improved visual clarity of feature localization through smoothing operations, generating more focused heat maps.	9. The Gaussian noise parameter σ needs to be carefully chosen to avoid generating overly generalized or noisy visualizations
Smooth Grad-CAM++ [166]	10. Reduces noise and improves visualization through smooth gradients.	10. Need to calculate additional gradient smoothing.

In response to problems ③④⑤, Wang et al. [160] proposed the Score-Weighted Visual Explanations for Convolutional Neural Networks (Score-CAM), a method distinct from previous CAM that does not rely on gradient representation. Instead, it is implemented by collecting the weight of each activation map of the target class in the forward propagation process. The linear combination of the weight and activation map obtains the final result. The characteristic of this method is that it can accurately measure the importance score of the feature map. The importance score of a feature map is defined as follows:

$$C(A_i^k) = f(X \circ H_i^k) - f(X_b) \text{ where } H_i^k = s(Up(A_i^k)) \quad (32)$$

The equation above A_i signifies the activation A of convolutional layer i , A_i^k denotes the k -th channel of A_i , $Up(\cdot)$ represents the up-sampling of A_i^k , and $s(\cdot)$ is the normalization function. The computed heat-map point is multiplied by the original image when inputted into the model, and the resulting score is subtracted from the original image's direct score, reflecting the feature map's importance.

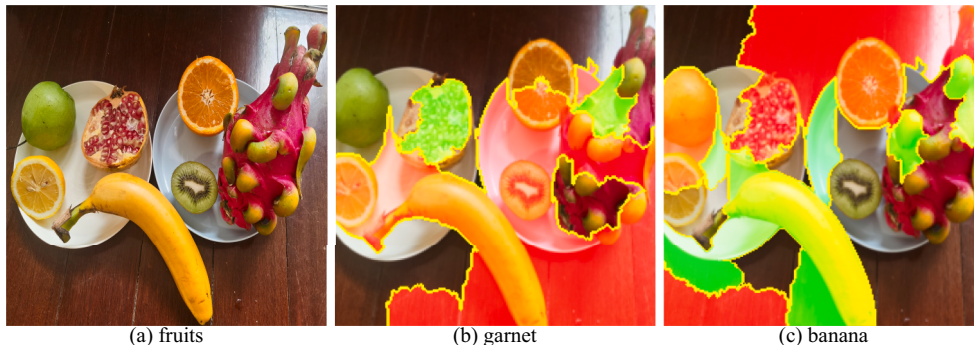


Fig. 17. Significance analysis of fruit category features based on local interpretable model-agnostic explanations.

Addressing issue ⑥, Jiang et al. [161] introduced the Layer-wise Class Activation Map (Layer-CAM), which initially multiplies the activation value at each position in the feature map by a weight to obtain $\hat{A}_{ij}^k = \text{relu}(g_{ij}^{kc}) \cdot A_{ij}^k$, and then linearly combines the results $\hat{A}_{ij}^k$ along the channel to produce a class activation map:

$$M^c = \text{ReLU}\left(\sum_k \hat{A}^k\right) \quad (33)$$

Furthermore, there is a variety of class activation mapping methods for performing the model's post-hoc explainability studies, including Ablation-based Class Activation Mapping (Ablation-CAM) [162], Ablation-CAM++ [163], (Integrated Score-CAM) (IS-CAM) [164], Smoothed Score-CAM (SS-CAM) [165], Smooth Grad-CAM++ [166], etc. The source code is available at <https://github.com/frgfm/torch-cam.git> and <https://github.com/jacobgil/pytorch-grad-cam.git>.

Meanwhile, In Fig. 16, Segformer [167] conducts semantic segmentation experiments on brain glioma. A series of class activation mapping methods provided by MMsegmentation would explore critical areas of concern for model training. The segmentation performed well on three randomly selected images containing gliomas. However, when conducting explanatory studies using the class activation map for visualization, the Class Activation Map using the Principal Components (Eigen-CAM) [168] method performed poorly. At the same time, EigenGrad-CAM and Layer-CAM could accurately identify the features of interest. Conversely, Grad-CAM++ and Axiom-based Gradient-CAM (XGrad-CAM) [169] mistakenly consider some non-glioma areas as regions of interest, leading to misinterpretation. The source code is available at <https://github.com/open-mmlab/msegmentation.git>.

Class activation mapping methods provide an intuitive display of the model's focus on key regions in the image, which aids in identifying the model's deficiencies in specific tasks, thereby fostering the adjustment and enhancement of model performance. However, these methods only provide superficial information and lack an in-depth understanding of the specific significance of the weights within the model. Meanwhile, Table 19 briefly summarizes the advantages and disadvantages of some class activation mapping methods.

4.3.3. Post-hoc explainability

Such methods are used directly on the trained model to explain and include LIME, SHAP, Captum, etc.

(1) LIME

LIME is a local interpretability model proposed by Ribeiro et al. [74], mainly used for interpretive research on models of text data, image data, and table data models. In LIME, the explanation model is denoted as $g \in G$, the function L serves as a metric, $\pi_x(x)$ is the distance between the instance z and x , defining the locality around x .

Firstly, an objective function ξ is defined, which describes how g approximates the complex model f through π_x within a local definition. When the explanation model $\Omega(g)$ complexity is low enough for humans

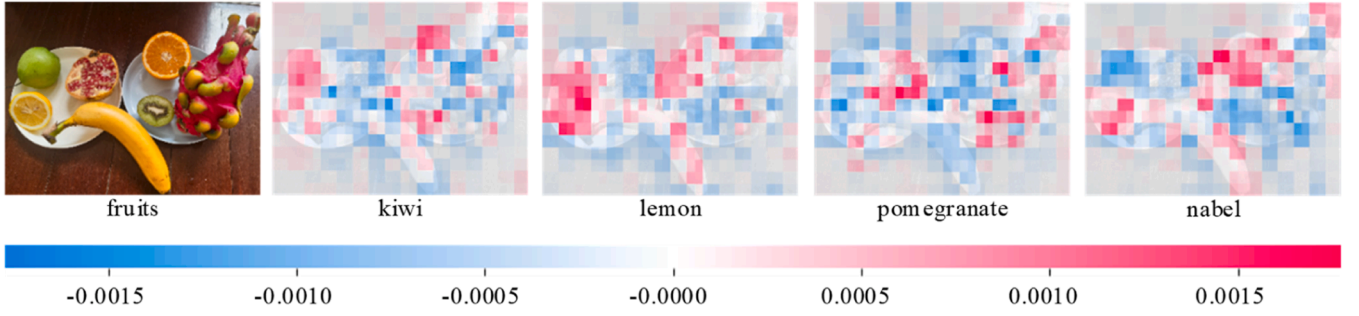


Fig. 18. Fruit recognition feature areas of interest using the Shapley Additive exPlanations method.

to comprehend it, function L is minimized to obtain the optimal solution for the objective function. The initial definition of the explanation is as follows:

$$\xi(x) = \operatorname{argmin}_{g \in G} L(f, g, \pi_x) + \Omega(g) \quad (34)$$

Subsequently, the similarity-based objective function is introduced. The distance π_x before and after the sample undergoes interpretable perturbation is further defined, and the calculation formula is as follows:

$$\pi_x(z) = \exp\left(-\frac{D(x, z)^2}{\delta^2}\right) \quad (35)$$

Ultimately, the objective function is optimized through linear regression, which π_x is utilized as the weight, integrating the influence of interpretable features into the predictions of the perturbed samples. The final objective function is obtained as follows:

$$\xi(x) = \sum_{z' \in Z} \pi_x(z') (f(z') - g(z'))^2 \quad (36)$$

where $f(x)$ represents the predicted value of the sample in the original feature space after the perturbation, $g(z')$ represents the predicted value of the perturbed samples in the interpretable feature space.

In this review, ResNet50 is initially employed for image recognition training on a data set containing 30 different fruits, and LIME is used for feature saliency analysis. The image containing various fruits, such as the garnet and banana, is randomly selected (see Fig. 17(a, b, c)), displaying the regional features the model focuses on when observing pomegranates and bananas, respectively.

Although LIME has achieved favorable results in many application scenarios, its computation process is relatively time-consuming. Moreover, LIME introduces perturbations through random interpolation to affect the interpretation results, leading to computation instability and inconsistency in interpretation results.

Based on this, Li et al. [170] proposed Global-LIME (G-LIME), which uses global prior knowledge, sparsity, and elastic network (ElasticNet) regularization. G-LIME solves the shortcomings of the LIME algorithm in terms of local interpretation stability and consistency. Specifically, G-LIME adopts an improved ElasticNet estimator to achieve global prior sparsity and informativeness:

$$\beta^{\text{mENet}}(\lambda_1, \lambda_2) \leftarrow \operatorname{argmin}_{\beta} \|X\beta - y\|_2^2 + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta - \beta^{\text{global}}\|_2^2 \quad (37)$$

where X represents the input data, λ_1 and λ_2 the tuning parameters are used to adjust sparsity and the β global centered Gaussian prior.

Concurrently, G-LIME utilizes least angle regression [171] to retrieve the complete path of coefficients, thereby efficiently filtering out potential local explanations that may apply to different L1 regularization strengths or feature selection subsets. By adjusting the appropriate parameters for L2 regularization, the modified ElasticNet estimator is reformulated as a quadratic minimization with L1 constraints.

$$\beta^g(t) \leftarrow \operatorname{argmin}_{\beta} \beta^T (X^T X + \lambda_2^* I) \beta - 2(X^T y + \lambda_2^* \beta^{\text{global}})^T \beta, \text{ s.t. } \|\beta\|_1 \leq t \quad (38)$$

Furthermore, Shankaranarayana and Runje [172] proposed the autoencoder-based local interpretable method to improve LIME's stability and local fidelity. The technique utilizes autoencoders as weight functions to enhance the performance of local explanation models in explaining DL model predictions. Recio-García et al. [173] proposed a Case-Based Reasoning-LIME (CBR-LIME) solution for interpreting image classifiers and addressed the challenge of configuring parameters for the LIME algorithm. This case library demonstrates how humans perceive the quality of image interpretations generated under different LIME algorithm parameter settings. These parameter settings can be applied to similar image inputs to optimize the interpretation results. Huang et al. [174] introduced the Graph Neural Networks Local Interpretable Model-agnostic Explanations (GraphLIME), a novel local interpretability framework to explain GNN model predictions. GraphLIME employs the HSIC Lasso nonlinear feature selection method to identify nonlinear interpretable models in subgraphs and generate local explanations for node predictions.

(2) SHAP

SHAP [79] is another post hoc explanation model. The core idea is to calculate the marginal contribution of features to model output and interpret the black-box model at both the global and local levels. Inspired by cooperative game theory, SHAP constructs an additive explanatory model that treats all features as "contributors." The model generates a prediction value for each prediction sample, and the SHAP values represent the value assigned to each feature in the sample. SHAP permits a comprehensive understanding of the impact of each feature on model predictions and explains the model's overall behavior.

Suppose the i -th sample is X_i , the j -th feature of the i -th sample is X_{ij} , the prediction value of the model for this sample is y_i , and the baseline of the entire model (usually the mean of the target variable for all samples) is y_{base} , then the SHAP values conform to the following equation:

$$y_i = y_{\text{base}} + f(X_{i_1}) + f(X_{i_2}) + \dots + f(X_{i_k}) \quad (39)$$

where $f(X_{i_j})$ represents the SHAP value for X_{i_j} . For instance, $f(X_{i_1})$ is the contribution of the first feature in the i -th sample to the final prediction value y_i .

When $f(X_{i_j}) > 0$, it indicates that the feature has increased the prediction value and has a positive effect. Conversely, it has a negative impact. Meanwhile, in Fig. 18, the SHAP method is employed to conduct an interpretability analysis on the nabel, lemon, pomegranate, and kiwi.

Moreover, Baptista et al. [175] studied the results of three algorithms: linear regression, multi-layer perceptron, and echo state network through SHAP, and they analyzed their relationships with prediction evaluation indicators such as monotonicity, trend, and predictability. Their findings demonstrate that SHAP values are closely related to these assessment indicators, revealing that model complexity is a significant factor when considering interpretability. Feng et al. [176] introduced the ML model that combines the eXtreme Gradient Boosting

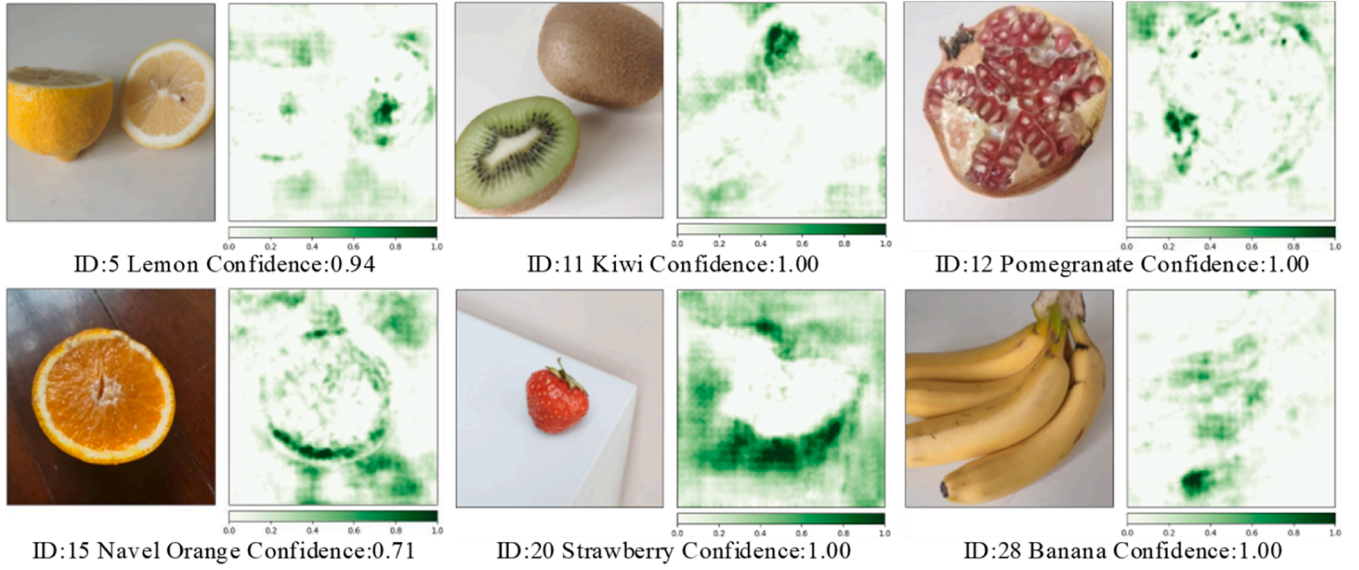


Fig. 19. Visualisation of important features learned by the fruit recognition model using the Captum library.

(XGBoost) algorithm with the SHAP, which provides a powerful and interpretable tool for structural engineering. The model improves prediction accuracy and enhances the model's transparency and credibility.

Fernando et al. [177] proposed a methodology based on DeepSHAP to interpret the outputs of neural retrieval models in information retrieval. DeepSHAP is an improved version of the DeepLIFT algorithm, designed to estimate the Shapley values for given input decisions. The Shapley value explains the difference in the contribution of input features to the prediction result, that is, the contribution of features to the difference between the prediction result and the average prediction value. However, as the number of features increases, the computation time for SHAP grows exponentially. Furthermore, the selection order of feature combinations may affect the calculation of SHAP values and the accuracy of the interpretation of results.

(3) Captum

Captum [178] is a model interpretability library released by Facebook based on PyTorch, which provides interpretive support for various modern network models. This library is designed to help users deeply understand the key, neurons, and neural network layers in model prediction results. Captum can quickly locate the pixels that affect the outcome of image classification tasks and display them intuitively. Captum can highlight the importance of different words and show the correlation between words through heat maps for tasks such as text translation. Captum is utilized to perform rapid localization and visualize feature pixels for individual fruit categories in Fig. 19. It is observed that different fruit models focus on markedly distinct areas in fruit recognition, which provides a basis for model interpretation.

Meanwhile, Stanchi et al. [179] utilized the Captum library in PyTorch to implement the Randomized Input Sampling for Explanation (RISE) method. RISE is an algorithm that generates explanations for the predictions of DL models by randomly masking portions of the input image and observing the changes in the model's output. The study focuses on implementation and performance rather than the RISE method. A series of experiments revealed that the number of masks used in the RISE method is crucial for the interpretability of the results.

Enguehard [180] expanded on Captum and proposed time-interpret, specifically used to enhance the interpretability design of time series data models. However, some of its components can also be applied to other applications, such as interpreting predictions from language models. Miglani et al. [181] introduced the latest features of the Captum library, specifically designed to analyze the behavior of Generative Language Models. An overview of these features and examples of their

potential applications in understanding LLMs are provided.

(4) CP-PD profiles

Gholami et al. [182] employed a DL model to map the spatial distribution of gully erosion and investigate the model's interpretability. The study utilized three innovative interpretability methods grounded in game theory—SHAP value, CP-PD profiles, and PFI—to elucidate the outputs of the deep learning model. The full name of CP-PD profiles is "Ceteris Paribus and Partial Dependence Profiles." In addition to tools for interpreting model output, an intriguing alternative is the Ceteris Paribus (CP), a model-independent interpretative method called Partial Dependence profiles (PDP). Researchers constructed CP profiles using the pyCeterisParibus library [183]. The PD value of the explanatory feature X^j at model $f(\cdot)$ and z can be expressed as follows:

$$\hat{g}_{PDP}^j(z) = E_{\underline{X}^{-j}}\{f(X^{j=z})\} \quad (40)$$

Thus, the expected value of the model's predicted value for $\underline{X}^{-j}$ is derived from the marginal distribution of $\underline{X}^{-j}$, which represents the joint distribution of all explanatory features excluding X^j , while fixing X^j at z . In general, the true distribution of $\underline{X}^{-j}$ remains unknown. The PD diagram, representing the mean value of the CP distribution of X^j , can be expressed in the following form:

$$\hat{g}_{PDP}^j(z) = \frac{1}{n} \sum_{i=1}^n f(\underline{x}_i^{j=z}) \quad (41)$$

Kuzba et al. [184] introduced a Python library named pyCeterisParibus, which offers a method grounded in CP analysis for interpreting the predictions of ML models. For the feature x_i and the observation x , the CP profile delineates the model's prediction regarding the response of $f(x)$ when x_i varies. This relationship is articulated as $f(x_i = v, x_{-i})$, where v signifies a potential value of x_i , and x_{-i} encompasses all feature values except for x_i . This approach is model-agnostic, allowing for understanding how changes in selected variables affect model responses. This work extends the concepts of PDP and individual conditional expectation graphs, facilitating a simultaneous interpretation of the effects of multiple variables on a single observation.

The advantage of the CP-PD profile lies in its model independence, allowing it to visually illustrate the impact of a single feature on model predictions. However, it may not fully capture the complex interactions in high-dimensional feature spaces and can yield inaccurate results when the data distribution is uneven.

(5) ALE plots

Table 20

Some characteristics and code links for post-hoc explainability.

Methods	Characteristics and code links
LIME [74]	Model explanation, local approximation, interpretability, faithfulness. https://github.com/marcotcr/lime
G-LIME [170]	High-dimensional Bayesian regression, sparsity, global priors, enhanced local interpretation. https://github.com/PaddlePaddle/InterpretDL
CBR-LIME [173]	Case-based reasoning, parameter optimization, personalized explanation, image classification. https://github.com/UCM-GAIA/CBR-LIME
GraphLIME [174]	HSIC Lasso, nonlinear feature selection, graph neural network explanation. https://github.com/WilliamCCHuang/GraphLIME
SHAP [79]	Feature importance, interpret model predictions, unique theoretical solution. https://github.com/shap/shap
Captum [178]	Model interpretability, attribution algorithms, multimodal, extensibility, visualization tool. https://github.com/pytorch/captum
CP-PD profiles [182]	Ceteris Paribus, partial dependence, model-agnostic, feature impact, interpretability tool. https://github.com/ModelOriented/pyCeterisParibus
ALE plots [185]	Accumulated local effects plot, black-box model interpretation, avoidance of extrapolation, high computational efficiency. https://github.com/blent-ai/ALEPython
DALE [187]	Feature effect estimation, high-dimensional scalability, out-of-distribution (OOD) sampling resistance. https://github.com/givasile/DALE

Apley and Zhu [185] proposed a novel visualization method, Accumulated Local Effects (ALE), to illustrate the influence of predictor variables on the prediction outcomes within black-box supervised learning models. The ALE plots demonstrate the impact of feature averaging on the predictions of the ML model [186]. ALE plots compute the average changes in features and subsequently accumulate these changes on a grid. ALE plots illustrate the "main effects" of each predictor variable and the "interaction effects" between pairs of variables by aggregating local effects. The theoretical formula is as follows:

$$\begin{aligned}\hat{f}_{X_S, M(X_S)} &= \int_{Z_{0,1}}^{X_S} E_{X_C|X_S}[\hat{f}(X_S, X_C)|X_S = z_s] dz_s - \text{constant} \\ &= \int_{Z_{0,1}}^{X_S} \int_{X_C} \hat{f}(X_S, X_C) P(x_c | z_s) dx_c dz_s - \text{constant}\end{aligned}\quad (42)$$

where X_S is selected for interpretation, while X_C comprises the other features. The algorithm computes the change in the average forecast

rather than the forecast itself. This change is defined as the gradient.

$$\hat{f}^s(X_S, X_C) = \frac{\partial \hat{f}(X_S, X_C)}{\partial X_S} \quad (43)$$

Compared to PDPs, this method avoids the challenges of unreliable extrapolation with highly correlated predictors and provides greater computational efficiency. Nevertheless, it is essential to note that ALE plots, reflecting the accumulation of local effects, may exhibit heightened sensitivity to local variations in the data, potentially overlooking broader global patterns.

(6) DALE

Gkolemis et al. [187] proposed a novel cumulative local effect approximation method, Differential Cumulative Local Effect (DALE), for efficient and accurate global interpretation. This method addresses the limitations of existing approaches, such as partial dependence graphs and cumulative local effects, which are vulnerable to out-of-distribution sampling, particularly in high-dimensional input spaces with small training sets. The first-order approximation of DALE is defined as follows:

$$\hat{f}_{DALE}(x_s) = \Delta x \sum_{k=1}^{k_x} \frac{1}{|S_k|} \sum_{i: x^i \in S_k} \left[\frac{\partial f}{\partial x_s}(x_i) \right] = \Delta x \sum_{k=1}^{k_x} \hat{\mu}_k \quad (44)$$

The second-order approximation of DALE is defined as follows:

$$\hat{f}_{DALE}(x_1, x_m) = \Delta x_1 \Delta x_m \sum_{p=1}^{p_x} \sum_{q=1}^{q_x} \frac{1}{|S_{p,q}|} \left[\frac{\partial^2 f}{\partial x_1 \partial x_m}(x_i) \right] = \Delta x_1 \Delta x_m \sum_{p=1}^{p_x} \sum_{q=1}^{q_x} \hat{\mu}_s \quad (45)$$

DALE employs automatic differentiation technology to derive gradient information for all samples in a single computation. This approach significantly enhances calculation efficiency compared to the traditional ALE method. By utilizing samples from the training set to assess feature effects, DALE mitigates the risk of misleading estimates arising from samples outside the distribution range. Under specific assumptions, the DALE method is demonstrated to provide an unbiased estimate of ALE, along with a framework for quantifying the standard error, thereby increasing the reliability of the estimates.

Based on the introduction of methods/models in articles by some researchers, some characteristics and code links corresponding to these methods/models are collected (see Table 20) to assist readers in their learning process.

4.3.4. Perturbation test

Robustness perturbation testing studies are conducted to analyze the model's performance in the face of perturbations, which can reveal the

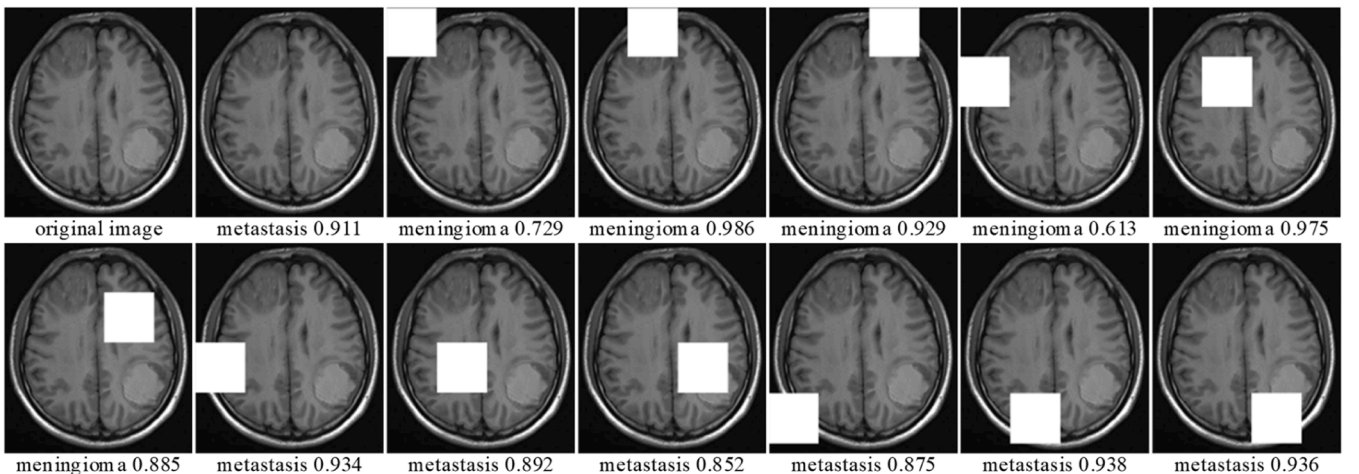


Fig. 20. Failure model discovery using occlusion perturbation tests.

Table 21

Derived expressions for sensitivity analysis methods.

Methods	Hidden layer activation function	Sensitivity coefficient of input variable to output variable	$f'(net_i)$ value
References [191]	$f(x) = \frac{1}{1 + e^{-x}}$	$s_i = f_i(1 - f_i) \sum_{j=1}^L \omega_{ij} v_{j1} f_j(1 - f_j)$	$f_j(1 - f_j)$
References [192]	$f(x) = \frac{1 - e^{-x}}{1 + e^{-x}}$	$s_i = \sum_{j=1}^L \omega_{ij} v_{j1} (1 + f_j)(1 - f_j)$	$0.5(1 + f_j)(1 - f_j)$
References [193]	$f(x) = \exp\left(\sum_{i=1}^N (x_i - u_{ji})^2 / -2v_j^2\right)$	$s_i = \sum_{j=1}^L w_j \frac{x_i - u_{ji}}{-v_j^2} \exp\left(\sum_{i=1}^N (x_i - u_{ji})^2 / -2v_j^2\right)$	$\frac{x_i - u_{ji}}{-v_j^2} \exp\left(\sum_{i=1}^N (x_i - u_{ji})^2 / -2v_j^2\right)$

internal logic and reasons behind the model's decisions. In the CV field, a common practice involves occluding various areas of the input image after training the model. Furthermore, changes in the output should be observed to identify the features that impact the results most. This method was introduced initially in the deconvolution network [153], allowing for better filtering of valid data and enhancing the model's accuracy and robustness.

As illustrated in Fig. 20, the image occlusion perturbation method has been employed to conduct an interpretability analysis on the modified tumor recognition model. This model primarily identifies gliomas, meningiomas, metastatic and pituitary. Initially, a metastasis image is identified, yielding an accuracy of 0.911. However, upon occluding brain regions with white pixels, it was observed that when the upper part of the brain was occluded, the model incorrectly identified the metastatic tumor. Conversely, occluding the lower part of the brain facilitated the correct identification of the metastatic tumor. This occlusion technique revealed that the model associates features of the upper brain region with metastatic tumors during its training phase. However, as can be seen from the figure, the growth location of the metastatic tumors is actually on the lower right side of the brain. This discrepancy suggests potential issues with the tumor identification model in accurately recognizing metastatic tumors, which may indicate a flawed model construction process. Despite the model's relatively high accuracy rate, there is a risk of misdiagnosis in the medical field, potentially leading to serious medical incidents.

Fel et al. [188] introduced an interpretability method, explaining using verified perturbation analysis. This method is the first to comprehensively employ validated perturbation analysis to explore the set of possible perturbations. The core idea involves assessing the importance of an input variable, denoted as the "u." It is achieved by measuring the impact of modifying "u" within a specific perturbation sphere, referred to as $B(x)$. This measurement aims to understand how changes in "u" affect the classification decision. A measure of adversarial overlap is introduced to achieve this goal:

$$AO(x, B) = \max_{c'} f_c(x + \delta) - \min_{c' \neq c} f_{c'}(x + \delta), 0 \quad (46)$$

where x is the set of input variables, δ represents the variable perturbation, c' is the adversarial category, c is the predicted category, $f_c(x + \delta)$ means the prediction for category c in $B(x)$, and $\min_{c' \neq c} f_{c'}(x + \delta)$ represents the minimum prediction for other categories in the perturbation sphere $B(x)$. Meanwhile, the method's formula for calculating the importance of a variable is as follows:

$$EVA(x, u, B) = \overline{AO}(x, B) - \overline{AO}(x, B_u) \quad (47)$$

Koh and Liang [75] utilized the influence function method to analyze the changes in perturbation $x \rightarrow x_\delta = x + \delta$ of an originally trained sample x affect the loss function of the output of a test sample x_{test} :

$$\frac{\partial L(x_{test}, x)}{\partial x} = -\frac{1}{n} \nabla_{\theta} L(x_{test}, \theta^*)^T H_{\theta}^{-1} [\nabla_x \nabla_{\theta} L(x, \theta^*)] \quad (48)$$

where $\theta^* = \operatorname{argmin}_{\theta} \left\{ \frac{1}{n} \sum_{i=1}^n L(x_i, \theta) \right\}$, $L(\cdot)$ is the loss function and $H_{\theta} = \frac{1}{n} \sum_{i=1}^n \nabla_{\theta}^2 L(x_i, \theta^*)$ is the positive definite Hessian matrix. The source code is available at <https://github.com/kohpangwei/influence-release>.

git.

This method can identify the sample features primarily relied upon by the model's predictive results, and experiments demonstrate the model's attribution to the decision-making features. Moreover, it can also be applied to generate adversarial samples and correct erroneous annotations.

Furthermore, Ross and Doshi-Velez [189] enhance the adversarial robustness and interpretability by regularizing the input gradients of the network model. This regularization can smooth the input gradients of the model and reduce the occurrence of extreme values, thereby enhancing the model interpretability and robustness of adversarial samples. Chen et al. [190] proposed a deep denoising non-negative matrix factorization model to learn deep, robust, and interpretable representations from big data. The model combines DL and non-negative matrix factorization to enhance the interpretability of deep representations and extract useful information from noisy data.

The robustness perturbation test interpretability provides security assurances for the model by explaining the degree of the impact on model predictions. However, these methods often have lower explanatory power.

4.3.5. Sensitivity analysis

Sensitivity analysis is an important research method, and its purpose is to quantitatively describe the impact of model $y = f(x_1, x_2, \dots, x_n)$ input variables on output variables. This method allows each attribute to vary within a possible range, thereby predicting the extent of the impact of these attribute changes on the model output values. The sensitivity analysis methods of neural networks include variable and sample sensitivity analyses. In variable sensitivity analysis, the derivative-based sensitivity analysis method employs the partial derivatives of different hidden layer activation functions [191–193] to assess the impact of input variables on the output. Specific details are shown in Table 21.

On the other hand, The average influence value method proposed by Dombi et al. [194] directly changes the eigenvalues of each independent variable. Moreover, the average according to the number of observation samples to obtain the corresponding output change value. The absolute size of these changes reflects the importance of the variable.

Koh and Liang [75] employed influence functions in sample sensitivity to characterize the changes in the loss function of a specific test sample after applying a small perturbation to the training samples, aiming to deepen understanding of the sample sensitivity of the model's prediction performance. Conversely, the LIME proposed by Ribeiro et al. [74] introduced minor disturbances around the sample points of interest and observed changes in model output. Furthermore, an interpretable simple model should be fit near the interest point based on this change to achieve local interpretability.

Dorie et al. [195] proposed a flexible and easily interpretable methodological framework for evaluating the impact of unobserved confounding factors on causal effect estimates. The framework integrates Bayesian Additive Regression Trees (BART) with a two-parameter sensitivity analysis strategy, which can assess the influence of posterior treatment effect distribution on the selection of sensitivity parameters. BART algorithm's packages and implementations are mentioned in the paper, which can be accessed through the following links: The "treatSens" package for sensitivity analysis in R is available at

Table 22

The implementation steps of permutation feature importance.

Step number	Detailed description
1. Data preparation.	Prepare a data set, including features and target variables, and divide the data set into a training set and a test set.
2. Model training.	Use the training set to train the ML model to ensure that the model fits the data.
3. Baseline performance evaluation.	Evaluate the model's baseline performance on the test set and calculate and record performance metrics. (such as accuracy, mean square error, etc.)
4. Feature replacement.	For each feature in the test set, the feature values are randomly replaced one by one, keeping other features unchanged.
5. Model re-evaluation.	Re-evaluate the model using the permuted data set and calculate new performance metrics.
6. Feature importance calculation.	Calculate the change in model performance and the difference between original and permutation performance. The larger the difference, the more significant the impact of this feature on model performance.
7. Repeat.	Each feature is replaced and evaluated multiple times to reduce the impact of randomness and obtain a more stable and reliable evaluation of one of the features.
8. Sorting features.	Ranking based on the importance value of each feature determines which features are most important to the model's predictive power.

<https://CRAN.R-project.org/package=treatSens>. The "BayesTree" package, which implements the BART algorithm, can be found at <https://CRAN.R-project.org/package=BayesTree>. Additionally, the "dbarts" package is another implementation of the BART algorithm. The source code is available at <https://CRAN.R-project.org/package=dbarts>.

Interpretability methods based on sensitivity analysis attribute explainability to input features or samples. Suitable for interpreting individual inputs, but often challenging to provide a comprehensive understanding of the network's overall performance.

4.3.6. Permutation feature importance

Permutation Feature Importance (PFI) is a method for interpreting ML models, often used for DL models [196]. The basic idea of PFI is to evaluate their impact on model performance by permuting input features. A key advantage of PFI is that it is model-independent, i.e., it can be applied to any fitted estimator. The steps of this method are shown in Table 22.

For DL models, PFI can be employed to elucidate the impact of input features on the model's output. It offers a relatively straightforward and intuitive approach to understanding how the model leverages input features. It is essential to note that PFI is computationally intensive, as multiple evaluations of the model are required for each feature. Furthermore, in the case of certain complex DL models, the interpretation of PFI may still encounter limitations, as the intricacies of the model can make it challenging to capture the effects of feature interactions. Moreover, it becomes less stable on small sample datasets.

Kaneko [197] proposed a novel method for evaluating feature importance, termed Cross-validated Permutation Feature Importance (CVPFI). This method addresses the instability issues associated with traditional PFI when the sample size is small and the potential underestimation of feature importance that can occur when there is a strong correlation among the explanatory variables. The steps are as follows:

Firstly, calculate the correlation coefficient between the features.

Subsequently, a Fisher z -transformation is applied to compute the confidence interval for the correlation coefficient.

$$z_{p,q} = \frac{1}{2} \log \frac{1 + r_{p,q}}{1 - r_{p,q}} \quad (49)$$

where $r_{p,q}$ represents the Pearson correlation coefficient between the p -

Table 23

The steps of incremental permutation feature importance explanation.

Require:	$\alpha \in (0, 1)$, sampling strategy φ_t , and $\hat{\phi}_{t-1}^{(s)}$, current model h_t , and current observation (x_t, y_t)
1:	procedure EXPLAINONE (h_t, x_t, y_t, j)
2:	$x_s \leftarrow \text{Sample}(\varphi_t)$
3:	$\hat{\lambda}_t^{(s)} \leftarrow \ h_t(x_t^{(s)}, x_s^{(s)}) - y_t\ - \ h_t(x_t) - y_t\ $
4:	$\hat{\phi}_t^{(s)} \leftarrow (1 - \alpha) \cdot \hat{\phi}_{t-1}^{(s)} + \alpha \cdot \hat{\lambda}_t^{(s)} \quad \alpha \in (0, 1)$
5:	$\varphi_{t+1} \leftarrow \text{UpdateSampler}(\varphi_t, x_t)$
6:	end procedure

Table 24

Feature summaries and code links for permutation feature importance and improvement methods.

Methods	Characteristics and code links
PFI [196]	Permutation importance, feature importance correction, model interpretability, P-value Estimation. https://scikit-learn.org/stable/modules/permutation_importance.html
CVPFI [197]	Cross-validation, PFI, consider correlation, stable assessment. https://github.com/hkaneko1985/dcekit
iPFI [198]	Incremental learning, model-agnostic, online estimation, feature importance, concept drift. https://github.com/mmschlk/iPFI.git
PermFIT [199]	Permutation test, feature importance, model-agnostic, computationally efficient, biomarker identification. https://github.com/SkadiEye/deepTL.git

th and q -th features.

Next, the model is constructed using the training data in each cross-validation iteration while the corresponding y -values are estimated from the validation data.

Finally, the CVPFI is calculated:

$$CVPFI_i = rscv - \frac{1}{J} \sum_{j=1}^J scv_{ij} \quad (50)$$

where $rscv$ represents the model score derived from the integrated y -values of all validation data during cross-validation, while scv_{ij} denotes the model score for each iteration j of feature i within the same cross-validation process.

CVPFI emphasizes the model's predictive capability and explanatory power, enabling researchers to comprehend its functioning and the impact of various features on the prediction outcomes.

Fumagalli et al. [198] proposed an efficient model-agnostic algorithm, Incremental Permutation Feature Importance (iPFI), for interpreting AI in online data streams. For a data stream at time t , with previous observations $(x_0, y_0), \dots, (x_t, y_t)$ and a sampling strategy $(\varphi_s)_{s=t_0, \dots, t}$ at a time $t_0 > 0$, the iPFI estimator is recursively defined as follows:

$$iPFI : \hat{\phi}_t^{(s)} := (1 - \alpha) \hat{\phi}_{t-1}^{(s)} + \alpha \hat{\lambda}_t^{(s)}(x_t, x_{\varphi_t}, y_t), \quad (51)$$

for $t > t_0$, $\hat{\phi}_{t-1}^{(s)} := 0$, and $\alpha \in (0, 1)$.

The interpretability of the iPFI for feature j at time t is presented in Table 23. By utilizing these features, the iPFI algorithm enhances the model's interpretability, enabling users to comprehend how the model formulates predictions at any specific moment and to identify the most significant features of the model's predictions. This understanding is essential for fostering user trust and ensuring transparency in the model.

Besides, Mi et al. [199] proposed a feature importance evaluation method based on the Permutation-based Feature Importance Test (PermFIT) for biomarker identification in complex diseases. The technique effectively evaluates and tests the importance of features without

refitting the model, thereby elucidating the role of each feature in complex models.

Rajbahadur et al. [200] conducted a case study to assess the performance of two feature importance methods, classifier-specific and classifier-agnostic, in determining feature importance rankings. Their findings underscore the necessity for caution when selecting feature-importance methods to ensure the stability and reproducibility of study results.

Based on the methods introduced in scholarly papers, characteristics are summarized, and code links corresponding to the agent models are collected, as shown in Table 24, to facilitate more efficient learning for readers.

4.4. Supplementary explanations

This article attributes interpretability methods based on information theory to supplementary explanations. Tishby et al. [201] introduced the Information Bottleneck (IB) theory that defines the relevant information in signal X as the information provided by this signal to another signal Y . Assuming X is the input random variable (e.g., facial images). At the same time, Y is the output random variable (e.g., person names). The conditional mutual information, $I(X; Y)$, is used to measure the amount of information about Y in X . The theory points out that identifying a compressed representation C of X aims to maximize the preservation of information about Y , denoted as $I_{\max}(C; Y)$, while minimizing the retention of information in X unrelated to Y , denoted as $I_{\min}(X; C)$. Therefore, the IB theory is articulated as $I_{\max}(C; Y) - I_{\min}(X; C)$ aligning with the DL interpretability requirement.

Subsequently, in 2015, Tishby and Zaslavsky [202] introduced the concept of the information plane, which is determined by quantifying the mutual information between layers within the network structure and between input-output variables. This concept reveals a possible explanation for the hierarchical network structures. The optimal architecture, the number of layers, and the characteristics and connections of each layer are all related to the bifurcation points in the IB theory.

Lan and Barner [203] proposed an improved probabilistic representation method for DL to enhance the information-theoretical interpretability of Multi-Layer Perceptrons (MLPs). Because the hidden layer activations of the mutual information estimation method when measuring the information flow in MLPs are not independently and identically distributed, a probabilistic interpretation is introduced, the MLPs as a whole are regarded as a Gibbs distribution, and the influence of the activation function on the probability measure is discussed.

A probabilistic interpretation was introduced in response to the issue that hidden layer activations in MLPs are not independently and identically distributed when measuring information flow using mutual information estimation methods. The MLPs are regarded as a Gibbs distribution, and the influence of the activation function on the probability measure is discussed. Furthermore, because of the limitations of IB theory when considering backpropagation, a new information theory explanation is proposed to explain the generalization ability of MLPs.

Liu et al. [204] introduced an explainable graph-level anomaly detection method in the Self-Interpretable Graph aNomaly dETection (SIGNET) model. It aims to identify graphs in the collection that are significantly different from most graphs and provide a meaningful explanation for each detected anomalous graph. The model operates within the multi-view subgraph information bottleneck framework. It evaluates the anomaly level of each graph. The model generates explanatory graphical reasons by extracting bottleneck subgraphs. These subgraphs are derived from the input graph and its corresponding dual hypergraph.

Pimentel et al. [205] used probes as an estimate of mutual information, introduced a control function to evaluate the effectiveness of probes, and provided corresponding explanations, refuting the view that "Prevent probes from learning the task itself, simpler probe models should be used" [206], and offered suggestions on how to select probes.

Table 25

Some characteristics and code links correspond to supplementary explanations.

Methods	Characteristics and code links
IB [201]	Signal compression, information preservation, relevance quantification, iterative algorithm. https://github.com/BioroboticsLab/IBA.git
SIGNET [204]	Self-interpretable, graph-level anomaly detection, multi-view learning, information bottleneck, subgraph explanation. https://github.com/yixinliu233/SIGNET.git
VIBI [207]	Deep variational, information bottleneck, concise and comprehensive, feature selection, black-box explanation. https://github.com/SeojinBang/VIBI.git

Bang et al. [207] used the information bottleneck principle as the standard and introduced a Variational Information Bottleneck for Interpretation (VIBI).

For each instance, the interpreter returns a probability, indicating whether to select a word, phrase, or sentence (NLP) or a set of pixels (CV) as the explanatory choice. The feature block chosen then acts as the information bottleneck to compress the information between input and output maximally. This method is system-independent and can explain any black-box model.

Applying relevant concepts from the field of information theory to neural networks can provide more informative explanations, but the effectiveness diminishes when generalizing to different network architectures.

Based on the introduction of some scholars' methods in their papers, some characteristics and code links corresponding to supplementary explanations are summarized, as shown in Table 25, so readers can learn more conveniently.

4.5. Integrated explanations

Integrated explanation refers to combining multiple explanation methods/models to enhance understanding and interpretability in the decision-making process. This approach integrates different interpretation techniques, such as feature importance analysis, gradient-based methods, and local sensitivity analysis, to achieve more comprehensive and reliable model interpretations. By integrating multiple explanation models or techniques, the limitations of a single explanation method can be compensated, which provides a richer model understanding.

Especially in Info. Fusion, integrated interpretation, and comprehensive analysis models are more transparent. For example, He et al. [208] proposed an interpretable and flexible hyperspectral imaging reconstruction method with a clear explanation path during the model optimization process. At the same time, an interpretable subspace distillation prior module was designed. Therefore, it can be seen as an integrated explanation method that combines "model optimization" and "interpretable module." The steps are shown below:

Step 1: In model optimization, the hyperspectral fusion theory first proposed transforms the reconstruction of the Full Hyperspectral Image (HSI) into the reconstruction of the subspace image. The method is achieved by taking advantage of the spectral low-rank characteristics of HSI. HSI can be expressed as the product of a few spectral basis vectors and subspace images, X is the original 3D HSI, U is the spectral basis matrix, and V is the subspace image matrix.

Step 2: The imaging process of the coded aperture snapshot spectral imaging system was modeled, including operations such as pixel shift and Hadamard product caused by the dispersion prism $Y = H(S(A \odot X))X = U \times_3 V$. Y is the 2D encoding measurement, A is the mask, S represents the shift operation caused by the dispersive prism, and H represents tensor addition.

Step 3: the interpretability module SP is proposed, directly embedded into the reconstruction process to extract subspace images and guide reconstruction.

$$X = SD(Y, \Phi, V) \quad (52)$$

where SD represents the subspace distillation operation, Φ is the sensing matrix composed of the mask and the subspace image, and V is the reconstructed subspace image.

Step 4: Using SP, the spectral basis can be indirectly optimized by optimizing the subspace image, thereby realizing the reconstruction of the HSI. The following equation illustrates restoring the spectral basis through the known subspace image and measured values.

$$U = [(\Phi_V)^T (\Phi_V)]^{-1} (\Phi_V)^T Y \quad (53)$$

Step 5: SP in inference (SP1) is tested. SP1 allows SP to be used directly in the test phase to improve the performance of existing models without retraining the existing model.

$$X = SD(Y, \Phi, A_{PCA}(A_{net}(Y, A)), k) \quad (54)$$

where A_{PCA} indicates that principal component analysis is used to extract subspace images, and k is the number of dimensions of the subspace image.

Step 6: The SP framework for HSI reconstruction (SP2). SP2 proposes a two-stage framework to transfer the task from high-dimensional HSI to low-dimensional subspace image reconstruction. The step provides a clear path to interpret and understand the model's decision-making process. The source code is available at <https://github.com/prowDIY>.

Meanwhile, Duan et al. [209] introduced a novel deep knowledge tracing model, deep knowledge tracing with multiple relations, that effectively incorporates various knowledge relationships to enhance the precision of student knowledge evaluation and the interpretability of the model.

Firstly, the graph representation learning layer uses two Generative Adversarial Networks (GANs): GraphGAN and DGGAN. GraphGAN is responsible for learning the representation vectors of undirected knowledge relationship graphs, while DGGAN focuses on directed knowledge relationship graphs. The DGGAN generator and discriminator engage in adversarial training to learn the source and target representations of nodes, while the formula for the GraphGAN generator is expressed as:

$$A_{ij} = \frac{\exp(a_i^T a_j)}{\sum_{k \neq j} \exp(a_i^T a_k)} \quad (55)$$

where A_{ij} represents the probability of connection between nodes i and j , a_i and a_j are the representation vectors of nodes i and j , respectively.

Next, the input layer of the model is formed by combining the representation vector of the knowledge point with the student's answer to the corresponding exercise, creating a complete input vector.

Thirdly, the GRU updates the student's knowledge status in the recurrent layer. The updated formula for GRU is defined as:

$$\begin{aligned} a_i^t &= \sigma(A_i a_i^t + A_{i-1} a_{i-1}) \\ r_i^t &= \sigma(A_i a_i^t + A_{i-1} a_{i-1}) \\ h_i^t &= r_i^t \odot h_{i-1} + (1 - r_i^t) \odot \tilde{h}_i^t \end{aligned} \quad (56)$$

where σ represents the sigmoid function used to output probability, a_i^t represents the input vector of node i at time t , A_i and A_{i-1} are the input weight matrix and the weight matrix from the previous state to the current state, respectively, r_i^t represents the reset gate of node i at time t , h_i^t represents the hidden state of node i at time t , $\tilde{h}_i^t$ represents the candidate's hidden state after reset.

Fourthly, In the fusion layer, the attention mechanism combines various types of knowledge relationships and autonomously determines each relationship's weight in the final prediction. The attention fusion formula is defined as:

$$\begin{aligned} z_i^t &= \alpha_i \cdot h_i^t + \beta_i \cdot \tilde{h}_i^t \\ \alpha_i, \beta_i &= \text{Softmax}(W_a h_i^t + b_a) \end{aligned} \quad (57)$$

where z_i^t represents the representation of node i at time t after fusion. α_i and β_i denote the attention weights corresponding to different representations, while W_a and b_a refer to the attention mechanism's weight matrix and bias term.

Finally, the prediction layer uses the mapping function to predict the likelihood of students correctly answering the new exercise.

The model employs the cross-entropy loss function to update the model parameters and dropout to mitigate overfitting. Therefore, the DL model, which incorporates multiple knowledge relationships and attention mechanisms, can be utilized to more accurately and comprehensively monitor students' knowledge status.

In addition, Zhang et al. [210] explained the adversarial robustness of deep models from a new perspective of neuron sensitivity, that is, by measuring the intensity of behavioral changes of neurons to benign and adversarial examples, and proposed the sensitive neuron stabilizing method. This approach combines explanatory methods based on sensitivity and robustness perturbation tests. It explains the robustness of deep models from the perspective of neuronal sensitivity by measuring the intensity of behavioral changes of neurons towards benign and adversarial examples.

Xu et al. [211] studied the model training process from the perspective of Fourier analysis. They demonstrated a general Frequency Principle on high-dimensional benchmark datasets (MNIST/CIFAR10) and DL models (VGG16): DL models typically fit the target functions from low to high frequencies. Based on this, an explanatory method that combines the frequency principle and optimization methods is proposed, utilizing the principle that low frequencies dominate DL models to accelerate the solution of the Poisson equation.

Zeng et al. [212] introduced a method based on absolute value class activation mapping. This method optimizes the gradients obtained from back-propagation and transforms them into gradients for positive activation. It enhances the visual features of activated output neurons and improves the localizing capability of the saliency map. The framework is divided into two phases: initial saliency mapping and final saliency mapping. The first stage improves the positioning ability of saliency mapping by optimizing gradients. The second stage linearly combines the optimized saliency map with the original image to enhance semantic information.

The integration of interpretations improves the understanding of the decision-making process in DL models by combining various explanatory methods or models. However, this can also introduce greater complexity and computational costs. It is crucial to select and integrate interpretation methods carefully.

5. Evaluation indicators

Interpretability evaluation indicators define a series of evaluation indicators, and the degree of model interpretability can be quantitatively measured. For example, the model's explanatory power can be assessed by comparing the difference between the actual output and the predictions of a simplified model. Due to the nature of the explanations generated for DL and the diversity of input data types, the interpretability evaluation indicators offered also differ [213]. So far, there is no unified and universal standardized index for interpretability. This section divides the interpretable assessment indicators into qualitative and quantitative assessments.

5.1. Qualitative assessment

The qualitative evaluation method relies on human visual perception to evaluate whether the interpretation results are consistent with human cognition. This method is simple, clear, easy to understand, and more

user-friendly. Qualitative evaluation mainly includes intuitiveness, understandability, etc.

(1) Visualization

Visualization presents the results of data, information, or models as graphics, images, or animations. Currently, for DNNs, visualization is more often used for interpretation, i.e., constructing saliency maps to highlight the most relevant information, thus providing interpretability [140,169]. Therefore, the qualitative evaluation method of interpretability is mainly considered from the aspect of visualization effect, including two aspects:

1)Observing the concentration and coverage of the target area, the saliency map needs to focus on the target area of interest and ignore other irrelevant regions. The more comprehensive the highlighted area in the saliency map covers the target of interest, the better the visualization effect.

2)Examine multi-objective visualization effects. When multiple targets of the same category appear in the image simultaneously, the visualization method can locate multiple targets without missing one.

(2) Explanation's basic unit

The explanation's basic unit [214] refers to the components that make up the explanation, and it may include feature importance values, instances from the training set, rule lists, pixels, etc. Consequently, it is essential to understand both the form and quantity of these explanation's basic units. If an explanation is based on features, it is crucial to determine whether it encompasses all features or only a select few. Additionally, when a counterfactual explanation is presented, assessing whether a causal relationship exists between the provided explanation and the decision outcome and the feasibility of the counterfactual outcome for the user is crucial.

(3) Understandability

It refers to human assessment of the extent to which an interpretable method is reasonable and understandable. Mohseni et al. [215] proposed a human-centered assessment method for evaluating feature saliency map interpretations generated by interpretable algorithms. This method is fast, replicable, and objective, enabling efficient assessment of the validity of explanations. However, this method may be subject to human bias due to human involvement in the process. Nevertheless, human consensus on a large number of data labels can counteract this bias to some extent.

(4) Explicitness

Alvarez Melis and Jaakkola [216] discuss explanations provided by Self-Explaining Neural Networks (SENN) versus common input-based explanation methods. When evaluating model interpretability, explicitness/intelligence is one of the main criteria. Explicitness refers to the clarity and directness of the explanation itself. Intelligibility refers to whether the explanation is easy for human users to understand. The paper does not give a precise formula to quantify explicitness/intelligibility but instead assesses it through the following steps:

1)Visualization of Explanations. The intuitiveness and comprehensibility of SENN's explanations are demonstrated by visually comparing the explanations provided by SENN with those of other methods.

2)Concept Prototypes. Each concept in SENN is associated with one or more prototypes, which can be images, feature maps, or other elements that intuitively represent the concept.

3)Simplicity of Explanation. The explanations constructed by SENN are more concise because they are based on concepts rather than raw input features, thereby reducing the complexity of the explanation.

(5) Faithfulness

Jacovi and Goldberg [217] discussed the increasing popularity of DL in NLP and the resulting necessity for interpretable systems. The article underscores the importance of evaluating the faithfulness of explanations, specifically whether these explanations accurately represent the reasoning underlying the model's predictions. The authors assess faithfulness by positing three hypotheses:

1)Model Assumption. Two models will make the same predictions using the same reasoning process. An interpretation system is unfaithful

if it results in different interpretations of models that make the same decisions.

2)Prediction Assumption. On similar inputs, the model makes similar decisions if and only if its reasoning is similar. An interpretation system is unfaithful if it interprets similar inputs and outputs differently.

3)Linearity Assumption. Certain input parts are more important to the model reasoning than others. Moreover, the contributions of different input parts are independent of each other. Under certain circumstances, heatmap interpretations can be faithful.

(6) Grounded Evaluation

Doshi-Velez and Kim [76] proposed a framework for evaluating interpretability, which consists of three main evaluation methods: Application-grounded Evaluation, Human-grounded Evaluation, and Function-grounded Evaluation.

1)Application-grounded Evaluation. This evaluation method involves conducting human experiments in real application scenarios to verify the interpretability of the model.

2)Human-grounded Evaluation. This method assesses the quality of explanations through simplified human experiments, focusing on maintaining the essence of the target application without necessarily being in a natural application environment.

3)Functionally-grounded Evaluation. This type of evaluation does not involve human experiments but uses proxy tasks or quantitative metrics to evaluate the interpretability of the model.

5.2. Quantitative assessment

Interpretability methods that rely on human subjective metrics to assess models are insufficient, and quantitative assessments objectively compare different interpretations.

Yeh et al. [218] proposed two objective indicators for evaluating the model interpretability: infidelity and sensitivity, and the optimal solutions of these two indicators were analyzed in detail.

(1) Infidelity

Infidelity assesses the degree to which an explanation reflects the actual response of the predictive function to significant perturbations in the input features. An explanation with low infidelity is one that closely captures the predictive function's reaction to changes in the features. Given a black-box function f , an explanation function Φ , and a random variable $I \in R^d$ (with a probability measure μ_I , representing the meaningful perturbations of interest), the distortion is defined as:

$$INFID(\Phi, f, x) = E_{I \sim \mu_I} \left[(I^T \Phi(f, x) - (f(x) - f(x - I)))^2 \right] \quad (58)$$

(2) Sensitivity

Sensitivity evaluates how much an explanation is affected by minor perturbations in the input features. An explanation with low sensitivity indicates that it is not overly reactive to minor variations in the input, which can be considered a form of simplicity. Given the black-box function f , the explanation function Φ , and a specified input neighborhood radius r , the max-sensitivity of the explanation is defined as:

$$SENS_{MAX}(\Phi, f, x, r) = \max_{\|y-x\| \leq r} \|\Phi(f, y) - \Phi(f, x)\| \quad (59)$$

(3) Fidelity

Fidelity refers to a measure of how closely the explanations approximate the predictions of the black-box model. Guo et al. [219] introduced a local explanation method using nonlinear approximation. This approach is designed to provide high-fidelity interpretations of DL models in security applications. Some key evaluation indicators are mentioned in the paper.

Local approximation accuracy is one of the quantitative metrics introduced in the paper. It directly assesses interpretation accuracy by comparing the approximate and original decision boundaries. Root Mean Square Error (RMSE) is used to quantify the accuracy of this approximation.

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (p_i - \hat{p}_i)^2}{n}} \quad (60)$$

where p_i represents a single prediction obtained from the target classifier, $\hat{p}_i$ represents the approximate prediction obtained from the explanatory method, and n is the total number of test data samples. A lower RMSE means that the approximated decision boundary ($\hat{p}$) is closer to the true boundary (P), indicating a higher fidelity of interpretation.

(4) Consistency

Consistency refers to a measure of whether the interpretation results of similar models produced by the same task will be different [220–222]. Han et al. [223] introduced the concept of local Lipschitz conditions [216] to analyze and explain the nature of consistency. This mathematical expression explains the relationship between consistency and output consistency. If the model output $h(x; \theta)$ is Lipschitz continuous locally in the interpretation I , $h(x; \theta)$ is locally difference-bounded by I , it satisfies the following:

$$\|h(x; \theta) - h(x_0; \theta)\| \leq L \|I(x) - I(x_0)\| \quad (61)$$

For current data point/instance x and reference data point/initial data point x_0 , there exist $L \in R$ and $\delta > 0$, when $\|x - x_0\| < \delta$, then the consistency of the interpretation will inevitably lead to the consistency of the output.

(5) Stability

Stability refers to the stability of the explanations generated by XAI when interpreting samples. In evaluating existing interpretation methods, the stability of the interpretation method is assessed by the similarity of multiple interpretations by repeating the method several times for the same sample [224]. The similarity of explanations is usually assessed using Jaccard's coefficient [225] with the following formula:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cap B|}{|A| + |B| - |A \cap B|} \quad (62)$$

where A and B represent the features used by the decision rules generated in different runs; the value of this coefficient lies between 0 and 1.

For the rule agent model, the author randomly selected 20 % of the samples from the covered set for perturbation. The average Jaccard coefficient was then calculated between the set of rule items in the matched rules before and after the sample perturbation, serving as a measure of the stability of the explanation.

(6) Model approximation degree

Lei et al. [226] proposed a convergent interpretation scheme for DNNs, which aims to provide a provably convergent and consistent interpretation for DNNs. In this article, the authors introduce the evaluation metric of the model approximation degree. It is used to measure the closeness of the predictions between the explanatory model D' and the original DNN model D , which is defined as:

$$\sigma(D, D') = \frac{1}{k(D, D')}, \quad k(D, D') = \frac{1}{n} \sum_{i=1}^n |y_i - y'_i| \quad (63)$$

where $k(D, D')$ is the sum of the absolute values of the difference between the prediction vectors of models D and D' . $y = [y_1, y_2, \dots, y_n]$ and $y' = [y'_1, y'_2, \dots, y'_n]$ are the prediction vectors for model D and model D' , respectively.

When $k(D, D')$ is small, it means that the predictions of the explanatory model D' are very close to the original model D . Therefore, $\sigma(D, D')$ is larger, indicating a higher model approximation.

(7) Causality

Causality pertains to the relationship between the generated explanation and the predictions made by the black box model. Specifically, it indicates that the features included in the explanation are the determinants of the predicted outcome. In evaluating causal explanation

Table 26

Summaries the advantages and disadvantages of qualitative and quantitative assessment.

Methods	Advantages	Disadvantages
Qualitative assessment	Flexibility, in-depth understanding, contextual adaptation.	Highly subjective, poorly repeatable, time and resource intensive.
Quantitative assessment	Objective, easily replicable, data driven.	Details may be overlooked, rely on specific metrics, ignore user experience.

Table 27

Some Interpretability tools and code links.

Interpretability tools libraries	Code links
AIX360	https://github.com/Trusted-AI/AIX360.git
TransformerLens	https://github.com/TransformerLensOrg/TransformerLens.git
lucid	https://github.com/tensorflow/lucid.git
automated-interpretability	https://github.com/openai/automated-interpretability.git
awesome-production-machine-learning	https://github.com/EthicalML/awesome-production-machine-learning.git
interpret	https://github.com/interpretml/interpret.git
BERTopic	https://github.com/MaartenGr/BERTopic.git
lit	https://github.com/PAIR-code/lit.git
stellargraph	https://github.com/stellargraph/stellargraph.git
shapash	https://github.com/MAIF/shapash.git
alibi	https://github.com/SeldonIO/alibi.git
ganspace	https://github.com/harskish/ganspace.git
Transformer-Explainability	https://github.com/hila-chefer/Transformer-Explainability.git
penzai	https://github.com/google-deepmind/penzai.git
interfacegan	https://github.com/genforce/interfacegan.git
DALEX	https://github.com/ModelOriented/DALEX.git
imodels	https://github.com/csinva/imodels.git
Awesome-explainable-AI	https://github.com/wangyongjie-ntu/Awesome-explainable-AI.git
xai	https://github.com/EthicalML/xai.git
responsible-ai-toolbox	https://github.com/microsoft/responsible-ai-toolbox.git

methods, it is essential to note that most current approaches are grounded in counterfactual explanations. Consequently, the effectiveness of these methods is assessed by examining the quality of the counterfactual explanations they produce. The counterfactual explanations have several evaluation metrics, such as proximity, diversity, etc.

Proximity refers to counterfactual explanations that should be as similar as possible to the original instance, and the formula is as follows:

$$Proximity = \frac{1}{k} \sum_{i=1}^k dist(c_i, x) \quad (64)$$

Diversity refers to counterfactual explanations generated for a data instance that should be different from each other, and the formula is as follows:

$$Diversity = \frac{1}{C_k^2} \sum_{i=1}^{k-1} \sum_{j=i+1}^k dist(c_i, c_j) \quad (65)$$

(8) Simplicity

Simplicity metrics can be quantified by counting the number of features utilized in the explanation or assessing the complexity of the logical rules. A lower number of features or more straightforward rules indicates greater simplicity. Simplicity can be expressed by counting the number of features K used by the explanation: $Simplicity = 1/K$. A higher value for the simplicity indicator typically indicates a more straightforward explanation.

Finally, in this article, the advantages and disadvantages of qualitative and quantitative assessment methods are summarised in Table 26

for comparison and use by the reader.

The current evaluation indicators for interpretability analysis are varied and intricate. Objects may receive different scores under different indicators, leading to difficulty establishing consistent reliability. While creating universal assessment standards across all domains may be impractical in the near term, there is potential to develop unified evaluation criteria for specific fields like medical care, education, cybersecurity, and intelligent manufacturing. This endeavor would necessitate collaboration from interpretability researchers, but it is feasible.

Finally, after introducing some interpretability evaluation indicators, links to some interpretability tools are given in this review in Table 27, and it is hoped that readers will be able to combine the theoretical knowledge mentioned above with the code events very soon to have a more in-depth understanding of model interpretability.

6. The trade-off between interpretability and model performance

In DNNs, the complexity of model structures poses a growing challenge to interpretability. When delving into research on model interpretability, researchers must consider not only the interplay between interpretability, model accuracy [227], and robustness [228] but also the connections to fairness [229], transparency [230], and other related factors. Striking the optimal balance between interpretability and these performance metrics is a pressing research issue that demands resolution.

6.1. Interpretability and accuracy

Model accuracy is the most fundamental performance indicator, directly affecting prediction capabilities. However, high accuracy does not always coincide with high interpretability. Sometimes, to pursue higher accuracy, researchers may design complex network structures with a large number of parameters, but this may cause the model's decision-making process to become opaque [231–233]. To address this issue, researchers began to explore ways to improve interpretability by simplifying the model structure or adopting more intuitive feature extraction methods while striving to maintain or improve model accuracy.

Dziugaite et al. [234] researched the statistical implications of incorporating interpretability constraints in ML models, focusing on the balance between interpretability and predictive accuracy. Formal modeling in ML often revolves around the Empirical Risk Minimization (ERM) framework, with interpretability as a key constraint in the learning process. One common approach is applying ERM to H_I and H , which are considered among the simplest models. When an algorithm performs empirical risk minimization on the hypothesis set H for all possible data sets S , the learned classifier $A(S)$ has the most minor risk:

$$\hat{R}_S(ERM_H(S)) = \min_{h \in H} \hat{R}_S(h) \quad (66)$$

The authors reference a group of classifiers trained by the general ERM algorithm using $ERM_H(S)$, with h_{ERM} denoting a component of $ERM_H(S)$. The interpretable learning algorithm can be formalized as ERM on a training data set, resulting in the classifier $h_{ERM}^0 \in ERM_{H_I}(S)$.

Gorczaczany et al. [235] utilized a Fuzzy Rule-Based Classification System (FRBCs) and multi-objective evolutionary optimization algorithms to optimize a decision support system for enhancing airline passenger satisfaction. This method has advantages in uncovering the significance of passenger satisfaction attributes and optimizing the balance between interpretability and accuracy in knowledge discovery systems.

The accuracy of FRBCs is measured as follows:

$$Q_{ACC}^{(lm)} = 1 - Q_{RMSE}^{(lm)}, \quad Q_{RMSE}^{(lm)} = \sqrt{\frac{1}{Kc} \sum_{k=1}^K \sum_{j=1}^c [\mu_{B_k^{(singl.)}(lm)}(y_j) - \mu_{B'_k}(y_j)]^2} \quad (67)$$

where $Q_{RMSE}^{(lm)} \in [0, 1]$, B'_k is the system's fuzzy set response to the learning data sample $x_k^{(lm)}$, $B_k^{(singl.)}(lm)$ is the fuzzy single response required for this sample.

The complexity-related interpretability of FRBCs is expressed as follows:

$$Q_{INT} = 1 - Q_{CPLX}, \quad Q_{CPLX} = \frac{Q_{RINP} + Q_{INP} + Q_{FS}}{3} \quad (68)$$

where

$$\left(Q_{RINP} = \frac{1}{R} \sum_{r=1}^R \frac{n_{INP}^{(r)} - 1}{n - 1}, n > 1 \right), \left(Q_{INP} = \frac{n_{INP} - 1}{n - 1} \right), \left(Q_{FS} = \frac{n_{FS} - 1}{\sum_{i=1}^n a_i - 1}, n > 1 \right) \quad (69)$$

The FRBC's complexity measure Q_{CPLX} is an average of three sub-indices, the average complexity of particular rules Q_{RINP} , average complexity of particular rules Q_{INP} , active fuzzy set Q_{FS} and $n_{INP}^{(r)}$ is the number of active input attributes in the r -th rule, n_{INP} and n_{FS} are the number of active inputs and fuzzy sets in the entire system respectively.

Coscato et al. [236] introduced the Neural Local Smoother (NLS), a neural network architecture designed to offer precise predictions while maintaining interpretability. NLS incorporates a smooth local linear layer onto a conventional neural network, enabling it to produce predictive results similar to current cutting-edge ML models but with enhanced interpretability. The steps are followed:

Step 1: The data set $(X_1, Y_1), \dots, (X_n, Y_n)$ is given, where $X_i \in R^d$ is the feature, and $Y_i \in R$ is the target to be predicted.

Step 2: NLS trains a neural network with an output layer of dimension $\Theta(x) = (\beta_0, \beta_1(x), \dots, \beta_d(x))$, where d represents the dimension of x . Predictions are generated by combining these outputs using the formula provided.

$$G_{\Theta}(x) := \beta_0 + \sum_{i=1}^d \beta_i(x)x_i \quad (70)$$

Step 3: Use the squared loss function to learn the network weights on the given training data set $(X_1, Y_1), \dots, (X_n, Y_n)$, where Γ is the set of all possible values of parameters (weights) related to the network.

$$\gamma^* = \operatorname{argmin}_{\gamma \in \Gamma} \sum_{i=1}^n [y_i - G_{\Theta}(x_i)]^2 \quad (71)$$

Step 4: The model is then interpreted locally by analyzing the value of $\beta_i(x^*)$, similar to, the interpretation of a linear model.

Step 5: NLS is extended to the classification problem to estimate the class probability:

$$P(Y=y|x) = \frac{\exp(\beta_0^{(Y=y)} + \sum_{i=1}^d \beta_i^{(Y=y)}(x)x_i)}{1 + \exp(\beta_0^{(Y=y)} + \sum_{i=1}^d \beta_i^{(Y=y)}(x)x_i)} \quad (72)$$

In addition, Rey et al. [237] investigated the optimization of accuracy and interpretability in Fuzzy Rule-Based Systems (FRBSs) by introducing the concept of rule correlation. The utilization of rule relevance can enhance this balance further. Nesvijejskaia et al. [238] introduced the balance between accuracy and interpretability in fraud detection models. They propose state-of-the-art methods in ML that are tailored to handle the challenges associated with fraud tasks, such as data imbalance, limited labeled data, and concept drift.

He et al. [239] emphasized balancing accuracy and interpretability in extracting rules from artificial neural networks. They cautioned that an excessive focus on interpretability could potentially lead to a

Table 28

Worst-case evaluation algorithm based on genetic algorithm.

Steps	Specific implementation process
1. Initialization.	1. The initial population is randomly generated, and each individual represents a potential input perturbation.
2. Selection.	2. Select individuals according to the fitness function $F(x_1) = I_c \cdot D(g(x), g(x_1))$, where I_c is an indicator function. When all feasible individuals $J(f(x), f(x')) < 0$ that satisfy the constraints, $I_c = 1, F > 0$. If the constraint is violated $I_c = -1, F < 0$. D represents the explained difference measure, $g(x)$ and $g(x_1)$ is the interpretation of the original input and perturbed input, respectively.
3. Crossover.	3. Genetic information is exchanged between parent individuals through crossover operations to produce offspring.
4. Mutation.	4. Randomly change some characteristics of the offspring individuals to maintain population diversity and ensure that the mutant samples are still within the specification ball $B_{x,r}$.
5. Termination.	5. The algorithm terminates When the maximum number of iterations is reached or the fitness is no longer significantly improved.
6. Optimization.	6. For $sol_p F$, the goal is to maximize the explained difference $D(g(x), g(x_1))$, while maintaining $J(f(x), f(x_1)) \geq 0$. For $sol_b F$, The goal is to minimize interpretation differences $D(g(x), g(x_1))$, while maintaining $J(f(x), f(x_1)) \leq 0$.

decrease in model performance. It means trying to make models that are both highly accurate in their predictions and transparent in their decision-making process.

6.2. Interpretability and robustness

The model robustness refers to its ability to maintain a stable performance level in the face of various uncertainties and disturbances. While improving model robustness, interpretability also plays a significant role [240].

Alvarez-Melis and Jaakkola [241] highlighted the importance of explanation robustness for interpretability, stating that consistent outputs should result from similar inputs. They argue that existing interpretation methods, including gradient-based and perturbation methods, lack robustness and are sensitive to minor input changes. To address this issue, they introduce the concept of local Lipschitz continuity as a quantitative measure of model robustness.

By defining local Lipschitz continuity, $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ is locally Lipschitz if for every x_0 there exist $\delta > 0$ and $L \in \mathbb{R}$ such that $\|x - x_0\| < \delta$ implies $\|f(x) - f(x_0)\| \leq L\|x - x_0\|$.

Table 29

Probabilistic evaluation algorithm based on subset simulation.

Steps	Specific implementation process
1. Design of intermediate events.	1. Decompose pF and rF into a series of intermediate events, such as pF_i and rF_i .
2. Estimating conditional probabilities.	2. Use the metropolis-hastings algorithm to estimate conditional probabilities $P(F_i F_{i-1})$.
3. Choices of intermediate event threshold.	3. Set the threshold $L_i = \{\beta_i, \gamma_i, \alpha_i\}$ of intermediate events to adjust the size of the conditional probability, usually set $P(F_i F_{i-1}) \approx \rho$, where ρ is a preset constant.
4. Simulating new samples.	4. New samples are simulated from $q_{x_i F_i}$ through the metropolis-hastings algorithm and added to the sample set.
5. Termination condition and returned estimation.	5. Terminates when $\hat{P}_F$ falls to $P_{\min}$, returning the estimated $\hat{P}_F$ and a measure of the estimated error (such as the coefficient of variation cov).
6. Probability measure.	6. Calculated by recursive formula $P_F := P(F_m) = P(\bigcap_{i=1}^m F_i) = P(F_1) \prod_{i=2}^m P(F_i F_{i-1})$, where m is the total number of intermediate events, $F = F_m \subset F_{m-1} \subset \dots \subset F_2 \subset F_1$.

The subsequent robustness estimation poses an optimization problem in estimating the local Lipschitz constant at each point:

$$\hat{L}(x_i) = \operatorname{argmax}_{x_j \in B_\epsilon(x_i)} \frac{\|f(x_i) - f(x_j)\|_2}{\|x_i - x_j\|_2} \quad (73)$$

where $B_\epsilon(x_i)$ is a ball centered on x_i and with a radius of ϵ .

At the same time, a local Lipschitz constant estimate for discrete samples is also proposed:

$$\tilde{L}_X(x_i) = \operatorname{argmax}_{x_j \in N_\epsilon(x_i) \leq \epsilon} \frac{\|f(x_i) - f(x_j)\|_2}{\|x_i - x_j\|_2} \quad (74)$$

where $N_\epsilon(x_i) = \{x_j \in X \mid \|x_i - x_j\| \leq \epsilon\}$.

The article employs a range of explanation methods and evaluates them across various data sets and prediction models, emphasizing the importance of striking a balance between explainability and robustness in the design of such methods. The decision on this balance may vary depending on the application scenario and the intended purpose of the explanation.

Huang et al. [242] proposed versatile and efficient evaluations for the robustness of interpretability, which quantifies and assesses the robustness of the DL model's explanatory (i.e., interpretable AI, XAI). The researchers point out that despite numerous efforts, existing XAI methods still lack robustness, and small input perturbations can lead to entirely different interpretation results. Therefore, the paper proposes two key evaluation indicators: worst-case explanation differences and a probabilistic measure of general robustness. The paper introduces two black-box evaluation methods, genetic algorithm, and subset simulation, to address the challenges existing methods cannot fully address. The steps for implementing the algorithm for two evaluation indicators are detailed in Table 28 and Table 29.

Crabbé and van der Schaar [243] introduced the concepts of explanatory invariance and equivariance and conducted a systematic robustness evaluation of interpretive methods in DNNs. The authors utilized the theoretical framework of geometric DL to propose two quantitative evaluation indicators for model invariance under symmetry groups: the explanation invariance measure Inv_g and the explanation equivariance measure $Equiv_g$. These indicators are employed to assess the consistency of interpretive methods in interpreting results when confronted with symmetric transformations of input data.

The explanation invariance metric defines a measure of the invariance of explanation e to G :

$$Inv_g(e, x) \equiv \frac{1}{|g|} \sum_{g \in G} s_e[e(\rho[g]x), e(x)] \quad (75)$$

The equivariance metric of explanation defines a metric that measures the equivariance of explanation e for G :

$$Equiv_g(e, x) \equiv \frac{1}{|g|} \sum_{g \in G} s_e[e(\rho[g]x), \rho'[g]e(x)] \quad (76)$$

where $s_e : \mathbb{R}^2 \rightarrow \mathbb{R}$ is the similarity score on the interpretation space $\mathbb{R}$.

Boychev [244] contended that adversarial training enhances the model's robustness to minor input alterations by incorporating adversarial samples during training, often imperceptible to humans. Research indicates a discrepancy between the interpretability and robustness of the model, but with a noted correlation. Adversarial training not only boosts model robustness but also improves interpretability.

Besides, using visualization techniques, one can intuitively observe the changes in the activation patterns of the model under adversarial sample attacks, which leads to the design of more robust feature extraction and defense mechanisms [245,246]. Interpretability can also help researchers identify and correct overfitting phenomena during model training, thus enhancing the model's generalization ability and robustness through understanding its learning process.

Table 30

The evaluation indicators and evaluation framework.

Evaluation indicators	
Fairness violation	Measures the extent to which a model deviates from fairness standards, encompassing both statistical equality and equality of opportunity.
Model complexity	Model complexity correlates with the feature count, where a smaller set enhances interpretability through reduced complexity.
Evaluation framework	
Generation of synthetic datasets	To study factors that impact the trade-off between explainability and fairness.
Classifier selection	The experiments encompassed a range of classifiers.
Trade-Off curves	Trade-off curves depict the relationship between model fairness and complexity, illustrating the balance between fairness violation and the number of features in a classifier.

6.3. Interpretability and fairness

Interpretability and fairness are closely linked concepts in the context of ML/DL. Interpretability refers to understanding how a model generates its predictions, while fairness is detecting and mitigating biases towards specific groups. These two aspects are often discussed in conjunction with responsible AI frameworks and at ML conferences. Alongside accuracy, businesses and researchers should prioritize explainability and fairness to ensure these models' ethical and transparent use [229].

Rule-based agent models are trained with the dual objectives of predictive accuracy and fairness. Wang and Saar-Tsechansky [247] introduced a model-agnostic method (A rule-based model) to address prediction bias in black-box decision-makers, Augmented Fairness (AuFair), particularly in human decision-making scenarios, as indicated below:

$$\begin{aligned} \text{Augmented decision-maker } f: & \text{ if } x_i \text{ obeys } R_+, Y = 1 \\ & \text{ else if } x_i \text{ obeys } R_-, Y = 0 \\ & \text{ else } Y = h(x_i) \end{aligned} \quad (77)$$

where R_+ is the positive rule set, R_- is the negative rule set, and h is the original decision maker.

When constructing R_+ and R_- , researchers strive to optimize two objectives: prediction accuracy and fairness. Prediction accuracy is assessed using prediction error, while fairness is measured using (un) equal opportunity, which defines fairness as:

$$\text{Error} : L(f; D, y_Q) = \sum_{i \in Q} I(y_i \neq f(x_i)) \quad (78)$$

$$\text{Bias} : \Phi(f; D, y_Q) = |P_r(\hat{Y}=1|z=0, Y=1) - P_r(\hat{Y}=1|z=1, Y=1)| \quad (79)$$

For convenience, researchers will overlook the dependence on D and y_Q in the notation. A multi-objective optimization was formulated to optimize both objectives simultaneously. This type of optimization addresses conflicting objectives that need to be maximized or minimized simultaneously, resulting in a set of trade-off solutions known as Pareto optimal sets. The researchers' multi-objective optimization can be stated as $\min(\Phi(f), L(f))$. The researcher aims to acquire a set of Pareto optimal solution sets, denoted as $F^* = \{f_1, \dots, f_{|G|}\}$, such that $\forall f \in F^*$ outperforms F . f is dominated by f' , then the term "advantage" is defined as follows:

$$\begin{aligned} f' \prec f, & \text{ if } \\ L(f') < L(f) & \text{ and } \Phi(f') \leq \Phi(f) \text{ or if } L(f') \leq L(f) \text{ and } \Phi(f') < \Phi(f) \end{aligned} \quad (80)$$

The AuFair approach achieves a superior trade-off between fairness and accuracy by providing interpretable decision rules to biased black-box human decision-makers. This approach offers essential

Table 31

Modules used in the transparency design network.

Module Type	Operation	Language Analogue
Attention	Attention $\times$ Stem $\rightarrow$ Attention	Which things are [property]?
Query	Attention $\times$ Stem $\rightarrow$ Encoding	What [property] is x ?
Relate	Attention $\times$ Stem $\rightarrow$ Attention	Left of, right of, in front, behind
Same	Attention $\times$ Stem $\rightarrow$ Attention	Which things are the same [property] as x ?
Comparison	Encoding $\times$ Encoding $\rightarrow$ Encoding	Are x and y the same [property]?
And	Attention $\times$ Attention $\rightarrow$ Attention	Left of x and right of y
Or	Attention $\times$ Attention $\rightarrow$ Attention	Left of x or right of y

flexibility in practical applications, enabling the identification of task-specific, advantageous trade-offs.

Jabbari et al. [248] delved into the balance between model interpretability and fairness within ML models. Their study introduced suitable metrics and a novel evaluation framework to assess the interpretability and fairness of various model categories, including linear and tree-based models, across synthetic and real-world datasets. The indicators and assessment framework are shown in Table 30.

Meng et al. [249] pointed out that algorithms in medical decision-making can be inequitable in patient populations of different races, genders, and ages, resulting in treatment disparities. Therefore, a comprehensive interpretability and fairness analysis was conducted using the MIMIC-IV dataset and DL models. The link between interpretable methods and fairness metrics is demonstrated, as well as how interpretable methods can be used to identify potential bias.

Besides, Jo et al. [250] emphasized the need to consider ML algorithms' interpretability, fairness, and high accuracy when using them in high-risk domains and explored the trade-offs between interpretability, fairness, and predictive accuracy. Pradhan et al. [251] discussed debugging fairness issues in ML models by generating interpretable data-driven explanations. On the other hand, Miroshnikov et al. [246] introduced a fairness interpretation framework that utilizes the Wasserstein metric to assess and elucidate bias in ML models. In short, good interpretability provides understandable reasoning and ensures that unbiased results are obtained.

6.4. Interpretability and transparency

Algorithm interpretability is an objective attribute of an algorithmic model, whereas algorithm transparency pertains to the relationship between the results of algorithmic operations and subjective expectations. Novakovsky et al. [252] introduced a novel DL model named ExplainNN, designed explicitly for genomics tasks and aims to deliver interpretable and transparent neural networks. The researchers achieve global interpretability by evaluating the weight of each unit within the ExplainNN model. Additionally, local interpretability is facilitated by calculating the contribution of each unit to the prediction result, referred to as the unit importance score.

Mascharka et al. [253] primarily investigated the balance between performance and interpretability in visual reasoning tasks. Their research introduces a neural module network named the transparent design network, which efficiently facilitates visual reasoning while preserving model transparency by incorporating an explicit attention mechanism.

As demonstrated in Table 31, the transparent design network constructs a model capable of executing complex reasoning tasks by integrating a modular design composed of a series of interpretable visual reasoning primitives. Among these primitives, "attention" and

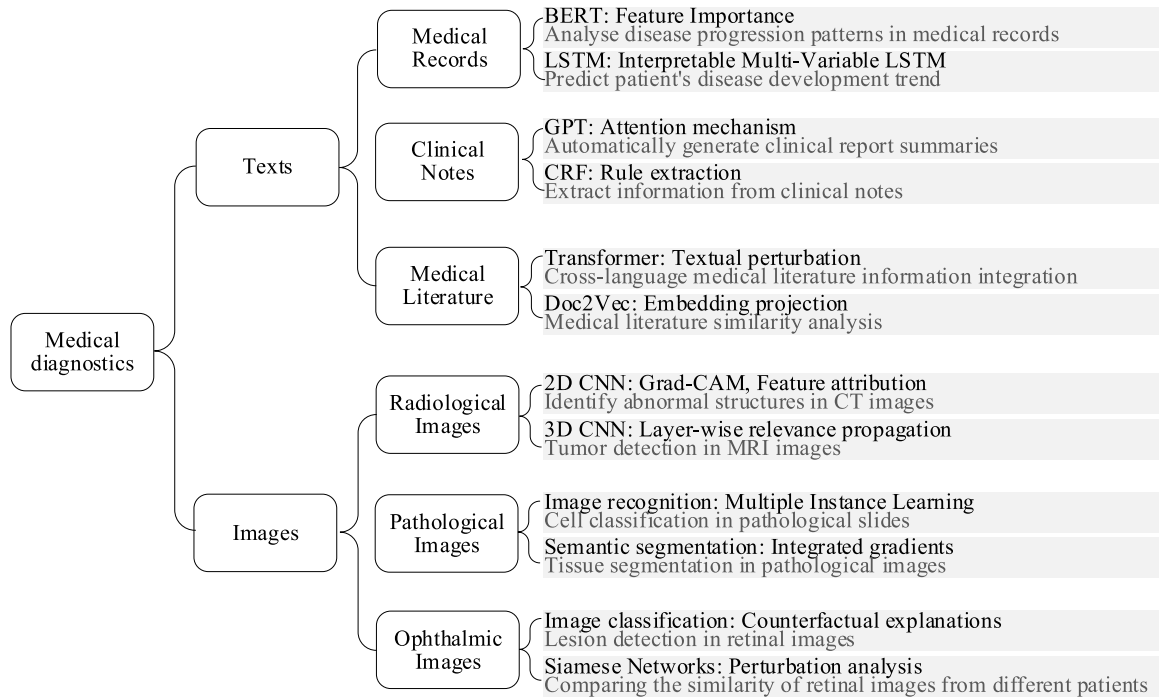


Fig. 21. Some subdivisions and methods of medical diagnosis.

"encoding" denote the single-dimensional and high-dimensional outputs from the module, respectively. "stem" refers to the image features generated by a trained neural network. The variables x and y represent different objects within the scene, while [property] signifies one of the following attributes: color, shape, size, or material.

Joyce et al. [254] primarily discussed the significance of applying XAI in the mental health domain and comprehensively understood algorithms through transparency and explainability. They note that within AI and ML, the definition of "explainability" lacks uniformity. The authors suggest that interpretability can be approximated by "understandability," a function of transparency and interpretability. They propose transparency and interpretability for the understandability framework, emphasizing the importance of ensuring human comprehension of model inputs, processes, and outputs when applying AI/ML models in mental health research. Furthermore, they argue that transparency and interpretability are interrelated, collectively contributing to the understandability of a model. Transparency pertains to how the input and feature space align with human understanding, while interpretability focuses on how humans can grasp the computational processes and structure of the model. These elements enhance the model's comprehensibility for users, such as medical professionals.

Xu et al. [255] proposed a hierarchical deep-learning system designed to automatically diagnose glaucoma, utilizing a limited number of samples while ensuring transparency and interpretability. This system allows for the visualization of intermediate steps in the prediction process, thereby providing diagnostic evidence comprehensible to ophthalmologists. It encompasses the segmentation of the OD, OC, and RNFLD, along with a two-dimensional feature display of the MCDR and ISNT scores. Additionally, it facilitates the comparison of currently evaluated fundus images with those with confirmed diagnoses in the database, specifically regarding MCDR, ISNT scores, and RNFLD. The performance indicators for model interpretability include sensitivity and specificity.

Interpretability focuses on understanding the model's decision-making process, while transparency emphasizes the openness of the model's internal mechanisms. High transparency can enhance interpretability but may also increase model complexity. Conversely, methods to improve interpretability may introduce computational

overhead and reduce accuracy. Therefore, performance, explainability, and transparency should be comprehensively assessed according to specific needs and scenarios to identify the optimal balance.

6.5. Trade-off strategies for models

In summary, alongside the considerations of model accuracy, robustness, fairness, and transparency, research on DL explainability must also address its relationship with trust, operability, user experience, and computing costs, among other significant factors. For instance, Lyu et al. [256] proposed a framework known as trustworthy decision-making, which seeks to enhance explainability to facilitate trustworthy decision-making.

Similarly, Tolomei et al. [257] introduced a technique that modifies the prediction outcomes of a tree-based ensemble classifier by adjusting features, allowing instances initially predicted as unfavorable (such as low-quality ads) to be reclassified as positive (such as high-quality advertisements). This approach emphasizes model interpretability, aiming to provide prediction results and offer specific action recommendations for altering those predictions.

Therefore, XAI is correlated and vital in modern AI systems with accuracy, robustness, fairness, transparency, etc. [258,259]. Because with the continuous development of AI and its broader application in various fields, the demand for model interpretable XAI has increased significantly in all aspects of trust and acceptance, regulation and compliance, user experience, and cross-disciplinary applications, from medical diagnosis [260] to financial decision-making [261] to autonomous driving [13] and smart manufacturing [262].

It is worth noting that while pursuing model performance, it is essential to recognize that interpretability is an additional feature and an indispensable component of model design and optimization. Specifically, consider the following three points.

- 1) Researchers should consider the requirements of interpretability at the early stages of model design, enhancing the model's transparency by selecting appropriate network architectures and activation functions.

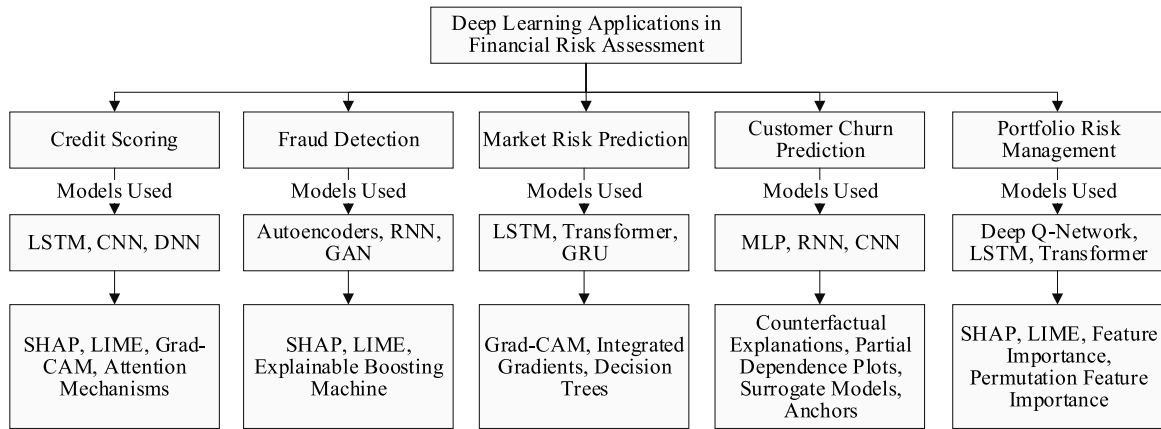


Fig. 22. Some application scenarios, models, and interpretability methods in financial risk assessment.

- 2)The model training process can be guided by introducing interpretability metrics, such as improving model interpretability by minimizing prediction uncertainty or maximizing the clarity of decision boundaries.
- 3)Interdisciplinary collaboration should be encouraged, combining knowledge from cognitive science, psychology, and other fields to develop explanatory frameworks and tools more consistent with human intuition.

By employing these strategies, we can move towards constructing DL models that are both efficient and comprehensible, thereby laying a solid foundation for the sustainable development of AI.

7. Application scenarios

With the development of interpretability research on DL models, more fields are using interpretability methods to solve problems, such as medical diagnostics, financial risk assessment, network security and management, recommendation systems, industrial modelling, Info. Fusion, etc.

7.1. Medical diagnostics

It is essential that the interpretability of AI medical diagnosis models, which can provide transparency and trust for people, contributes to the rationality and patient acceptance of the medical decision-making process [263–265]. In Fig. 21, medical diagnosis is subdivided into two major areas: text diagnosis and image diagnosis. Each domain is divided into three subdomains, and examples, corresponding models, and interpretability methods are provided for each subdomain. At the same time, in the following paragraphs, this review introduces some relevant examples in detail.

Wu et al. [266] introduced an interpretable retinal image captioning system based on expert-defined keywords and a specialized attention mechanism strategy. This system aims to enhance the interpretability of medical reports generated by ML models, particularly in the automatic analysis and report generation of retinal images, thereby increasing the trust of medical professionals in automated systems.

Wang et al. [267] introduced a new interpretable network, the two-branch bi-coordinate network, which aims to enhance the accuracy of classifying benign and malignant thyroid nodule ultrasound images. This study efficiently translates regional and boundary knowledge in thyroid imaging reporting and data systems into multi-granularity labels. It uses a bi-branch network to learn this knowledge from cartesian and polar images separately. An integrated attention mechanism then fuses features for classification prediction. Additionally, it elucidates the model's decision-making process through knowledge attention graphs,

thereby significantly improving model interpretability.

Wang et al. [268] have proposed a novel Multimodal Prototype Network (MProtoNet) designed to enhance the accuracy and interpretability of medical image classification. The model employs a multimodal attention mechanism and prototype activation restriction loss to improve prototype generation and activation precision. It also leverages position embedding and a multi-factor similarity calculation method to utilize positional information in medical images. These features enable MProtoNet to offer increased interpretability during disease classification tasks. The authors have conducted experiments on two public chest X-ray datasets, MIMIC-CXR and Open-I, using the MProtoNet framework to classify a variety of thoracic diseases from chest X-ray images. These diseases include conditions such as Atelectasis, Cardiomegaly, and Consolidation.

Wani et al. [269] proposed an interpretable DL approach named "DeepXplainer" for lung cancer detection, explaining its predictions. This method is based on a combination of CNN and XGBoost, integrated with XAI techniques, particularly the SHAP method, to enhance the transparency and credibility of the model.

Besides, Nematzadeh et al. [270] proposed an Ensemble-based Genetic Algorithm Explainer (EGAE), an interpretable AI approach grounded in genetic algorithms, for the automated segmentation and interpretation of predictive outcomes from DL models, specifically targeting melanoma detection datasets. The EGAE is designed to minimize user intervention during the image interpretation process, enhance the precision of explanations, and reduce the number of superpixels required. The authors also discussed the advantages of EGAE over LIME, including automation, interpretation accuracy, and reduction in hyperparameters.

7.2. Financial risk assessment

The accuracy and validity of financial risk assessment rely heavily on models' interpretability, which provides transparency and confidence in risk management decisions. Fig. 22 illustrates various application scenarios in financial risk assessment, detailing the DL models employed and the corresponding interpretability methods, along with case studies to aid readers' understanding.

Liu et al. [271] investigated the application of hybrid ML methods to improve the performance and interpretability of models in the financial time-series forecasting field, specifically focusing on predicting Bitcoin prices. A comparative study has been proposed that encompasses linear regression models (ordinary least squares and LASSO), LSTM networks, and DT regressors, emphasizing the importance of interpretability in predictive models. The paper also outlines several tangible application contexts, including investment decision support, risk management strategies, and policy and economic planning. Additionally, it discusses

Necessity of Explainability Analysis	Transparency		Debugging and Improvement		Compliance	
	• Explain model decision-making processes • Help security experts understand and trust model judgments		Identify and rectify potential model flaws		• Meet regulatory requirements • Ensure system decision-making complies with standards	

(a)

Practical Application Scenarios	Network Traffic Monitoring		Malware Detection		Intrusion Detection Systems	
	Application Examples: Employ explainability analysis to identify abnormal traffic and network attacks	Model Explainability: Explain how models detect abnormal behaviour and support investigations	Application Examples: Use explainability tools to analyse feature importance and uncover hidden malware patterns	Model Explainability: Explain judgement logic to assist in identifying new malware features	Application Examples: Utilise LIME, SHAP to enhance attack detection accuracy and explainability	Model Explainability: Use explainability analysis to understand alert sources and decision basis

(b)

Fig. 23. Applications and case studies of model interpretability in network security and management.

several techniques to improve the interpretability of models, including the use of DT, linear regression, and Lasso regression, ensuring that the models not only predict accurately but also provide clear and understandable insights into their decision-making processes.

Lin and Gao [272] have explored and implemented the interpretability of financial fraud detection models. The authors utilized the Shapley value and introduced a novel method known as group SHAP to assess the marginal contribution of each financial feature or group of features to the predictive outcome. This approach not only enhances the interpretability of the models but also significantly reduces computational time. Additionally, it uncovers common characteristics within the same industry and distinctive features between different sectors. The authors ultimately suggest that regulatory bodies or investors can leverage the interpretive results of the models to guide the focus of due diligence. Risk managers can also utilize the models' risk alerts when monitoring stocks in a portfolio to make informed risk management decisions.

Wang et al. [273] developed a network-based risk assessment framework for assessing the financial distress of small and medium-sized enterprises. This framework leverages network and event data, employing an adaptive interpretable graph contrastive learning model to determine the propagation effects of indirect events and generate embeddings of heterogeneous graphs. The model has been applied to companies listed on China's National Equities Exchange and Quotations (NEEQ) market to predict whether these firms will encounter financial distress in the following year.

Chen et al. [274] proposed a selective learning framework designed to discern which dataset segments can be explained by linear models, such as logistic regression, and which require more complex machine learning models to enhance accuracy without compromising interpretability. The authors strengthen transparency in the model's decision-making process through global explanations, local instance-based explanations, and local feature-based explanations, meeting the financial regulatory requirements for model interpretability and assisting financial institutions in better understanding and interpreting their credit assessment models' decisions. Specifically, in credit scoring, the authors utilized two datasets—the Taiwan payment system dataset and the 2011 Kaggle competition dataset—for testing and validating credit scoring models, thereby determining the creditworthiness of borrowers.

Zhang et al. [275] proposed an alternating minimization-based sparse least-squares classifier to enhance the accuracy and interpretability of credit risk assessment. Compared to traditional support vector machine classifiers, AMSLC (Alternating Minimization-based Sparse Least Squares Classifier) can simultaneously identify significant instances and features within the data, thereby providing more interpretable predictive results. For example, in financial risk assessment, recognizing which instances (such as loan applications) and features (such as economic indicators) are most critical for classification outcomes can assist risk assessors in making wiser decisions.

7.3. Network security and management

Network security and management are closely linked to interpretability studies because ensuring networked systems' security and management efficiency requires a clear understanding and traceability of the system's decision-making processes and behavioral patterns [276,277]. Fig. 23(a) illustrates the necessity of model interpretability in network security and management, while Fig. 23(b) provides specific examples of its application in the industry. Furthermore, the text elaborates on several practical cases to further elucidate its applications.

Nascita et al. [278] utilized XAI techniques to enhance the performance, reliability, and feasibility of multimodal multitask DL for network traffic classification. The researchers conducted experiments using the ISCX VPN-NONVPN dataset, which includes traffic encrypted through VPN and non-VPN traffic, simulating real-world network security environments. Moreover, they discussed the application of multitask learning in addressing multiple issues related to network traffic, such as malware detection/classification and traffic prediction. By employing XAI techniques, such as Deep SHAP and Integrated Gradients, they provided a global explanation of the model's predictions, aiding in understanding how the model makes classification decisions based on input data.

Morichetta et al. [279] introduced a methodology called EXPLAIN-IT, an explainable AI framework for unsupervised network traffic analysis. Researchers have emphasized using LIME technology to provide local explanations for model predictions, enhancing the model's interpretability. Meanwhile, in the application to the YouTube video quality classification problem, the EXPLAIN-IT method was used to analyze the encrypted network traffic of YouTube video streams to automatically identify different and relevant video resolution groupings caused by poor network performance. Experiments demonstrate that EXPLAIN-IT can be applied to network security, network traffic management, and service quality monitoring (Quality of Experience, QoE).

Guarino et al. [280] have employed DL techniques for the packet-level traffic prediction of mobile communication and collaboration apps like Skype, Teams, Webex, and Zoom. The authors discovered that the surge in traffic during the COVID-19 pandemic was notably associated with remote work, online education, e-commerce, and entertainment. The research team gathered traffic data from these applications during different activities, including audio calls, chats, and video calls, and used it to train and assess their DL models. Additionally, the research team incorporated XAI techniques, particularly the Deep SHAP approach, to elucidate the outcomes of the deep learning models' forecasts, thus bolstering confidence in the models' predictive results.

Dias et al. [281] proposed and implemented an interpretable and explainable hybrid intrusion detection system, a cybersecurity mechanism for analyzing network traffic, identifying suspicious activities, and issuing alerts to protect systems and data from cyber threats. Specifically, the authors used a simple .csv file containing test network data to simulate a distributed denial of service attack, particularly an HTTP flood attack. Additionally, the paper employed a naturally transparent

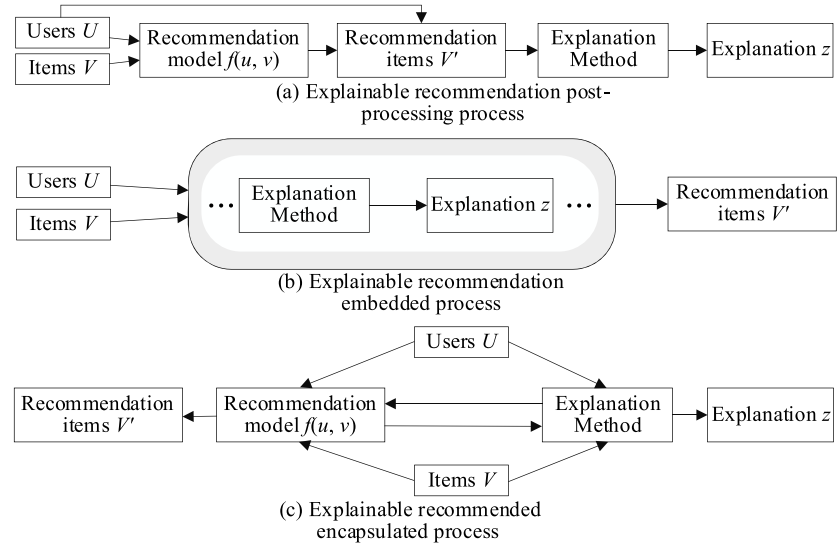


Fig. 24. Three explainable recommendation processes.

DT model capable of providing a reasonable level of explainability. Furthermore, the system can generate explanatory rules that provide reasons for each diagnosis and visually display this information through a user interface, thereby enhancing the system's interpretability and user trust.

Besides, Nkoro et al. [282] introduced an explainable DNN-based network intrusion detection system designed to detect and mitigate IoT-based cyberattacks targeting Metaverse learning platforms. The study utilized the latest cybersecurity IoT datasets and employed effective feature selection methods. The model was provided with visual, trustworthy, and quantitative explanations using SHAP and LIME explainability methods. The paper explored the application of Metaverse technology in virtual learning platforms, discussing how these platforms enhance traditional teacher-centered education with immersive and interactive learning environments. The paper also mentioned that during the COVID-19 pandemic, when most learning activities were conducted remotely, some educational institutions used VR-based platforms to teach and improve students' learning experiences.

7.4. Recommendation systems

Interpretability in recommender systems is a crucial factor in enhancing the user experience, helping to build trust between the user and the system, increasing user satisfaction, and facilitating more effective personalized services. In the context of explainable recommendations, the primary process of the explainable method encompasses several key elements: the user set U , the item set V , the explained recommendation system $f(u, v)$, the set of recommended items V' , the interpretation module, and the interpretation z of its output. Fig. 24 presents three prevalent processes for explainable recommendations: the explainable recommendation post-processing process, the explainable recommendation embedded process, and the explainable recommendation encapsulated process.

1)The explainable recommendation post-processing process. The post-processing method for generating explanations is independent of the recommendation results provided by the system. Regardless of any changes to the recommendation system, the explanation will remain consistent for the same user and item pair. Post-processing methods primarily focus on generating explanatory text, which can be categorized into three main approaches: rule-based, retrieval-based, and generative model-based.

2)The explainable recommendation embedded process. The embedded method integrates the interpretation module into the development of the recommendation system. The explanation module typically identifies the characteristics of items and selects those that have the most significant impact on recommendation accuracy as explanations. Most explanation modules in embedded methods are relatively shallow, utilizing techniques such as topic models [283], matrix factorization [284], and single-layer attention networks [285]. Additionally, a deep network $f(u, v)$ is constructed [286].

3)Encapsulated interpretable methods are implemented using reinforcement learning [287]. Specifically, the reinforcement learning framework is utilized to elucidate any recommendation model, thereby ensuring both the interpretability of the model and the quality of the explanations provided.

Ghazimatin et al. [288] proposed a new mechanism called PRINCE (Provider-side Interpretability with Counterfactual Explanations in Recommender Systems) to provide user interpretability in recommendation systems. PRINCE calculates a minimal set of user actions that, when altered, would change the recommendation outcome, thereby offering end-users explanations that are actionable, concise, and based on counterfactual evidence. In the context of online store recommendations, the PRINCE method can explain why a user received a specific product recommendation, for instance, based on the user's past purchasing behavior, reviews, or other activities related to the recommended item.

Wang et al. [289] proposed a novel recommendation system algorithm called the tree-enhanced embedding model, which aims to combine the generalization capability of embedding-based models with the interpretability of tree-based models to provide accurate and explainable recommendations. The paper mentions specific application scenarios, such as tourist attractions and restaurant recommendations on TripAdvisor. Specifically, the authors conducted experiments using two London and New York City datasets, corresponding to tourist attraction and restaurant recommendation tasks.

Wang et al. [290] introduced a novel method, large language model reasoning graphs, which leverages LLMs to construct personalized reasoning graphs to enhance the logic and interpretability of recommendation systems. The paper mentions specific application scenarios, such as modeling and reasoning about user behavior and preferences on e-commerce platforms (Amazon's beauty and apparel categories) and movie recommendations (MovieLens-1M dataset), to provide more

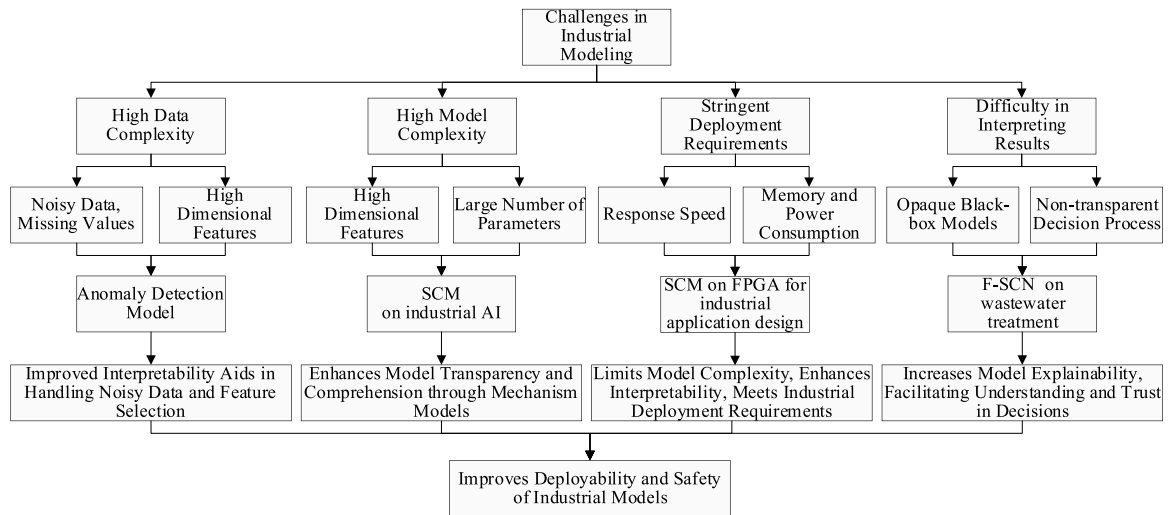


Fig. 25. The vital role of interpretability research in industrial modeling.

accurate personalized recommendations.

Zhang et al. [291] proposed an enhanced GNN model (Multi-Aspect enhanced GNNs) for personalized recommendation systems. The specific application scenario primarily involves e-commerce websites, where the recommendation system serves as a core component to address the issue of information overload. The recommendation system assists users in finding products that interest them among a vast array of goods, enhancing user experience and purchase efficiency. This model aims to improve the performance and interpretability of recommendations by fine-grained user preference modeling. Furthermore, the paper also conducted case studies using heat maps to demonstrate how preferences in different aspects affect users' final purchasing decisions, further proving the model's interpretability.

Shams et al. [292] introduced an algorithm named XAI-CROP, an innovative application of XAI principles that aims to enhance the transparency and interpretability of crop recommendation systems. Specifically, these systems aim to assist farmers in making informed decisions about crop selection based on data such as soil characteristics, historical crop performance, and current weather patterns to optimize yields. Additionally, it utilizes XAI techniques, particularly the LIME technique, to provide clear explanations for its recommendations.

7.5. Industrial modelling

There is a close and complex interrelationship between industrial modeling and interpretability studies, and the depth and breadth of industrial modeling require that the results be highly interpretable to ensure the model's reliability, validity, and transparency, which in turn drives the development of accurate and innovative industrial decision-making. Fig. 25 summarizes the critical role of interpretability research in addressing industrial modeling challenges, specifically how to improve model transparency and deployment security.

Wang and Felicetti [293] proposed a new stochastic learning model called Stochastic Configuration Machines (SCM) for industrial artificial intelligence applications. A vital feature of the SCM model is that it contains a mechanism model, a simulated or fuzzy expert system. Mechanism models are essential in cognitive learning and industrial informatics and help provide interpretable results. In addition, the concept of model complexity is discussed in the paper, which helps to understand the capabilities of the learning model and guides the development of learning algorithms and the setting of hyperparameters in deep learning, thus improving the interpretability of the model.

Li et al. [152] proposed the F-SCN, which enhances the model's transparency and interpretability by employing a TS-based fuzzy

inference system. The system estimates the total phosphorus concentration in discharged water during wastewater treatment. The system enhances the connection between parameters and their practical significance by specifying the relationship between inputs and outputs through predefined rules and determining model parameters based on data distribution characteristics. In addition, the F-SCN structure allows direct observation of the effect of fuzzy rules on the outputs and the extraction of nonlinear information through an enhancement layer, further deepening the understanding of the model processing mechanism.

Felicetti and Wang [294] explored the implementation of a neural network model, SCM, on FPGAs (Field Programmable Gate Arrays). SCM is a stochastic learner designed for industrial applications that improve learning performance and interpretability of results by restricting weights to binary values and using a mechanism model. The model is particularly suitable for scenarios in industrial environments with specific requirements for response speed, memory size, and power consumption. In this paper, the use of SCM in the production of hot-rolled steel is described.

Jiang et al. [139] introduced an autoencoder-based approach to identifying defects in various industrial products, including textiles, semiconductors, pharmaceuticals, and construction materials. The algorithm incorporates interpretability scores throughout the training and testing phases and integrates an innovative interpretability-aware loss component during training to bolster the explainability of the anomaly detection process. The resultant attention maps aid in directly elucidating detection outcomes, offering valuable insights for those working in the industry.

Furthermore, In the hydrometallurgical purification process of nonferrous metals, Sun et al. [295] proposed an interpretable method for partitioning operating conditions in complex industrial processes. This approach, grounded in a Global Spatial Structure Compensation-Local Temporal Information Aggregation Self-Organizing Map (GSFC-LTA-SOM), reformulates the partitioning challenge into one of spatiotemporal feature extraction. The method segments industrial variables into interpretable, time-specific groups by introducing a locally separable aggregation model supported by an informational backbone network. This segmentation enables the model to discern the underlying factors contributing to condition occurrences, thereby offering a solid foundation for root cause analysis that is both reliable and interpretable.

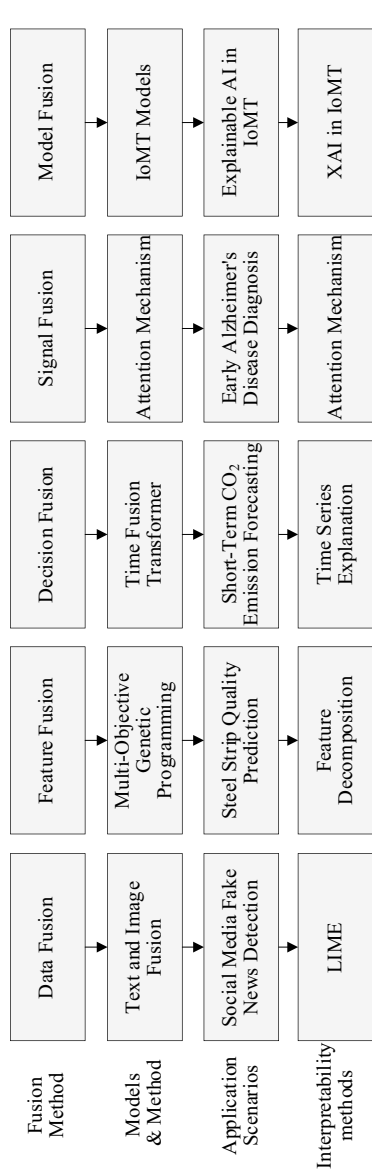


Fig. 26. Several interpretable methods/models and application scenarios relevant to the field of Info. Fusion.

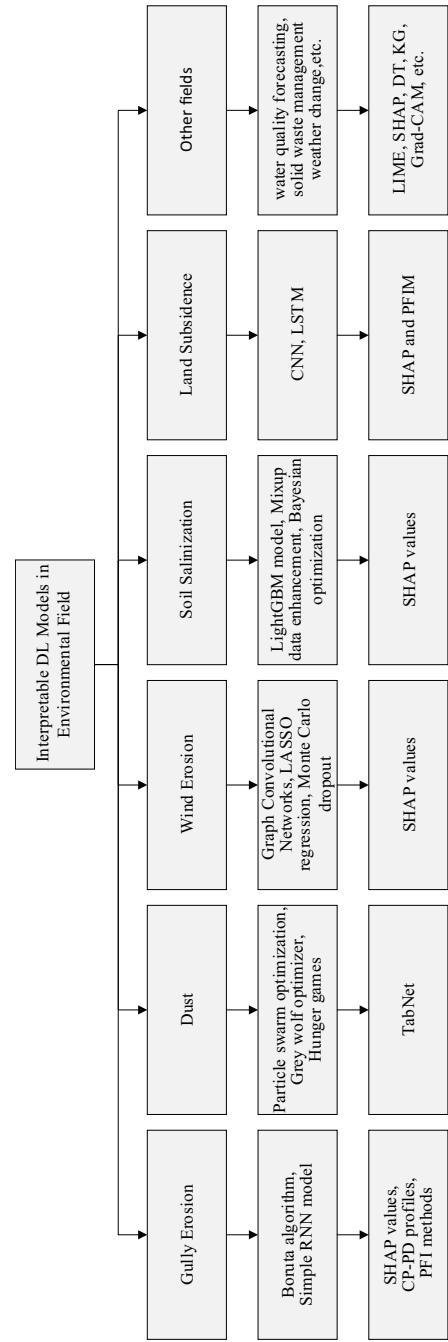


Fig. 27. Some interpretable methods/models and applications in environmental sciences.

7.6. Info. Fusion

Info. Fusion is a dynamic and ongoing process that involves detecting, associating, estimating, and integrating information from multiple sources across various levels and aspects. The goal is to accurately understand the target being measured, ensure consistency in estimation, and provide real-time evaluation [296]. With the development of AI, when this kind of multi-source information is used to build multi-modal neural networks, the complexity of the enormous model makes the autonomous training and learning process extremely opaque. Therefore, Info. Fusion is, to some extent, a crisscrossing field involving many aspects, such as medical systems [297], network security [298], visual intelligence [299], and other fields. Interpretability analysis for such multimodal models becomes particularly important. Fig. 26 visualises the practical application of the interpretable model in Info. Fusion.

Zamil and Charkari [300] proposed a fusion approach for improving the detection and interpretability of fake news on social media. The researchers developed a fake news detection model based on DL techniques that improve detection reliability by fusing text and image features. In addition, to enhance the interpretability of the model, LIME was used to identify hyperpixels that contribute to the interpretability of the model.

Wang et al. [301] introduced an interpretable modeling approach that utilizes multi-objective genetic programming for the quality prediction of steel strips in the continuous annealing production line. A novel method called feature decomposition is proposed, which fuses mechanism knowledge with data analysis to tackle the challenge of high-dimensional input features. The paper also highlighted that IMA can develop interpretable models, illustrating the relationships between process parameters and product quality. The interpretability of the method is validated through the visualization and analysis of the final analytical model.

Xu et al. [302] introduced an interpretable medical DL framework for early diagnostic analysis of Alzheimer's disease through an attention mechanism based on logit constraints and Graph-based Multi-scale Fusion (GMF), or LA-GMF for short. The model improves the accuracy and interpretability of early AD diagnosis. It is a standard Info. Fusion method. With the LA module, the model can identify and focus on local brain regions associated with AD pathology, which helps improve the model's transparency and credibility.

Wu et al. [303] proposed a novel and interpretable short-term CO_2 emission forecasting model called ICEEMDAN-EWT-JADE-TFT. This model incorporates a flexible two-stage decomposition method, an adaptive differential evolutionary algorithm (JADE), and a Time Fusion Transformer (TFT). Researchers mention using the TFT model's explanatory power to analyze the impact of the sub-sequences. It provides an interpretable analysis of the model's forecasting results, which helps policymakers to understand the dynamics behind the results better and make more informed decisions accordingly. This composite forecasting model, which integrates multiple methods and algorithms, works together to improve the accuracy of forecasts of short-term CO_2 emissions.

On the other hand, Wani et al.'s [304] perspective focuses on applying XAI to healthcare, especially in the Internet of Medical Things (IoMT). The paper mentions several specific application scenarios, such as using XAI to improve the transparency and credibility of medical decisions in an IoMT environment by integrating data collected by smart devices, using XAI models for disease diagnosis and prediction, improving patient monitoring and treatment processes through XAI technologies, such as remote health monitoring and personalized medicine, using Info. Fusion technologies to collect, analyze, and share healthcare data to enhance the understanding of patient conditions.

7.7. Environmental sciences

At the same time, we note that the application of interpretable DL

models in the environmental field is increasing significantly. As shown in Fig. 27, interpretive DL models have demonstrated outstanding potential in data analysis and decision support, especially in gully erosion, dust, and subsidence. By providing interpretability of model decision-making processes, these models enhance researchers' understanding of complex environmental data and improve the transparency and credibility of relevant policy formulation.

Gholami et al. [182] employed DL models, including the simple RNN and GRU, to map gully erosion susceptibility in the Shamil-Minab plain, Hormozgan province, Iran. The authors first used the Boruta algorithm to identify seven key features that control gully erosion, and then they utilized these features along with an inventory map of gully erosion to generate spatial maps of gully erosion. Performance assessment indicated that the Simple RNN model outperformed the GRU model regarding predictive accuracy. Furthermore, the authors applied SHAP values, CP-PD profiles, and PFI methods to interpret the output of the DL models, finding that soil silt content, vegetation cover (NDVI), and land use type had the most significant impact on the model predictions. The findings contribute to a better understanding of the distribution of gully erosion and aid in the development of effective land management and conservation strategies to mitigate the adverse effects of gully erosion, supporting sustainable watershed management.

Razavi-Termeh et al. [305] proposed a method that combines satellite remote sensing technology with an optimized Attentive Interpretable Tabular Learning (TabNet) model, using three metaheuristic algorithms—particle swarm optimization, grey wolf optimizer, and hunger games search—to enhance the model's predictive accuracy. The TabNet model is a core component of the attention mechanism and interpretability. It provides transparency in the model's decision-making process by using the attention mechanism to select and weigh relevant features dynamically. This approach addresses the issue of identifying and mapping the susceptibility of dust events in Bushehr province, Iran, to mitigate the impacts of dust on human health, air quality, and food safety.

Gholami et al. [306] also studied the first attempt to model wind erosion hazards in the arid central region of Iran, Semnan Province, using a novel approach that combines Graph Convolutional Networks (GCNs) with LASSO regression and Monte Carlo dropout. The authors identified and mapped 15 potential factors controlling wind erosion and then used LASSO regression to select the eight most important ones, including soil properties, vegetation cover, land use, wind speed, and evaporation. The GCN model predicted that about 65 % of the province is highly or very highly susceptible to wind erosion. They further evaluated the performance and interpretability of the GCN model using ROC-AUC and SHAP values from game theory, demonstrating excellent predictive accuracy. This research method is highly recommended for future spatial predictions of wind erosion hazards in other arid environments worldwide.

Wang et al. [307] proposed an integrative modeling approach for heterogeneous soil salinity based on sparse ground samples and remote sensing images. This method uses machine learning technology to rapidly monitor large-scale spatial soil salinity, especially in arid and semi-arid areas where it is difficult to obtain soil salinity information. An integrated framework was proposed in the study, using a lightweight gradient boosting machine (LightGBM) to predict soil salt content (SSC). The LightGBM model combined with Mixup data enhancement and Bayesian optimization can not only accurately predict soil salinity content but also, through the application of SHAP values, make the decision logic of the model clear and interpretable. This interpretability is crucial for researchers and decision-makers to understand the logic behind the model's predictions, helping to enhance the model's acceptance and effectiveness in practical applications.

Liu et al. [308] utilized an interpretable CNN to predict the susceptibility of land subsidence areas in China, and employed the SHAP theory to interpret the predictive results of the CNN model. The purpose of the study is to understand better and prevent land subsidence

disasters, and to assess the exposure of the population, Gross Domestic Product (GDP), railways, and roads in areas with high susceptibility to land subsidence in the future. Moreover, Rahmani et al. [309] identify critical factors affecting land subsidence by combining field surveys with particle swarm optimization. They utilize DL models, namely CNN and LSTM, to predict the susceptibility to land subsidence hazards. Research demonstrates the efficiency of CNN and LSTM models in land subsidence risk assessment. It also improves model interpretability through interpretation techniques such as SHAP and PFIM. The paper provides an accurate and transparent prediction tool that can land subsidence risk management.

In addition, interpretability DL is also extensively applied in various domains such as water quality forecasting [310], solid waste management [311], weather change [312], etc. In these applications, the interpretable DL models are essential, enabling us to comprehend the decision-making processes of the models. This fusion of DL with explainability plays a crucial role in fostering the sustainable and healthy development of the ecological environment.

8. Challenges and future developments

Interpretability research mainly focuses on the underlying reasons for the model's decision-making. However, it still faces a series of challenges. These challenges not only test the boundaries of existing technologies but also drive the direction of future research. Firstly, a lack of uniform standards for assessing interpretability methods' effectiveness makes direct comparisons difficult between different studies. Secondly, enhancing interpretability without sacrificing model performance is a technical challenge. Thirdly, with the growing awareness of data privacy protection, ensuring user data security while providing model explanations has become an urgent issue. Fourthly, the model's black-box nature makes internal decision-making processes challenging to understand, which adds to the difficulty of interpretability. Finally, the current interpretability methods lack generalizability and are often limited to specific models or tasks.

It is noteworthy that SCN possesses potential contributions to interpretability research. This model configures the hidden layer node parameters through a supervision mechanism and can track each node's parameter value, thereby understanding the model's decision-making process on different tasks, which offers local interpretability. However, globally, the complexity of the SCN structure makes its internal decision-making process challenging to understand. For instance, balancing model performance and interpretability in SCN, generalizing it to different types of tasks, and integrating it with traditional explanation methods are all challenging aspects of current research. Info. Fusion also encounters challenges, particularly the absence of standardized criteria for assessing interpretability methods within the field. Given the diverse fusion strategies and model architectures utilized across studies, comparing the effectiveness of various methods proves challenging. For instance, one study may employ method A to elucidate the model's decision-making process with an 80 % accuracy rate, while another study may use method B with an accuracy of 85 %. However, determining the superior method becomes problematic due to differences in evaluation indicators. Balancing model interpretability with performance remains a technical hurdle. Complex structures in Info. Fusion models are often necessary to handle the intricate relationships among multiple data sources, yet this complexity can hinder interpretability. DNN models may achieve high diagnostic accuracy in specific medical diagnosis scenarios, but their internal decision-making processes are inscrutable to medical professionals and patients, impacting trust and practical utility.

In the face of these challenges, future research must explore multiple directions. Firstly, establishing a unified evaluation framework is crucial for comparing and selecting interpretability methods. If consistency cannot be achieved in all areas, efforts must be made to achieve uniformity in specialized areas. Secondly, researchers should endeavor to

Table 32

List all abbreviations for professional terms.

Full name	Abbreviation
Ablation-based Class Activation Mapping	Ablation-CAM
Accumulated Local Effects	ALE
AND-OR Graph	AOG
Artificial Intelligence	AI
Augmented Fairness	AuFair
Average Decision path Length	APL
Axiom-based Gradient-CAM	XGGrad-CAM
Bayesian Additive Regression Trees	BART
Case-Based Reasoning-LIME	CBR-LIME
Ceteris Paribus	CP
Ceteris Paribus and Partial Dependence Profiles	CP-PD profiles
Class Activation Map using the Principal Components	Eigen-CAM
Class Activation Mapping	CAM
Classification And Regression Tree	CART
Computer Vision	CV
Continuous/discrete Rule Extractor via Decision tree induction	CERD
Convolutional Neural Networks	CNNs
Cross-validated Permutation Feature Importance	CVPFI
Decision Tree	DT
Deep Belief Networks	DBN
Deep Learning	DL
Deep Learning Important Features	DeepLIFT
Deep Neural Networks	DNNs
Differential Cumulative Local Effect	DALE
Empirical Risk Minimization	ERM
Ensemble-based Genetic Algorithm Explainer	EGAE
Evolving Stochastic Configure Network	E-SCN
Explainable Artificial Intelligence	XAI
eXtreme Gradient Boosting	XGBoost
Forest Based Tree	FBT
Full Hyperspectral Image	HIS
Fuzzy Frees and Forests	FTF
Fuzzy Rule-Based Classification System	FRBCs
Fuzzy Rule-Based Systems	FRBSs
Fuzzy Stochastic Configuration Network	F-SCN
Gated Recurrent Unit	GRU
Generalized Additive Model	GAM
Generalized Additive Models with Pairwise Interactions	GA ² Ms
Generalized Additive Models with Structured Interactions	GAMI-Net
Generalized Gradient-based Visual Explanations for Deep Convolutional Networks	Grad-CAM++
Generative Adversarial Networks	GANs
Genetic Algorithm	GA
Global Average Pooling	GAP
Global-LIME	G-LIME
Gradient Boosting	GB
Gradient-weighted Class Activation Mapping	Grad-CAM
Graph Capsule Networks	GraCapsNets
Graph Neural Networks	GNNs
Graph Neural Networks Local Interpretable Model-agnostic Explanations	GraphLIME
Graph Convolutional Networks	GCNs
Incremental Permutation Feature Importance	iPFI
Information Bottleneck	IB
Information Maximizing Generative Adversarial Networks	InfoGAN
Integrated Score-CAM	IS-CAM
Interpretable Generalized Additive Neural Networks	IGANN
Interpretable Linguistic Fuzzy Rules	ILFRs
Jacobian norm and Selective Input Gradient Regularization	J-SIGR
Knowledge Graph	KG
Knowledge Guided Interpretable Disease Prediction	KGxDP
Knowledge-aware Path Recurrent Network	KPRN
Large Language Models	LLMs
Layer-wise Class Activation Map	Layer-CAM
Local Interpretable Model-agnostic Explanations	LIME
Long Short-Term Memory	LSTM
Machine Learning	ML
Model Gradient Similarity	MGS
Multi-Layer Perceptrons	MLPs
Multimodal Prototype Network	MProtoNet
Natural Language Processing	NLP
Neural Additive Model	NAM
Neural Local Smoother	NLS
Neural-Backed Decision Trees	NBDTs

(continued on next page)

Table 32 (continued)

Full name	Abbreviation
Partial Dependence profiles	PDP
Path-enhanced Recurrent Network	PeRN
Permutation Feature Importance	PFI
Permutation-based Feature Importance Test	PermFIT
Policy-Guided Path Reasoning	PGPR
Provider-side Interpretability with Counterfactual Explanations in Recommender Systems	PRINCE
Random Forest	RF
Randomized Input Sampling for Explanation	RISE
Recurrent Neural Networks	RNNs
Referential Expression Grounding	REF
Regional Tree Regularization	RTR
Research Questions	RQs
Root Mean Square Error	RMSE
Score-Weighted Visual Explanations for Convolutional Neural Networks	Score-CAM
Self-Explaining Neural Networks	SENN
Self-Interpretable Graph aNomaly dETection	SIGNET
Shallow-to-Deep Feature Interaction for Knowledge Graph Embedding	SDFormer
Shapley Additive Explanations	SHAP
Smoothed Score-CAM	SS-CAM
Stacked Auto-Encoders	SAE
Stochastic Configuration Machines	SCM
Stochastic Configuration Network	SCN
Takagi-Sugeno	TS
Time Fusion Transformer	TFT
Tree Supervision Loss	TSL
Variational Information Bottleneck for Interpretation	VIBI
Visual Question Answering	VQA

develop new model architectures and training strategies to optimize model performance and interpretability synergistically. Applying causal inference theory or domain knowledge and knowledge distillation [313–316] is expected to provide deeper explanations for DL. Developing user-friendly explanation methods will enable non-experts to understand and trust the model’s decisions. As the importance of interpretability grows in legal and ethical aspects, researching how to implement and assess interpretability within these frameworks becomes particularly important. Finally, improving the knowledge and skills of researchers and practitioners in interpretability through education and training will support the development of interpretability research and application. These research directions will advance the maturity of DL model interpretability technologies and lay a solid foundation for the healthy development of AI.

9. Conclusions

With the DL’s widespread application in various fields, the mistrust in its black-box nature has gradually deepened, which makes the model’s un-interpretability a significant challenge, which has led more and more scholars to begin research on DL interpretability. This review provides a comprehensive review of DL interpretability research. Firstly, the importance of interpretability is emphasized, divided into four categories: active, passive, supplementary, and integrated explanations, introducing some typical interpretability algorithms. Subsequently, the commonly used evaluation indicators of interpretability models are introduced. Next, the relationship between interpretability and model performance is discussed. Then, the application of interpretive methods/models in several application scenarios is explored. Finally, given the current challenges, the great potential for developing future interpretability research is pointed out. The full review systematically examines the issue of DL interpretability, with a particular focus on interpretability research in CV tasks, with depth and breadth. The review provides valuable references for future research, promotes the development of XAI technologies, and accelerates the implementation and deployment of AI in various industries. Eq. (1)–(80)

Abbreviation

Finally, for the convenience of readers, Table 32 collects the abbreviations of all professional terms used throughout the review.

CRedit authorship contribution statement

Biao Xu: Investigation, Formal analysis, Software, Writing – original draft. **Guanci Yang:** Investigation, Funding acquisition, Supervision, Resources, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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